



Radiation Tolerant Versal AI Edge Series: Data Sheet

DS955 (v1.0) January 13, 2025

Product Specification

General Description

The radiation tolerant (RT) AMD Versal™ XQR devices are the industry's first adaptive SoCs, combining customized, heterogeneous hardware solutions for a wide variety of space-flight applications. Versal XQR adaptive SoCs extend the benefits of commercial silicon with ruggedized organic packaging, tested and qualified to the stringent AMD class B flow, with full M-grade operating temperature range support. Radiation effects are characterized for total ionizing dose and single-event effects. Versal AI Edge XQR devices contain transformational features like an integrated silicon host interconnect shell and AI and DSP engines, programmable logic, and processing system, providing superior performance/watt over conventional FPGAs, CPUs, and GPUs. The Versal AI Edge XQR device delivers high performance, low latency AI inference for multi-mission payloads in space. More than just AI, the Versal AI Edge XQR device accelerates the whole application from sensor to AI to real-time control. As an adaptive SoC, the Versal AI Edge XQR device allows developers to rapidly evolve their sensor fusion and AI algorithms. Unlimited on-orbit reconfiguration allows Versal XQR adaptive SoCs to target space applications such as on-board processing, digital payloads, and remote sensing, enabling high-volume data processing and autonomous decision-making in flight. The [AMD Space Secure Site](#) provides access to design guidelines and resources specific to space applications. This data sheet is part of an overall set of documentation on the Versal architecture-based devices available on the AMD website at <https://docs.amd.com>.

Summary of Radiation Parameters

This product is intended for use in space environments and offered in the AMD class B manufacturing and process flow. The AMD 7 nm technology is developed with innovative configuration memory and block RAM design for single-event effects (SEE) mitigation, with optimized SEU and SEL design rules and strategic implementation of SEU enhanced cells. The Versal adaptive SoCs use more than 50 proprietary, patented circuit design and layout techniques to reduce the SEU cross-section. Block RAM, UltraRAM, and accelerator RAM feature error detection and correction (EDAC) capability for high-performance SEU mitigation.

AMD Adaptive Computing is creating an environment where employees, customers, and partners feel welcome and included. To that end, we're removing non-inclusive language from our products and related collateral. We've launched an internal initiative to remove language that could exclude people or reinforce historical biases, including terms embedded in our software and IPs. You may still find examples of non-inclusive language in our older products as we work to make these changes and align with evolving industry standards. Follow this [link](#) for more information.

Table 1: Summary of Radiation Rates for GEO

Symbol	Description	Min	Typ	Max	Units
TID	Total ionizing dose	–	100	120	Krad (Si)
SEL	Single-event latch-up immunity ¹	–	80	–	MeV-cm ² /mg
SEU _{CRAM}	Single-event upset in configuration RAM ^{2, 3, 4}	–	6.5×10^{-12}	–	Event/bit/day
SEU _{BRAM}	Single-event upset in block RAM ^{2, 3, 4}	–	9.4×10^{-10}	–	Event/bit/day
SEU _{URAM}	Single-event upset in UltraRAM ^{2, 3, 4}	–	1.8×10^{-10}	–	Event/bit/day
SEU _{XRAM}	Single-event upset in accelerator RAM ^{2, 3, 4}	–	–	–	Event/bit/day

Notes:

1. SEL immunity at maximum recommended operating voltage and junction temperature.
2. Error bar is $\pm 40\%$ at 90% confidence interval.
3. GEO estimate is based on CREME96 orbital models for worst-case conditions (solar minimum) and 100 mils of Al shielding.
4. SEU rates are calculated without deployment of EDAC encoding.

Table 2: Summary of Radiation Rates for LEO

Symbol	Description	Min	Typ	Max	Units
TID	Total ionizing dose	–	100	120	Krad (Si)
SEL	Single-event latch-up immunity ¹	–	80	–	MeV-cm ² /mg
SEU _{CRAM}	Single-event upset in configuration RAM ^{2, 3, 4}	–	3.5×10^{-9}	–	Event/bit/day
SEU _{BRAM}	Single-event upset in block RAM ^{2, 3, 4}	–	1.2×10^{-7}	–	Event/bit/day
SEU _{URAM}	Single-event upset in UltraRAM ^{2, 3, 4}	–	2.9×10^{-8}	–	Event/bit/day
SEU _{XRAM}	Single-event upset in accelerator RAM ^{2, 3, 4}	–	–	–	Event/bit/day

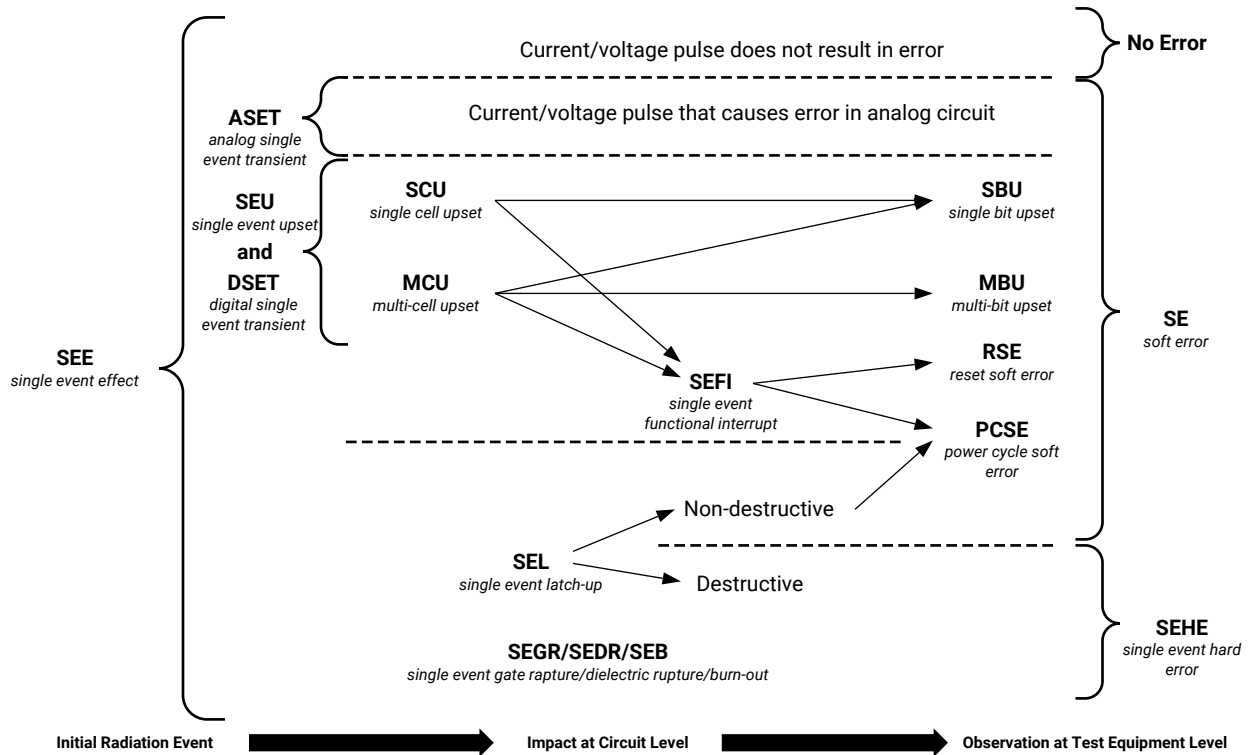
Notes:

1. SEL immunity at maximum recommended operating voltage and junction temperature. Proton test total fluence at 10^{12} p/cm².
2. Error bar is $\pm 40\%$ at 90% confidence interval.
3. LEO estimate is based on CREME96 AP8_{MAX} orbital models for worst-case conditions (solar minimum) at 500 Km and 51.6° inclination.
4. SEU rates are calculated without deployment of EDAC encoding.

Table 3: Single Event Effect Type Descriptions

Single Event Effect (SEE) Type	SEE Signature	Target/Comments
Single event upset (SEU)	Corruption of the information stored in a memory element	Memories, latches in logic devices. Composed of single bit upset (SBU) or multiple bit upset (MBU).
Single event transient (SET)	Impulse response of certain amplitude and duration	Analog and mixed-signal circuits. Can lead to SEU if latched in memory cell.
Single event latchup (SEL)	High-current conditions	CMOS devices. Might lead to hard failure.
Single event functional interrupt (SEFI)	Corruption of a datapath leading to loss of normal operation	Complex devices with built-in state machine/control sections or systems.

Figure 1: Types of Alpha Particle and Terrestrial Cosmic Ray Induced Soft Error Event Types



X27079-091422

This figure is reproduced, with permission, from the JEDEC document *JESD89B*, Figure 1.

Weibull Fit Parameters

Weibull curves were fitted to the measured test data. See the radiation test summary data in the radiation reports in the XQRVE2302 area of the [AMD Space Secure Site](#). A summary of the best fit Weibull parameters for the various components of the XQRVE2302 is shown in the following table.

Table 4: Weibull Fit Parameters for Protons

Parameter	SEE Type	MeV	W	S	σ_{sat}
Configuration RAM	SEU per bit	5	32	2.8	3.0×10^{-17}
Block RAM	SEU per bit	5	28	2.7	1.2×10^{-15}
UltraRAM	SEU per bit	5	26	2.1	3.1×10^{-16}
XiIEM IP	SEFI per block	N/A	N/A	N/A	N/A
Processor (PS)	SEFI	4	18	0.9	2.2×10^{-11}
AI Engine-ML (AIE-ML)	SEFI				
GTYP transceiver	SEFI per quad				

Table 5: Weibull Fit Parameters for Heavy Ions

Parameter	SEE Type	LET _{th}	W	S	σ _{sat}
Configuration RAM	SEU per bit	0.8	34	1.65	1.6 x 10 ⁻¹⁰
Block RAM	SEU per bit	0.8	12	1.80	1.0 x 10 ⁻⁹
UltraRAM	SEU per bit	0.8	24	1.30	3.6 x 10 ⁻¹⁰
XiISEM IP	SEFI	1.0	18	1.65	2.5 x 10 ⁻⁶
Processor (PS)	SEFI	0.8	18	1.68	7.0 x 10 ⁻⁵
AI Engine-ML (AIE-ML)	SEFI				
GTYP transceiver	SEFI per quad				

Single Event Latch-up Results

Table 6: SEL Results

Junction Temperature (°C)	Proton Kinetic Energy (MeV)	Cumulated Fluence	SEL Events	LET Range	Cumulated Fluence	SEL Events
		(p/cm ²)	(Protons)	(MeV-cm ² /mg)	(ions/cm ²)	(Heavy-ions)
30	64-400	8.7 x 10 ¹¹	0	1-80	> 1 x 10 ⁷	0
50		1.4 x 10 ¹²	0			0
100		3.7 x 10 ¹¹	0			0
120		1.1 x 10 ¹²	0			0

Key Ruggedized Package Features

- Ruggedized packaging
- MIL-STD-883 group D Qualification testing
- Military (M) temperature support of -55°C to +125°C
- Full range extended temperature testing
- Mask set control
- Full compliance with MIL-PRF-38535 Pb content for microcircuit finishes
- Longer-term availability
- Anti-counterfeiting features
- Available information assurance (IA) methodology
- Available anti-tamper (AT) technology

Ruggedized Packaging

XQR ruggedized Versal devices are offered in lidless packages, which simplifies the board-level assembly process for applications requiring conformal coating. In the conformal coating process, boards go through a caustic etching process to achieve the required conformal coating adherence. The caustic etching material or other corrosive chemicals can become trapped inside of non-ruggedized forged or stamped lid packaging, leading to reliability concerns with flip-chip packaging. With the XQR ruggedized packages, the lidless packages significantly simplify the cleaning and manufacturing process, allowing the device to be fully flushed prior to sealing the device/board with conformal coating.

MIL-STD-883 group D specification stress testing is performed prior to production release of the XQR devices. Qualification reports are available for the Versal architecture XQR ruggedized devices. MIL-STD-883 group D Qualification testing for defense-grade products includes the following:

- Physical dimensions (TM 2016)
- Thermal shock (TM 1011 condition B 15 cycles)
- Temperature cycling (TM 1010 condition C 100 cycles)
- Moisture resistance (TM 1004)
- Vibration—Variable frequency (TM 2007 condition A minimum)
- Salt atmosphere (TM 1009 condition A minimum)

Full Range Extended Temperature Testing

Ruggedized XQR devices are offered in Military (M) temperature grades (–55°C to +125°C). Full-range extended temperature testing is offered on XQR ruggedized devices and includes full functional and parametric testing at room temperature as well as hot and cold temperature extremes. AMD tests 100% of all die at wafer sort and 100% of all devices at final production testing. AMD continuously improves the test coverage of its products through advancements in design for test (DFT) methods spanning digital logic, IP cores, memory elements, I/O cells, and many other areas. AMD achieves very high test coverage with industry-leading manufacturing and foundry processes, as confirmed by a low PPM failure rate and low customer return rates; for more information see [Quality and Reliability](#).

Mask Set Control

Mask set control is valuable for secure and critical applications where a mask set change can trigger a detailed silicon-level analysis, re-verification, and/or re-certification process. Ruggedized XQR products have a locked mask set throughout the production life cycle. In the event that a technical change must be made, a formal customer notification process is required for these XQR devices.

Tin Whisker Mitigation with MIL-PRF-38535 Practices

Tin whisker growth concerns are addressed by the ruggedized package used for the Versal XQR devices. All solder connections are alloyed with a minimum of 3% lead by weight or are encapsulated with epoxy per MIL-PRF-38535M requirements. The BGA balls are tin/lead and are compatible with a leaded solder reflow profile during board mounting.

Anti-Counterfeiting Features

Versal architecture-based XQR devices offer multiple levels of anti-counterfeiting protection using unique laser marking, which uses micro-watermarking characters and patterns, whereby certain elements can be verified by the end-user. Other elements can only be verified by AMD.

Summary of Features

Architecture

Versal AI Edge XQR devices are built around a programmable network on chip (NoC), which enables seamless memory-mapped access to the full height and width of the device. Adaptive SoCs comprise: a multicore scalar processing system, SIMD VLIW AI Engine accelerators for artificial intelligence and complex signal processing, and programmable logic (PL). Together, these form a platform for fast time-to-market (TTM) compute acceleration for cloud, edge, and networking applications. The platform management controller (PMC), adjacent to the PS, is responsible for booting and configuring the device. Versal AI Edge XQR devices have I/O and memory controllers on the north and south edges of the device and serial transceivers on the east and west edges. The NoC spans full height and width of the device.

Compute and Acceleration

Versal AI Edge XQR devices have an array of signal processing cores that are highly optimized for functions in machine learning, convolutional neural networks, image processing, radio, radar, spectrometry and other instrumentation applications. The array consists of AI Engine-ML tiles with increased throughput for ML/AI inference workloads, Bfloat16 and INT4 for application precision, and local memory tiles.

Every Versal AI Edge XQR device has processing systems (PS) that comprise a dual-core Arm® Cortex®-A72 (APU) and a dual-core Arm Cortex-R5F (RPU). The processing systems includes a number of peripherals for communication standards and can communicate with DDR memory via the programmable NoC as well as on-chip memory and local cache.

The PL is made up of configurable logic blocks, containing 6-input look-up tables (LUTs) and flip-flops; different-sized memory blocks; 36 Kb block RAM and 288 Kb UltraRAM; digital signal processing (DSP) blocks; and a wealth of interconnect, switches, and MUXes to connect blocks together. All resources are arranged in columns. The PL is divided into regions that are a fixed height. Each region has its own clocking capabilities and NoC access points.

Platform Management

The platform management controller resides adjacent to, but is independent from, the processing system. It is responsible for the boot and configuration of the device from the primary boot source. The platform management controller is also responsible for configuring the PL, which can be configured before or after the processing system. It also controls encryption, authentication, system monitoring, and device debug capabilities of the platform.

Connectivity

The south edge of the Versal AI Edge XQR devices contain a number of XPIO banks and associated memory controllers to read from and write to DDR4 and LPDDR4 memory. XPIO can be used independently from the dedicated memory controllers for many functions, including any with soft memory controllers created in the PL. The PL can also contain integrated blocks for high-value functions, such as the integrated block for PCIe (PL PCIE) and 40G Ethernet MAC.

XQRVE2302-1MSBSSRA784 Features

Table 7: XQRVE2302-1MSBSSRA784 (AI Edge Series)

Feature	Resource	Specification
AI Engines and DSP Engines	AIE-ML tiles	34
	AI Engine data memory	17 Mb
	AIE-ML shared memory	68 Mb
	DSP Engines	464
Programmable logic (PL)	System logic cells	329,000
	LUTs	150,272
	NoC to PL primary/secondary ports	5
	Distributed RAM	4.6 Mb
Memory	Total block RAM	5.4 Mb
	UltraRAM	43.6 Mb
	Accelerator RAM	32 Mb
	Total PL memory	86 Mb
	DDR memory controllers (DDRMC)	1
	DDR memory bus width	64
Processing system (PS)	Application processing unit (APU)	Dual-core Arm Cortex-A72, 48 KB/32 KB L1 cache w/ECC, and 1 MB L2 cache w/ECC
	Real-time processing unit (RPU)	Dual-core Arm Cortex-R5F, 32 KB/32 KB L1 cache, and 256 KB TCM w/ECC
	Memory	256 KB on-chip memory w/ECC
	Connectivity	Ethernet (x2), UART (x2), CAN-FD (x2), USB 2.0 (x1), SPI (x2), I2C (x2)
Serial transceivers	GTYP transceivers	8 GTYP (26.5625 Gb/s)
Integrated protocol IP	PCI Express	1 x Gen4x8
	Multirate Ethernet MAC	1
Platform	Platform management controller (PMC)	Boot, security, safety, monitoring, high-speed debug, SEU mitigation (XiI SEM)
Ruggedized Package	Organic BGA, Sn-Pb solder balls	SSRA784, 23 mm x 23 mm, 0.8 mm pitch
I/O	I/O	216 XPIO, 22 HDIO, 78 MIO, 8 GTYP

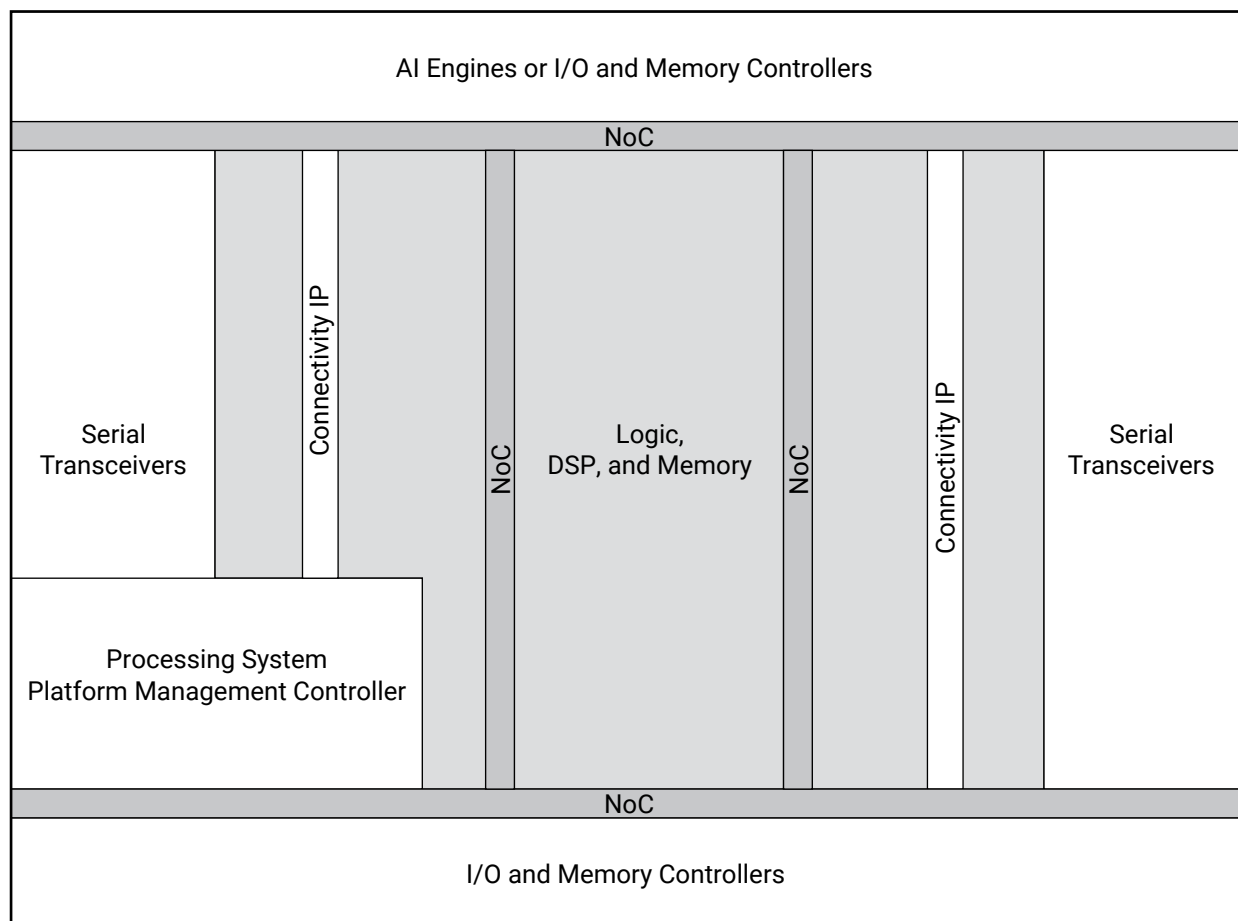
Table 8: XQRVE2302-1MSBSSRA784 I/O

I/O Type	Maximum I/O
XPIO DDR only	132
XPIO DDR+PLL	30
XPIO PL only	54
HDIO	22
MIO	78
GTYP	8

Device Layout (Architecture and Interconnect)

Versal AI Edge XQR devices are built from a library of building blocks dedicated to processing, compute, acceleration, and connectivity. The following figure shows the layout of an adaptive SoC.

Figure 2: Versal AI Edge XQR Device Layout



X29558-082624

Serial transceivers are located on the east and west edges of the device with XPIO and memory controllers on the south of the device. In the Versal AI Edge XQR device, an acceleration array is on the north edge of the device in place of the XPIO and memory controllers. Connectivity IP is located in columns close to the serial transceivers. Resources are connected together through a matrix of programmable interconnect routes for local and regional signal connectivity as well as the NoC for high bandwidth and long distance communication around the device.

NoC

The programmable NoC is an AXI4 based network of interconnect within the Versal architecture that easily enables high-bandwidth connections to be routed around the device. The NoC extends in both horizontal and vertical directions to the edges of the device. It exists to connect together areas of the device that demand and use large quantities of data alleviating any resource burden on the local and regional device interconnect. The NoC is a full blocking crossbar between memory controllers, programmable logic, processing system, AIE-ML tiles, and platform management controller. Examples of NoC connections include:

- Sharing device access to DRAM (DDR memory)
- PL to PL connections
- Memory mapped access to the AIE-ML array
- Connecting between the processing system and PL
- Connecting between the processing system and DDR memory

Platform Management Controller

The platform management controller is responsible for managing the Versal AI Edge XQR device with the following main categories of responsibility: securely booting and configuring the platform; and life-cycle management, which includes device integrity and debug, and system monitoring.

Boot and Configuration

The platform management controller is responsible for booting the Versal AI Edge XQR device from the primary boot source in a multi-stage boot process that supports both a non-secure and a secure boot. For a secure boot, the AES-GCM, SHA3-384 decryption/authentication, and ECDSA/RSA blocks decrypt and authenticate the image. Upon reset, the mode pins are read to determine the primary boot device, such as quad SPI, octal SPI, SD, or eMMC. The platform management controller then proceeds to execute the code out of on-chip BootROM and copies the platform loader and manager (PLM) from the boot device to the on-chip memory while undergoing authentication and decryption. The configuration of the PL is also undertaken by the PLM. The device image is loaded from its storage medium, and after authentication and decryption, is sent to the PL configuration interface.

It is also possible to reconfigure portions of the PL using Dynamic Function eXchange (DFX). A new device image for a portion of the PL can be loaded from the PS, through the primary or secondary boot interfaces, e.g., PCIe or Ethernet. Upon reconfiguration, a portion of the PL provides the new functionality determined by the new device image, enabling users to quickly adapt the functionality of their design to changing system requirements.

System Monitoring

The platform management controller contains system monitoring capability for monitoring voltage and temperature in the processing system and PL to enhance the overall safety, security, and reliability of the system. The core of the system monitor is a 10-bit 200kSPS ADC, which can be accessed via JTAG, PMBus, or I2C interfaces, via the processing system directly, and via the PL through the NoC.

Device Integrity and Debug

JTAG is the primary interface for Versal AI Edge XQR device debug features. The JTAG architecture has two IEEE Std 1149.1 compliant TAP controllers that are connected in series: the Arm DAP controller and the PMC TAP controller. The Arm DAP controller is the main controller for debug functions supporting: processing system CoreSight debug architecture, debug of the PL, programming of supported external flash memory, and eFUSE/BBRAM programming. The platform management controller TAP controller supports: reading the device IDCODE, programming of the PL, and boundary scan.

The platform management controller also contains a high-speed debug port (HSDP) that can be used as a faster debug method than the primary JTAG interface. The HSDP interface is a high-throughput interface consisting of separate ingress and egress simplex Aurora 64B/66B channels that leverage the transceivers to the north of the processing system. The HSDP allows daisy-chaining of channels from different devices. The HSDP can also be accessed by the serial transceivers in the PL via an Aurora bridge also in the PL.

External Flash Memory Interfaces

The SD/eMMC controller supports 1- and 4-bit data interfaces at low, default, high-speed, and ultra-high-speed (UHS) clock rates. This controller also supports 1-, 4-, or 8-bit-wide eMMC interfaces that are compliant to the eMMC 4.51 specification. eMMC is one of the primary boot modes and supports boot from managed NAND devices. The controller has a built-in DMA for enhanced performance.

The quad SPI controller is one of the primary boot devices. It supports 4-byte and 3-byte addressing modes. In both addressing modes, single, dual-stacked, and dual-parallel configurations are supported. Single mode supports a quad serial NOR flash memory, while in double stacked and double parallel modes, it supports two quad serial NOR flash memories.

The octal SPI controller is one of the primary boot and configuration devices. It has an 8-pin interface and provides up to 400 MB/s of bandwidth in double data rate mode and up to 166 MB/s in single data rate mode. It has two chip-selects to support deeper memory and a built-in DMA for enhanced performance.

Boot Modes

In addition to JTAG, SelectMAP is an available secondary boot mode. SelectMAP is a high bandwidth, stream oriented, parallel interface that can be configured as 8-, 16- or 32-bit wide. It runs up to 200 MHz.

Compute and Acceleration Engines

AI Engine Array

The Versal AI Edge XQR devices contain a two-dimensional array of AI Engine-ML tile that includes a high-performance VLIW (SIMD) processor; integrated data memory; and interconnects for streaming, configuration, and debug. Alongside the tile is the AI Engine array that provides the necessary logic to connect the AI Engine array to the other resources in the PL, processing system, and the NoC. Devices with an AIE-ML array include additional rows of 512 KB memory tiles.

AI Engine

The AI Engine contains a scalar unit, a vector unit, load units, and a memory interface. The scalar unit contains: a 32-bit scalar RISC processor with register files for general-purpose, pointer, configuration, and backup registers; and a 32 x 32-bit scalar multiplier. The AI Engine also supports non-linear functions including sine/cosine (AIE only), squareroot, and inverse-squareroot.

Three address generator units (AGUs) are available: two dedicated as load units, and one dedicated as a store unit. The vector unit contains a 512-bit vector fixed-point/integer unit. Devices with AIE-ML and v2 contain a floating-point vector unit also used for Bfloat16 and MX6/MX9 (v2 v2 only) support. The vector units in AIE-ML support concurrent operation on multiple vector lanes.

Within each AI Engine is a dedicated, single-port, 16 KB program memory 128-bit wide and 1k deep. The program memory supports instruction compression and has ECC protection and reporting.

AI Engine Data Memory

Separate from the AI Engine, each tile contains 64 KB of data memory for the AIE-ML, divided into eight single-port banks. This structure allows up to eight parallel memory access transactions every clock cycle, with five cycle access latency. Stall signals identify memory access conflicts during which time any outstanding memory operations are buffered. Each data memory module supports memory error detection (parity) and reporting.

The data memory also contains DMA logic that supports incoming stream to local memory, outgoing stream from local memory, and buffered streams in local memory. Support for two-dimensional stride access enables any AI Engine to access data memories in adjacent AI Engine tiles in the north, south, east, and west directions, allowing a single AIE to access up to 128 KB of data memory and a single AIE-ML to access up to 256 KB of data memory.

AIE-ML Memory Tiles

Devices with AIE-ML array also include, at the bottom of the array, memory tiles that contain high-density (512 KB) and high bandwidth memory, divided into eight banks, and an integrated DMA to access local memory and neighboring memories.

Digital Signal Processing (DSP)

DSP applications use many binary multipliers and accumulators, best implemented in dedicated DSP Engines. Versal AI Edge XQR devices have many dedicated, low-power DSP Engines, combining high speed with small size while retaining system design flexibility.

Each DSP Engine fundamentally consists of a dedicated 27×24 bit two's complement multiplier and a 58-bit accumulator. The multiplier can be dynamically bypassed, and two 58-bit inputs can feed a single-instruction-multiple-data (SIMD) arithmetic unit (dual 24-bit add/subtract/accumulate or quad 12-bit add/subtract/accumulate), or a logic unit that can generate any one of ten different logic functions of the two operands.

The DSP Engine includes an additional pre-adder, typically used in symmetrical filters. This pre-adder improves performance in densely packed designs and reduces the DSP Engine count by up to 50%. The 116-bit-wide XOR function, programmable to 12, 22, 24, 34, 58, or 116-bit widths, enables performance improvements when implementing forward error correction and cyclic redundancy checking algorithms.

The DSP Engine also includes a 58-bit-wide pattern detector that can be used for convergent or symmetric rounding. The pattern detector is also capable of implementing 116-bit-wide logic functions when used with the logic unit.

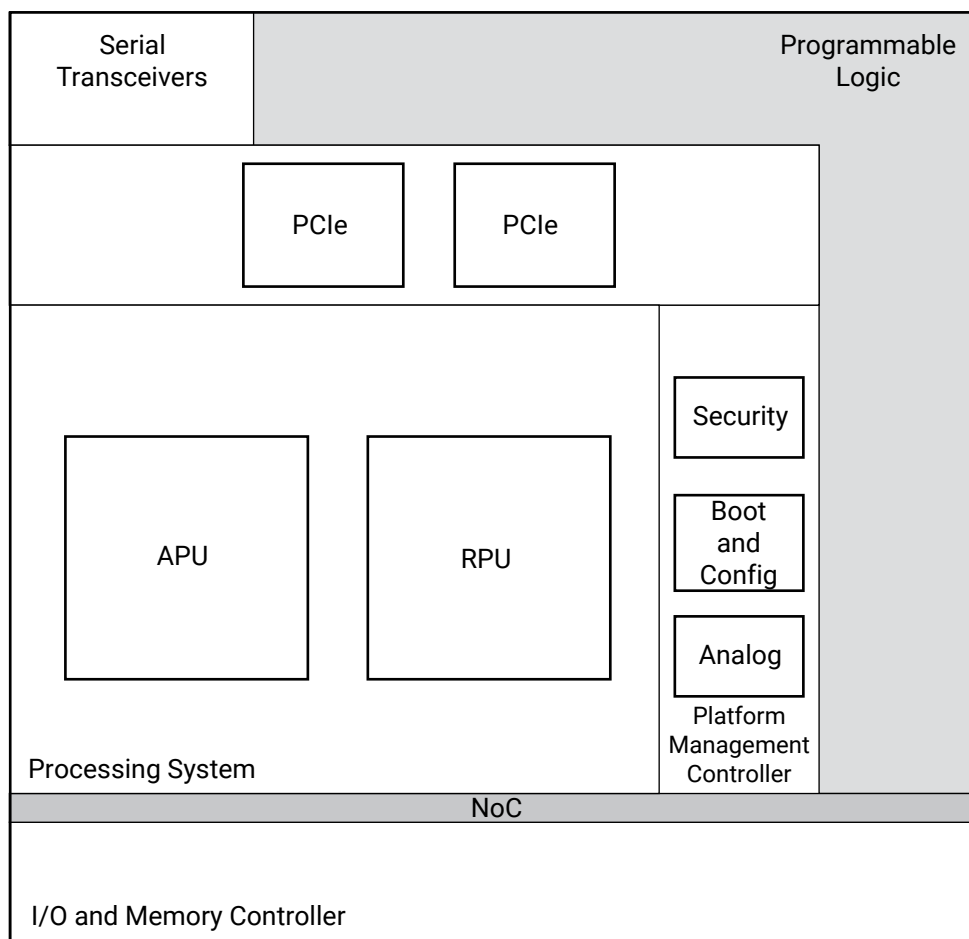
The DSP Engine provides extensive pipelining and extension capabilities that enhance the speed and efficiency of many applications beyond digital signal processing, such as wide dynamic bus shifters, memory address generators, wide bus multiplexers, and memory-mapped I/O register files. The accumulator can also be used as a synchronous up/down counter. The DSP Engine layout enables new modes of operation in addition to the conventional fixed-point operation.

- **Three element vector/INT8 dot product:** The DSP Engine can be used in vector fixed-point ALU mode in which the 27×24 bit multiplier is replaced by a three-dimensional vector dot-product unit. The dot-product unit supports element-wise product negation with negate pins.
- **Complex 18b x 18b:** Using two back to back DSP Engines, the Versal architecture enables creation of an $18 \times 18 + 58$ two's complement complex multiply accumulator in which each of the two complex inputs can be optionally conjugated.
- **Single precision floating point:** The DSP Engine contains a floating-point multiplier and a floating-point adder with separate outputs in binary32 format. Each floating-point multiplier input can be in either binary32 (single-precision or FP32) or binary16 (half-precision or FP16) format.

Processing System

All Versal AI Edge XQR devices contain a processing system (PS) consisting of an APU, RPU, and peripherals. The processing system is part of a group of architectural elements that include the platform management controller (PMC), NoC, and integrated memory controllers that are tightly coupled, but are also capable of operating independently from each other. The simplified layout is shown in the following figure.

Figure 3: Processing System and Surrounding Blocks



X29559-082624

The platform management controller is responsible for booting the processing system from one of its primary boot sources. The processing system also has direct access to the features inside the CPM, which talks to the serial transceivers directly to the north for implementation of high-performance interconnect based on PCI-SIG technologies. Programmable logic can be configured at any stage of the process and can be performed before or after the processing system is booted.

Application Processing Unit (APU)

The APU has a feature-rich dual-core Arm Cortex-A72 processor. Cortex-A72 cores are 64-bit-wide application processors based on the Arm-v8A architecture, which supports hardware virtualization. Each of the Cortex-A72 cores has: 48 KB of instruction L1 cache and 32 KB of data L1 cache, with parity and ECC protection respectively; a NEON SIMD engine; and a single and double precision floating point unit. In addition to these blocks, the APU consists of a snoop control unit and a 1 MB L2 cache with ECC protection to enhance system-level performance. The snoop control unit keeps the L1 caches coherent thus eliminating the need of spending software bandwidth for coherency. The APU also has a built-in interrupt controller supporting virtual interrupts.

The APU communicates to the rest of the PS through the 128-bit AXI4 coherent extension (ACE) port via cache coherent interconnect (CCI) block, using the system memory management unit (SMMU). The APU is also connected to the PL through the 128-bit accelerator coherency port (ACP), providing a low latency coherent port for accelerators in the PL. To support real-time debug and trace, each core also has an embedded trace macrocell (ETM) that communicates with the Arm CoreSight debug system.

Real-Time Processing Unit (RPU)

The RPU in the processing system contains a dual-core Arm Cortex-R5F processor. Cortex-R5F cores are 32-bit real-time processor cores based on the Arm-v7R architecture. Each of the Cortex-R5F cores has 32 KB of Level 1 (L1) instruction and data cache with ECC protection. In addition to the L1 caches, each of the Cortex-R5F cores also has a 128 KB tightly coupled memory (TCM) interface for real-time single cycle access. The RPU also has a dedicated interrupt controller and floating point unit. The RPU can operate in either split or lock-step mode. In split mode, both processors run independently of each other. In lock-step mode, they run in parallel with each other, with integrated comparator logic, and the TCMs are used as a 256 KB unified memory.

The RPU communicates with the rest of the processing system via the 128-bit AXI4 ports connected to the low power domain switch. It also communicates directly with the PL through 128-bit low latency AXI4 ports. To support real-time debug and trace, each core also has an embedded trace macrocell (ETM) that communicates with the Arm CoreSight debug system.

Connectivity Peripherals

In the processing system, many peripherals are used to connect to external devices over industry-standard protocols, including CAN-FD, SPI, USB, Ethernet, I2C, and UART. Many of the peripherals support clock gating and power gating modes to reduce dynamic and static power consumption. These peripherals either use multiplexed I/O (MIO) to connect to the external components, or if required, they can also be routed into and through the PL using the extended multiplexed I/O (EMIO).

With the platform management controller providing access to the high-speed and boot configuration interfaces respectively, the number of peripherals required directly in the I/O unit is relatively small, containing:

- 2 gigabit Ethernet controllers
- 2 SPI controllers
- 2 I2C controllers
- 2 CAN/CAN-FD controllers
- 2 UARTs
- GPIO
- 1 USB 2.0 (device and host) controller

The following functions are included in the I/O unit so they can share MIO:

- 4 triple-timer counters
- 1 watchdog timer

All peripherals within the I/O unit have Trustzone support through system control registers. The I/O unit has master and slave AXI4 interface ports to the LPD interconnect. One APB bus at top level is used to control AXI4 bridges within the I/O unit.

USB 2.0

The USB controller can be configured as host or device. The controller is compliant to the USB 2.0 specification and supports high, full, and low-speed modes in all configurations. In host mode, the USB controller is compliant with the Intel XHCI specification. In device mode, it supports up to the 12 endpoints. The universal low peripheral interface (ULPI) is used to connect the controller to an external PHY operating up to 480 Mb/s.

Ethernet MAC

The pair of tri-speed Ethernet MACs support 10 Mb/s, 100 Mb/s, and 1 Gb/s operations. They also support jumbo frames and timestamping through the interfaces based on IEEE Std 1588 v2. Time Sensitive Network (TSN), which either uses IEEE Std 1588 or 802.1AS-REV, is also supported. The Ethernet MACs can be connected through the MIO (RGMII), or through EMIO (GMII). The GMII interface can be converted to a different interface within the PL.

Programmable Logic

Configurable Logic Block (CLB)

Every configurable logic block (CLB) contains 32 look-up tables (LUTs) and 64 flip-flops. The LUTs can be configured as either one 6-input LUT with one output, or as two 5-input LUTs with separate outputs but common inputs. Each LUT can optionally be registered in a flip-flop. In addition to the LUTs and flip-flops, the CLB contains arithmetic carry logic and multiplexers to create wider logic functions. Within each CLB, 16 LUTs can be configured as a 64-bit RAM, as a 32-bit shift register (SRL32), or as two 16-bit shift registers (SRL16s). For every group of 64 flip-flops, there are four clocks signals, four set/reset signals, and 16 clock enables. Within every CLB are dedicated interconnect paths for connecting LUTs together without having to exit and re-enter a CLB and cascade MUXes. This enables a flexible carry logic structure that allows a carry chain to start at any bit in the chain.

Internal Memory

Each Versal AI Edge XQR device contains several programmable, internal storage capabilities. In addition to the distributed RAM capability in the CLB, there are dedicated blocks for building various size storage elements.

On-Chip Memory (OCM)

In addition to the 32 KB of L1 data cache, the RPU contains 256 KB OCM with ECC. The OCM is accessed through two 128-bit AXI4 interfaces with one AXI4 interface dedicated to the two Cortex-R5F processors and the other AXI4 interface available to the APU and other masters. Memory accesses from the RPU are treated with higher priority than memory accesses through the general 128-bit AXI4 interface.

The Versal AI Edge XQR device includes accelerator RAM, an additional 4 MB of on-chip memory with ECC located outside of the processing system. This memory provides direct access from the RPU via a 128-bit AXI4 interface and can also be accessed from the PL through two 256-bit AXI4 interfaces. The memory is divided into three banks supporting concurrent read or write accesses from the PL and RPU to different banks.

Block RAM

True dual-port block RAMs, each having 36 Kb of storage capacity, can be configured as either one 36 Kb RAM, or two completely independent 18 Kb RAMs. Each port can be configured as $4K \times 9$, $2K \times 18$, $1K \times 36$, or 512×72 in simple dual-port mode. The two ports can have different aspect ratios. Also, the read port width can be different from the write port width for each port.

- **Synchronous operation::** Each memory access, read, and write is controlled by the clock. All inputs, data, address, clock enable, and write enable are registered. The data output is always latched, retaining data until the next operation. An optional output data pipeline register allows higher clock rates at the cost of an extra cycle of latency. During a write operation, the data output can be made to reflect the previously stored data, the newly written data, or remain unchanged. There is independent reset control of output latches and registers.
- **Asynchronous operation:** The data outputs can also be set/reset asynchronously. The sleep input, which places the array in a low-power state, can be optionally asynchronous.
- **True dual-port operation:** The block RAM has two completely independent ports that share nothing but the stored data.
- **Simple dual-port operation:** One port is dedicated as a write port and the other as a read port. The data width can thus be extended to 72 bits for the 36 Kb full block RAM or 36 bits for the *split* 18 Kb block RAM.

Cascade mode supports all configurations available in 36 Kb RAM or 18 Kb RAM. Cascading refers to combining multiple block RAMs to build larger ones, without using additional logic resources.

Each 64-bit-wide block RAM can generate, store, and use eight additional bits to perform single-bit error correction and double-bit error detection (ECC) during the read process. The ECC logic can also be used without the memory array to support the implementation of ECC on user designed internal datapaths or memory controllers. Block RAM contents can be initialized or cleared by the configuration device image.

UltraRAM

Dual-port UltraRAMs, each having 288K bits of storage capacity, can be configured as one 288 Kb RAM. Each port can be configured as $32K \times 9$, $16K \times 18$, $8K \times 36$, or $4K \times 72$. The two ports can have different aspect ratios.

- **Synchronous operation only:** Each memory access, read, and write is controlled by the clock. All inputs, data, address, clock enable, and write enable are registered. The data output is always latched, retaining data until the next operation. An optional output data pipeline register allows higher clock rates at the cost of an extra cycle of latency.
- **Asynchronous control:** The data outputs can also be set/reset asynchronously. The sleep input, which places the array in a low power state, can be optionally asynchronous.

- **Pseudo dual-port operation:** There are two ports on the memory. Each is capable of reading or writing in a single cycle. The ports are sequenced in a fixed order, allowing up to two transactions per cycle. (Both ports write, both ports read, or one port reads while the other writes.) This necessitates that the two ports share a common clock. During a write operation, the data output remains unchanged on a given port. There is independent reset control of output latches and registers.

ECC logic in the UltraRAM supports error checking and correction. Both ports have dedicated ECC for either read or write. The ECC logic is organized for 64-bit-wide data, which can generate, store, and use eight additional bits to perform single-bit error correction and double-bit error detection (ECC) during the read process.

It is possible to cascade the address and data of adjacent blocks to build deeper memories. Optional pipelining is also available to maintain the clock rate through tall cascades of UltraRAM.

Connectivity

GTYP Transceivers

GTYP transceivers support data rates up to 26.5625 Gb/s. The minimum data rate for all transceivers is 1.2 Gb/s, but lower data rates can be achieved by using oversampling in the programmable logic.

The serial transmitter and receiver are independent circuits that use an advanced phase-locked loop (PLL) architecture to multiply the reference frequency input by certain programmable numbers between 4 and 25 to become the bit-serial data clock. Each transceiver has a large number of user-definable features and parameters. All of these can be defined during device configuration, and many can also be modified during operation.

GTYP Transmitter

The transmitter is fundamentally a parallel-to-serial converter with a conversion ratio of 16, 20, 32, 40, 64, 80, 128, or 160. This allows the designer to trade off datapath width against timing margin in high-performance designs. These transmitter outputs drive the PC board with a single-channel differential output signal. TXOUTCLK is the appropriately divided serial data clock and can be used directly to register the parallel data coming from the internal logic. The incoming parallel data is fed through an optional FIFO and has additional hardware support for the 8B/10B, 64B/66B, or 64B/67B encoding schemes to provide a sufficient number of transitions. The bit-serial output signal drives two package pins with differential signals. This output signal pair has programmable signal swing as well as programmable pre- and post-emphasis to compensate for PC board losses and other interconnect characteristics. For shorter channels, the swing can be reduced to reduce power consumption.

GTYP Receiver

The receiver is fundamentally a serial-to-parallel converter, changing the incoming bit-serial differential signal into a parallel stream of words, each 16, 20, 32, 40, 64, 80, 128, or 160. This allows the designer to trade off internal datapath width against logic timing margin. The receiver takes the incoming differential data stream, feeds it through programmable DC automatic gain control, linear and decision feedback equalizers (to compensate for PC board, cable, optical and other interconnect characteristics), and uses the reference clock input to initiate clock recognition. There is no need for a separate clock line. The data pattern uses non-return-to-zero (NRZ) encoding and optionally ensures sufficient data transitions by using the selected encoding scheme. Parallel data is then transferred into the device logic using the RXUSRCLK clock. For short channels, the transceivers offer a special low-power mode (LPM) to reduce power consumption by approximately 30%.

The receiver DC automatic gain control and linear and decision feedback equalizers can optionally "auto-adapt" to automatically learn and compensate for different interconnect characteristics. This enables even more margin for 10G+ and 25G+ backplanes.

Out-of-Band Signaling

The transceivers provide out-of-band (OOB) signaling, often used to send low-speed signals from the transmitter to the receiver while high-speed serial data transmission is not active. This is typically done when the link is in a powered-down state or has not yet been initialized. This benefits PCIe and SATA/SAS applications.

Integrated Block for PCI Express

The Versal architecture uses integrated blocks to enable PCIe designs. Versal AI Edge XQR devices contain one or more instances of a programmable logic integrated block for PCIe designs (PL PCIE), which reside in the PL as illustrated in [Device Layout](#). Multiple versions of these integrated blocks exist in the Versal architecture.

The PL PCIE communicates with the adjacent serial transceivers and supports the protocols, data rates, and link widths shown in the following table. Each PL PCIE can be configured as an Endpoint or Root port. The root-port configuration can be used to build the basis for a compatible Root Complex, to allow custom chip-to-chip communication via the PCI Express protocol, and to attach endpoint devices, such as Ethernet controllers or Fibre Channel HBAs, to the SoC. For high-performance applications, the PL PCIE advanced buffering techniques offer a flexible maximum payload size. The PL PCIE interfaces to the integrated high-speed transceivers for serial connectivity and to PL memory resources for data buffering. Combined, these elements implement the physical layer, data link layer, and transaction layer of the PCI Express protocol.

Table 9: Supported Protocols, Data Rates, and Link Widths

	PLPCIE4
Governing specifications	PCI Express Base Specification Rev 4.0
Maximum PCIe link configurations	Gen4x8 Gen3x16
Key PCIe features	SRIOV 4PF/252VF

Multirate Ethernet MAC

The Versal architecture contains integrated blocks for Ethernet functionality capable of operating at different data rates.

The multirate Ethernet MAC (MRMAC) provides high-performance, low latency Ethernet ports supporting a wide range of customization and statistics gathering. Supported configurations are: 1 x 40GE; 4 x 25GE; and 4 x 10GE.

The MRMAC supports the following FECs defined and required by IEEE standards: Clause 91 RS(528, 514) KR4 FEC, for 25 GE NRZ support; and Clause 74 FEC for 10/25/40GE low-latency support. The MRMAC has a rich set of bypass modes to enable access to FEC-only mode (for custom protocols) and FEC+PCS (for protocol testers).

The MRMAC also supports a high-precision timestamping feature to enable sub-nanosecond accuracy on IEEE Std 1588 timestamps. This provides hardware support for new IEEE Std 1588-based time-sensitive networks (TSN) as well as the next generation Ethernet-based wireless fronthaul protocol (eCPRI).

I/O

As listed in the following table, three types of programmable I/Os exist in the programmable logic with additional I/Os available in the processing system.

Table 10: Programmable I/O

I/O Type	XPIO	HDIO	MIO
Voltage	1.0V–1.5V	1.8V–3.3V	1.8V–3.3V
Purpose	Highest performance, DDR memory interface	Lower performance, wider voltage range	Supports PS peripherals

XPIO

XPIO are located on the bottom of the device. XPIOs are optimized for high-performance communication including, but not limited to, interfacing to DDR4 memory through the integrated memory controller blocks. XPIO are arranged in banks of 54 I/O and organized as nine 6-bit nibbles. XPIO support standards with maximum supply voltage of 1.5V. Every XPIO bank includes a physical layer interface (PHY) that can operate in 4:1 mode for use with the integrated memory controllers or 8:1 mode for use with custom circuitry.

HDIO

High-density I/O (HDIO) banks are designed to be a cost-effective method for supporting lower speed, higher voltage range I/O standards. Arranged in banks of 22, HDIOs offer single-ended I/O including 3.3V and 2.5V LVTTTL and LVCMOS. HDIOs also offer differential receivers for low-speed clock inputs and pseudo-differential transmitters. There is an internal V_{REF} support. The system designer can specify the slew rate and the output strength. The input is always active but is usually ignored while the output is active. Each pin can optionally have a weak pull-up resistor, a weak pull-down resistor, or weak keeper.

MIO

Multiple banks of general-purpose I/O are implemented within the PS and PMC, each with a dedicated power supply. The main category of I/O are the three banks of multiplexed I/O (MIO), which can be accessed by the PS, the PMC, and the PL. Fixed-function I/O are also available for control and configuration functions.

Clocking

Multiple clock generation blocks are used to synthesize clock frequencies. Clock buffers and routing connect the signals to their destinations.

Processing System Clocking

All clocks in the processing system belong to one of three groups: the main PLL clocks; the internal ring oscillator clock, and interface clocks.

Main PLL Clocks

The majority of the logic in the processing system is clocked from the three PLLs in the processing system and one PLL in the platform management controller through user configurable clock divider circuits. These divider circuits generate clocks to all CPUs, main interconnects, the platform management controller, and all peripherals. The clocks and their associated PLLs are spread across three power domains: the platform management controller domain, containing the platform management controller; the low-power domain, containing the RPU and all peripheral clocks; and the full-power domain, containing all other clocks and their PLLs.

Internal Ring Oscillator

The platform management controller operates as the security manager for the device and uses a clock provided by an internal ring oscillator.

Interface Clocks

This category includes clocks that are directly supplied from outside the processing system and includes clocks for the external interfaces, including Ethernet, USB, SWDT and CAN-FD.

PL Clocking

Clock signals travel around the Versal AI Edge XQR device on a network of bidirectional, horizontal, and vertical routing tracks that support many independent clock networks. The vertical tracks reside adjacent to the NoC columns. The programmable logic is divided into clock regions that each have a horizontal clock spine through the middle that can carry 24 clock signals. Clock signals travel along these horizontal clock spines and are then driven into the individual clocked elements within the PL such as flip-flops, DSP Engines, block RAM, and UltraRAM. Clock buffers and clock management components reside adjacent to the XPIO rows on the south edges of the device.

Clock Management

To generate multiple clock frequencies and phases from an input clock source, Versal AI Edge XQR devices contain mixed-mode clock managers (MMCMs) and phase-locked loops (PLLs). MMCMs reside adjacent to the horizontal NoC row adjacent to the XPIO and PLLs reside in the XPIO banks. The MMCM and PLL share many characteristics. Both can serve as a frequency synthesizer for a wide range of frequencies and as a jitter filter for incoming clocks. At the center of both components is a voltage-controlled oscillator (VCO), which speeds up and slows down depending on the input voltage it receives from the phase frequency detector (PFD).

There are three sets of programmable frequency dividers: D, M, and O. The predivider D, programmable by configuration and afterwards via the dynamic reconfiguration port (DRP), reduces the input frequency and feeds one input of the traditional PLL phase/frequency comparator. The feedback divider M (programmable by configuration and afterwards via DRP) acts as a multiplier because it divides the VCO output frequency before feeding the other input of the phase comparator. D and M must be chosen appropriately to keep the VCO within its specified frequency range. The VCO has eight equally spaced output phases (0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315°). Each can be selected to drive one of the output dividers (six for the PLL, O0 to O5, and seven for the MMCM, O0 to O6), each programmable by configuration to divide by any integer from 1 to 128.

- **MMCM additional programmable features:** The MMCM has a fractional counter in either the feedback path (acting as a multiplier) or in one output path. Fractional counters allow non-integer increments of 1/8 and can thus increase frequency synthesis capabilities by a factor of 8. The MMCM can also provide fixed or dynamic phase shift in small increments that depend on the VCO frequency

Memory Controllers

Dedicated controllers that can support either DDR4 or LPDDR4 are located on the south edges of the device. Each memory controller has four bidirectional 128-bit system ports and contains a scheduler with transaction reordering capability to improve memory access efficiency. The memory controller operates at half the DRAM clock rate. For example, if the DRAM data rate per bit is 3200 Mb/s, then the DRAM clock rate is 1600 MHz, and the memory controller clock rate is 800 MHz. The memory controllers talk to the dedicated memory PHY (XPHY) in the XPIO banks which, in turn, interface with the I/O pins.

DC Characteristics

Absolute Maximum Ratings

Table 11: Absolute Maximum Ratings—Power, Voltages, and Current

Symbol	Description ¹		Min	Max	Units
V _{CCAUX}	Programmable logic auxiliary power supply		–0.500	1.650	V
V _{CCAUX_PMC}	Platform management controller (PMC) auxiliary power supply voltage		–0.500	1.650	V
V _{CCAUX_SMON} ²	PMC system monitor power supply relative to GND_SMON		–0.500	1.650	V
V _{CC_BATT}	Battery power supply		–0.500	1.650	V
V _{CC_FUSE}	eFUSE programming power supply		–0.500	2.000	V
V _{CCINT}	Programmable logic primary power supply		–0.500	0.970	V
V _{CC_IO} ³	XPIO power supply		–0.500	0.970	V
V _{CCO} ⁴	XPIO bank 7## output driver power supply Where the bank number is the three-digit bank number beginning with 7		–0.500	1.650	V
	HDIO bank 3## and bank 4## output driver power supply Where the bank number is the three-digit bank number beginning with 3 or 4		–0.500	3.63	V
	PSIO bank 5## power supplies Where the bank number is the three-digit bank number beginning with 5		–0.500	3.63	V
V _{CC_PMC}	PMC primary power supply		–0.500	0.970	V
V _{CC_PSF}	PS full-power domain power supply		–0.500	0.970	V
V _{CC_PSLP}	PS low-power domain power supply		–0.500	0.970	V
V _{CC_RAM}	PL RAM and clocking network power supply		–0.500	0.970	V
V _{CC_SOC} ³	Network on Chip (NoC) and integrated DDR memory controller power supply		–0.500	0.970	V
V _{IN} ^{4, 5}	I/O input voltage for PSIO, HDIO, and XPIO banks		–0.550	V _{CCO} + 0.550	V
V _{REF}	PMC system monitor reference input (V _{REFP} /V _{REFN}) relative to GND_SMON		–0.500	1.120	V
GTYP Transceivers					
V _{GTYP_AVCC}	GTYP transceiver primary analog power supply ⁶		–0.500	0.97	V
V _{GTYP_AVCCAUX}	GTYP transceiver auxiliary analog (PLL) power supply ⁶		–0.500	1.650	V
V _{GTYP_AVTT}	GTYP transceiver termination power supply ⁶		–0.500	1.320	V
V _{GTYP_AVTTTRCAL}	GTYP transceiver analog resistor calibration power supply ⁶		–0.500	1.320	V
V _{GTYP_REFCLK_DC}	Transceiver reference clock absolute input DC voltage		–0.500	1.000	V
V _{GTYP_REFCLK_AC}	Transceiver reference clock absolute input AC voltage ⁷		–0.500	1.350	V
V _{IN_DC}	Receiver (RXP/RXN) and transmitter (TXP/TXN) absolute DC voltage	Unpowered	–0.800	0.800	V
		Powered	–0.300	1.000	V
V _{IN_AC+DC}	Receiver (RXP/RXN) and transmitter (TXP/TXN) absolute AC plus DC voltage	Unpowered	–0.800	0.800	V
		Powered	–0.500	1.250	V
I _{DCIN_FLOAT}	DC input current for receiver input pins DC coupled RX termination = floating ⁸		–	10	mA
I _{DCIN_GTYP_AVTT}	DC input current for GTYP receiver input pins DC coupled RX termination = or V _{GTYP_AVTT}		–	16	mA
I _{DCIN_GTYP_GND}	DC input current for GTYP receiver input pins DC coupled RX termination = GND ¹⁰		–	16	mA
I _{DCIN_PROG}	DC input current for receiver input pins DC coupled RX termination = programmable ⁹		–	0	mA
I _{DCOUT_FLOAT}	DC output current for transmitter pins DC coupled RX termination = floating		–	6	mA
I _{DCOUT_GTYP_AVTT}	DC output current for transmitter pins DC coupled RX termination = or V _{GTYP_AVTT}		–	6	mA

Table 11: Absolute Maximum Ratings—Power, Voltages, and Current (cont'd)

Symbol	Description ¹	Min	Max	Units
Other Ratings				
I _{DC}	Available output current at the pad	-20	20	mA
I _{RMS}	Available RMS output current at the pad	-20	20	mA

Notes:

- Stresses beyond those listed under Absolute Maximum Ratings might cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those listed under Operating Conditions is not implied. Exposure to Absolute Maximum Ratings conditions for extended periods of time might affect device reliability.
- V_{CCAUX_SMON} must be connected to V_{CCAUX_PMC}. See *Versal Adaptive SoC System Monitor Architecture Manual (AM006)* for V_{CCAUX_SMON} information and filter considerations.
- V_{CC_IO} must be connected to V_{CC_SOC}.
- When operating outside of the recommended operating conditions, refer to the specifications in the [V_{IN} Maximum Allowed AC Voltage Overshoot and Undershoot](#) section.
- The lower absolute voltage specification always applies.
- For more information on supported GTYP transceiver terminations see the *Versal Adaptive SoC GTY and GTYP Transceivers Architecture Manual (AM002)*.
- The or GTYP_REFCLK AC signal operates in the reference clock frequency range. See [GTYP Transceiver Reference Clock Switching Characteristics](#).
- AC coupled operation is not supported for RX termination = floating.
- DC coupled operation is not supported for RX termination = programmable.
- For DC coupled links with RX termination set to ground, the transmitter or the board should insure that the RX input common mode voltage is set between 0.2V and 0.8V.

Table 12: Absolute Maximum Ratings—Temperature

Symbol	Description ¹	Min	Max	Units
T _{STG}	Storage temperature (ambient, unpowered)	-55	150	°C
T _{SOL} ²	Maximum dry rework soldering temperature	-	235	°C
T _j	Maximum junction temperature	-	125	°C

Notes:

- Stresses beyond those listed under Absolute Maximum Ratings might cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those listed under Operating Conditions is not implied. Exposure to Absolute Maximum Ratings conditions for extended periods of time might affect device reliability.
- See the *Soldering Guidelines* and *Thermal Specifications* sections in this data sheet.

Recommended Operating Conditions

Table 13: Recommended Operating Conditions

Symbol	Description ^{1, 2, 3}	Min	Typ	Max	Units
V _{CCAUX}	Auxiliary power supply	1.455	1.500	1.545	V
V _{CCAUX_PMC}	PMC auxiliary power supply voltage	1.455	1.500	1.545	V
V _{CCAUX_SMON} ⁴	PMC system monitor power supply relative to GND_SMON	1.455	1.500	1.545	V
V _{CC_BATT} ⁵	Battery power supply to the battery-backed RAM and battery-backed real-time clock (RTC)	1.200	-	1.500	V
V _{CC_FUSE} ⁶	eFUSE programming power supply	1.745	1.800	1.854	V
V _{CCINT}	PL primary power supply	0.775	0.800	0.825	V
V _{CC_IO} ⁷	XPIO power supply voltage	0.775	0.800	0.825	V

Table 13: Recommended Operating Conditions (cont'd)

Symbol	Description ^{1, 2, 3}	Min	Typ	Max	Units
V _{CCO} ⁸	XPIO bank 7## output driver power supply Includes V _{CCO} of 1.0V, 1.1V, 1.2V, 1.35V, 1.5V at ±5%	0.950	–	1.575	V
	HDIO bank 3## and bank 4## output driver power supply Includes V _{CCO} of 1.8V, 2.5V at ±5%, and 3.3V at +3/–5%	1.710	–	3.400	V
	PSIO bank 5## power supplies Includes V _{CCO} of 1.8V, 2.5V at ±5%, and 3.3V at +3/–5%	1.710	–	3.400	V
V _{CC_PMC} ⁹	PMC primary power supply	0.775	0.800	0.825	V
V _{CC_PSFP}	PS full-power domain power supply	0.775	0.800	0.825	V
V _{CC_PSLP}	PS low-power domain power supply	0.775	0.800	0.825	V
V _{CC_RAM}	PL RAM and clocking network power supply	0.775	0.800	0.825	V
V _{CC_SOC} ⁷	Network on Chip (NoC) and integrated DDR memory controller power supply	0.775	0.800	0.825	V
V _{GTYP_AVCCAUX} ¹⁰	GTYP transceiver auxiliary analog (PLL) power supply	1.455	1.500	1.545	V
V _{GTYP_AVTT} ¹⁰	GTYP transceiver termination power supply	1.164	1.200	1.236	V
V _{GTYP_AVTTTCAL} ¹⁰	GTYP transceiver analog resistor calibration power supply	1.164	1.200	1.236	V
V _{IN} ^{11, 12}	I/O input voltage for PSIO, HDIO, and XPIO banks	–0.200	–	V _{CCO} + 0.200	V
I _{IN} ¹³	Maximum current through any PL, PMC, or PS pin in a powered or unpowered bank when forward biasing the clamp diode	–	–	10	mA
T _J ¹⁴	Junction temperature operating range for Class B, flow B, (M) temperature devices	–55	–	125	°C
	Junction temperature operating range for eFUSE programming	–40	–	125	°C

Notes:

- All voltages are relative to GND and in relation to the BGA package ball.
- For the design of the power distribution system consult the *Versal Adaptive SoC PCB Design User Guide* (UG863) and Power Design Manager (PDM) tool (download at www.amd.com/power).
- Each voltage listed requires decoupling as described in the *Versal Adaptive SoC PCB Design User Guide* (UG863). Refer to the Power Design Manager (PDM) tool (download at www.amd.com/power) for design specific decoupling recommendations.
- V_{CCAUX_SMON} must be connected to V_{CCAUX_PMC}. See *Versal Adaptive SoC System Monitor Architecture Manual* (AM006) for V_{CCAUX_SMON} information and filter considerations.
- When the battery-backed RAM and the real-time clock are not used, connect to GND.
- V_{CC_FUSE} must be within the recommended operating conditions during eFUSE programming. When V_{CC_FUSE} is not used, connect to GND.
- V_{CC_IO} must be connected to V_{CC_SOC}.
- For XPIO and HDIO operation, see the *Versal Adaptive SoC SelectIO Resources Architecture Manual* (AM010). PSIO includes PMC dedicated I/O banks, PMC MIO banks, and LPD MIO banks. For PSIO operation, see the *Versal Adaptive SoC Technical Reference Manual* (AM011).
- The physical unclonable function (PUF) is only supported when using a nominal V_{CC_PMC} of 0.70V on certain Versal devices. Refer to the *Versal Adaptive SoC Security Manual* (UG1508) in the [Design Security Lounge](#) for a list of affected devices and for detailed information on PUF usage, including a V_{CC_PMC} 0.70V solution. When using a V_{CC_PMC} 0.70V solution, performance is reduced for functions based on the V_{CC_PMC}.
- Each voltage listed requires filtering as described in the *Versal Adaptive SoC GTY and GTYP Transceivers Architecture Manual* (AM002). The noise beyond the recommended operating conditions at the power pins must not exceed 10 mVpp over the band from 10 kHz to 80 MHz.
- The lower absolute voltage specification always applies.
- In XPIO banks, V_{IN} overshoot above V_{CCO} and undershoot below GND can reduce the performance of V_{REF}-based receivers within the same nibble.
- A total of 200 mA per bank should not be exceeded.
- Junction temperature specification is in relation to the absolute value reported by the PMC system monitor.

Available Speed Grades and Operating Voltages

The following table describes the speed grades per device and operating supply voltages for the PMC, PS low power, PS full power, and PL domains.

Table 14: Available Speed Grades and Operating Voltages

Speed Grade	Mode	Vivado Design Tools Device Code ¹	V _{CCINT}	V _{CC_PSLP}	V _{CC_PSFP}	V _{CC_CPM5}	V _{CC_PMC} ²	V _{CC_SOC} and V _{CC_IO}	V _{CC_RAM}	Units
-1MSB	Standard	-1MM-b-S	0.80	0.80	0.80	0.80	0.80	0.80	0.80	V

Notes:

1. The specified portion of the Vivado design tools device selection code includes speed grade (-1), operating voltages (MM), temperature grade (-b), and maximum static power screen (-S).
2. The physical unclonable function (PUF) is only supported when using a nominal V_{CC_PMC} of 0.70V. Refer to the *Versal Adaptive SoC Security Manual* (UG1508) in the [Design Security Lounge](#) for detailed information on PUF usage, including a V_{CC_PMC} 0.70V solution. When using a V_{CC_PMC} 0.70V solution, performance is reduced for functions based on the V_{CC_PMC}.

DC Characteristics Over Recommended Operating Conditions

Table 15: DC Characteristics Over Recommended Operating Conditions

Symbol	Description	Min	Typ ¹	Max	Units
$C_{IN}^{2,3}$	HDIO and PSIO die input capacitance at the pad	–	–	3.50	pF
	XPIO die input capacitance at the pad	–	–	1.75	pF
I_L	HDIO, XPIO, and PSIO input or output leakage current per pin (sample-tested)	–	–	20	μA
I_{RPU}	Pad pull-up (when selected) at $V_{IN} = 0V$, $V_{CCO} = 3.3V$	60	–	200	μA
	Pad pull-up (when selected) at $V_{IN} = 0V$, $V_{CCO} = 2.5V$	50	–	169	μA
	Pad pull-up (when selected) at $V_{IN} = 0V$, $V_{CCO} = 1.8V$	29	–	120	μA
	Pad pull-up (when selected) at $V_{IN} = 0V$, $V_{CCO} = 1.5V$	30	–	120	μA
	Pad pull-up (when selected) at $V_{IN} = 0V$, $V_{CCO} = 1.2V$	10	–	100	μA
I_{RPD}	Pad pull-down (when selected) at $V_{IN} = 3.3V$	60	–	200	μA
	Pad pull-down (when selected) at $V_{IN} = 1.8V$	29	–	120	μA
I_{CC_FUSE}	V_{CC_FUSE} supply current during eFUSE programming	–	–	165	mA
Battery Supply Current					
$I_{CC_BATT}^{3,4}$	Battery supply current at $V_{CC_BATT} = 1.20V$, RTC disabled	–	–	160	nA
	Battery supply current at $V_{CC_BATT} = 1.50V$, RTC disabled	–	–	320	nA
	Battery supply current at $V_{CC_BATT} = 1.20V$, RTC enabled	–	–	1360	nA
	Battery supply current at $V_{CC_BATT} = 1.50V$, RTC enabled	–	–	1930	nA
Calibrated programmable on-die termination (DCI) in XPIO banks⁵ (measured per JEDEC specification)					
R^7	Thevenin equivalent resistance of programmable input termination where x = target impedance of 40, 48, 60, 120, or 240	–20% ⁶	ODT = RTT_x	+20% ⁶	Ω
Uncalibrated programmable on-die termination in HDIO banks (measured per JEDEC specification)					
R^7	Thevenin equivalent resistance of programmable input termination to $V_{CCO}/2$ where ODT = RTT_48	–50%	48	+50%	Ω
Differential termination	Programmable differential termination (TERM_100) for XPIO banks	–35%	100	+35%	Ω

Notes:

- Typical values are specified at nominal voltage, 25°C.
- This measurement represents the die capacitance at the pad, not including the package.
- Maximum value specified for worst case process at 25°C.
- Battery-backed RAM (BBRAM) is always enabled and included in I_{CC_BATT} .
- VR resistor tolerance is (240Ω ±1%).
- The tolerance limits are specified after calibration with stable voltage and temperature.
- On-die input termination resistance, for more information see the *Versal Adaptive SoC SelectIO Resources Architecture Manual* (AM010).

V_{IN} Maximum Allowed AC Voltage Overshoot and Undershoot

Table 16: V_{IN} Maximum Allowed AC Voltage Overshoot and Undershoot for HDIO and PSIO Banks

AC Voltage Overshoot ¹	% of UI ² at -55°C to 125°C	AC Voltage Undershoot ¹	% of UI ² at -55°C to 125°C
$V_{CCO} + 0.20$	100%	-0.20	100%
$V_{CCO} + 0.25$	100%	-0.25	75%
$V_{CCO} + 0.30$	75%	-0.30	75%
$V_{CCO} + 0.35$	75%	-0.35	75%
$V_{CCO} + 0.40$	75%	-0.40	60%
$V_{CCO} + 0.45$	50%	-0.45	30%
$V_{CCO} + 0.50$	30%	-0.50	0%
$V_{CCO} + 0.55$	20%	-0.55	0%
$V_{CCO} + 0.60$	20%	-0.60	0%
$V_{CCO} + 0.65$	10%	-0.65	0%
$V_{CCO} + 0.70$	10%	-0.70	0%
$V_{CCO} + 0.75$	10%	-0.75	0%
$V_{CCO} + 0.80$	10%	-0.80	0%
$V_{CCO} + 0.85$	8%	-0.85	0%
$V_{CCO} + 0.90$	6%	-0.90	0%
$V_{CCO} + 0.95$	2%	-0.95	0%

Notes:

1. A total of 200 mA per bank should not be exceeded.
2. For UI smaller than 20 μ s.

Table 17: V_{IN} Maximum Allowed AC Voltage Overshoot and Undershoot for XPIO Banks

AC Voltage Overshoot ¹	% of UI ² at -55°C to 125°C	AC Voltage Undershoot ¹	% of UI ² at -55°C to 125°C
$V_{CCO} + 0.20$	100%	-0.20	100%
$V_{CCO} + 0.25$	100%	-0.25	100%
$V_{CCO} + 0.30$	100%	-0.30	10%
$V_{CCO} + 0.35$	100%	-0.35	0%
$V_{CCO} + 0.40$	96%	-0.40	0%
$V_{CCO} + 0.45$	2%	-0.45	0%

Notes:

1. A total of 200 mA per bank should not be exceeded.
2. For UI smaller than 20 μ s.
3. In XPIO banks, V_{IN} overshoot above V_{CCO} and undershoot below GND can reduce the performance of V_{REF} -based receivers within the same nibble.

DC Input and Output Levels

Values for V_{IL} and V_{IH} are recommended input voltages. Values for I_{OL} and I_{OH} are guaranteed over the recommended operating conditions at the V_{OL} and V_{OH} test points. Only selected standards are tested. These are chosen to ensure that all standards meet their specifications. The selected standards are tested at a minimum V_{CCO} with the respective V_{OL} and V_{OH} voltage levels shown. Other standards are sample tested.

PSIO Levels

Table 18: PSIO DC Input and Output Levels

I/O Standard ^{1, 2}	V_{IL}		V_{IH}		V_{OL}	V_{OH}	I_{OL}	I_{OH}
	V, Min	V, Max	V, Min	V, Max	V, Max	V, Min	mA	mA
LVC MOS18	-0.300	35% V_{CCO}	65% V_{CCO}	$V_{CCO} + 0.300$	0.450	$V_{CCO} - 0.450$	Note 3	Note 3
LVC MOS25	-0.300	0.700	1.700	$V_{CCO} + 0.300$	0.400	$V_{CCO} - 0.400$	Note 3	Note 3
LVC MOS33	-0.300	0.800	2.000	3.400	0.400	$V_{CCO} - 0.400$	Note 3	Note 3

Notes:

1. Tested according to relevant specifications.
2. Standards specified using the default I/O standard configuration. For details, see the *Versal Adaptive SoC Technical Reference Manual* (AM011).
3. Supported drive strengths of 4, 8, or 12 mA.

SelectIO Levels

Table 19: SelectIO DC Input and Output Levels For HDIO Banks

I/O Standard ^{1, 2}	V_{IL}		V_{IH}		V_{OL}	V_{OH}	I_{OL}	I_{OH}
	V, Min	V, Max	V, Min	V, Max	V, Max	V, Min	mA	mA
HSTL_I_18	-0.300	50% $V_{CCO} - 0.100$	50% $V_{CCO} + 0.100$	$V_{CCO} + 0.300$	0.400	$V_{CCO} - 0.400$	8.0	-8.0
LVC MOS18	-0.300	35% V_{CCO}	65% V_{CCO}	$V_{CCO} + 0.300$	0.450	$V_{CCO} - 0.450$	Note 3	Note 3
LVC MOS25	-0.300	0.700	1.700	$V_{CCO} + 0.300$	0.400	$V_{CCO} - 0.400$	Note 3	Note 3
LVC MOS33	-0.300	0.800	2.000	3.400	0.400	$V_{CCO} - 0.400$	Note 3	Note 3
LVTTL	-0.300	0.800	2.000	3.400	0.400	2.400	Note 3	Note 3
SSTL18_I	-0.300	50% $V_{CCO} - 0.125$	50% $V_{CCO} + 0.125$	$V_{CCO} + 0.300$	$V_{CCO}/2 - 0.470$	$V_{CCO}/2 + 0.470$	8.0	-8.0

Notes:

1. Tested according to relevant specifications.
2. Standards specified using the default I/O standard configuration. For details, see the *Versal Adaptive SoC SelectIO Resources Architecture Manual* (AM010).
3. Supported drive strengths of 4, 8, or 12 mA in HDIO banks.

Table 20: SelectIO DC Input and Output Levels for XPIO Banks

I/O Standard ^{1, 2, 3}	V_{IL}		V_{IH}		V_{OL}	V_{OH}	I_{OL}	I_{OH}
	V, Min	V, Max	V, Min	V, Max	V, Max	V, Min	mA	mA
HSTL_I	-0.300	$50\% V_{CCO} - 0.100$	$50\% V_{CCO} + 0.100$	$V_{CCO} + 0.300$	0.400	$V_{CCO} - 0.400$	5.8	-5.8
HSTL_I_12	-0.300	$50\% V_{CCO} - 0.080$	$50\% V_{CCO} + 0.080$	$V_{CCO} + 0.300$	$25\% V_{CCO}$	$75\% V_{CCO}$	4.1	-4.1
HSUL_12	-0.300	$50\% V_{CCO} - 0.130$	$50\% V_{CCO} + 0.130$	$V_{CCO} + 0.300$	$20\% V_{CCO}$	$80\% V_{CCO}$	0.1	-0.1
LVC MOS12	-0.300	$35\% V_{CCO}$	$65\% V_{CCO}$	$V_{CCO} + 0.300$	0.400	$V_{CCO} - 0.400$	Note 4	Note 4
LVC MOS15	-0.300	$35\% V_{CCO}$	$65\% V_{CCO}$	$V_{CCO} + 0.300$	0.450	$V_{CCO} - 0.450$	Note 5	Note 5
LVDCI_15	-0.300	$35\% V_{CCO}$	$65\% V_{CCO}$	$V_{CCO} + 0.300$	0.450	$V_{CCO} - 0.450$	7.0	-7.0
SSTL12	-0.300	$50\% V_{CCO} - 0.100$	$50\% V_{CCO} + 0.100$	$V_{CCO} + 0.300$	$V_{CCO}/2 - 0.150$	$V_{CCO}/2 + 0.150$	8.0	-8.0
SSTL135	-0.300	$50\% V_{CCO} - 0.090$	$50\% V_{CCO} + 0.090$	$V_{CCO} + 0.300$	$V_{CCO}/2 - 0.150$	$V_{CCO}/2 + 0.150$	9.0	-9.0
SSTL15	-0.300	$50\% V_{CCO} - 0.100$	$50\% V_{CCO} + 0.100$	$V_{CCO} + 0.300$	$V_{CCO}/2 - 0.175$	$V_{CCO}/2 + 0.175$	10.0	-10.0

Notes:

1. Tested according to relevant specifications.
2. Standards specified using the default I/O standard configuration. For details, see the *Versal Adaptive SoC SelectIO Resources Architecture Manual* (AM010).
3. POD10 and POD12 DC input and output levels are shown in Table 21, Table 26, and Table 27.
4. Supported drive strengths of 2, 4, 6, or 8 mA in XPIO banks.
5. Supported drive strengths of 2, 4, 6, 8, or 12 mA in XPIO banks.

Table 21: DC Input Levels for Single-ended POD10, POD12, LVSTL06_12, and LVSTL_11 I/O Standards

I/O Standard ^{1, 2}	V_{IL}		V_{IH}	
	V, Min	V, Max	V, Min	V, Max
POD10	-0.300	$70\% V_{CCO} - 0.068$	$70\% V_{CCO} + 0.068$	$V_{CCO} + 0.300$
POD12	-0.300	$70\% V_{CCO} - 0.068$	$70\% V_{CCO} + 0.068$	$V_{CCO} + 0.300$
LVSTL06_12	-0.300	$V_{CCO}/8 - 0.100$	$V_{CCO}/8 + 0.100$	$V_{CCO} + 0.300$
LVSTL_11	-0.300	$V_{CCO}/6 - 0.100$	$V_{CCO}/6 + 0.100$	$V_{CCO} + 0.300$

Notes:

1. Tested according to relevant specifications.
2. Standards specified using the default I/O standard configuration. For details, see the *Versal Adaptive SoC SelectIO Resources Architecture Manual* (AM010).

Table 22: Differential SelectIO DC Input and Output Levels for MIPI_DPHY

I/O Standard	V _{ICM} (V) ¹			V _{ID} (V) ²			V _{ILHS} ³	V _{IHHS} ³	V _{OCM} (V) ⁴			V _{OD} (V) ⁵		
	Min	Typ	Max	Min	Typ	Max	Min	Max	Min	Typ	Max	Min	Typ	Max
MIPI_DPHY for operation <1.5 GB/s ⁷	0.070	–	0.330	0.070	–	–	–0.040	0.460	0.150	0.200	0.250	0.140	0.200	0.270
MIPI_DPHY for operation at >1.5G GB/s ⁷	0.070	–	0.330	0.040	–	–	–0.040	0.460	0.150	0.200	0.250	0.140	0.200	0.270

Notes:

1. V_{ICM} is the input common mode voltage.
2. V_{ID} is the input differential voltage (Q – $\overline{Q}$).
3. V_{IHHS} and V_{ILHS} are the single-ended input high and low voltages, respectively.
4. V_{OCM} is the output common mode voltage.
5. V_{OD} is the output differential voltage (Q – $\overline{Q}$).
6. LVDS15 is specified in Table 28.
7. High-speed option for MIPI_DPHY. The V_{ID} maximum is aligned with the standard's specification. A higher V_{ID} is acceptable as long as the V_{IN} specification is also met.

Table 23: Complementary Differential SelectIO DC Input and Output Levels for HDIO Banks

I/O Standard	V _{ICM} (V) ¹			V _{ID} (V) ²		V _{OL} (V) ³	V _{OH} (V) ⁴	I _{OL}	I _{OH}
	Min	Typ	Max	Min	Max	Max	Min	mA	mA
DIFF_HSTL_I_18	0.300	0.900	1.425	0.100	–	0.400	V _{CCO} – 0.400	8.0	–8.0
DIFF_SSTL18_I	0.300	0.900	1.425	0.100	–	(V _{CCO} /2) – 0.47	(V _{CCO} /2) + 0.47	8.0	–8.0
LVDS_25	0.300	1.200	1.425	0.100	0.600	–	–	–	–
SUB_LVDS	0.500	0.900	1.300	0.070	–	–	–	–	–
LVPECL	0.300	1.200	1.425	0.100	0.600	–	–	–	–
SLVS_400_25	0.070	0.200	0.330	0.140	0.450	–	–	–	–

Notes:

1. V_{ICM} is the input common mode voltage.
2. V_{ID} is the input differential voltage (Q – $\overline{Q}$).
3. V_{OL} is the single-ended low-output voltage.
4. V_{OH} is the single-ended high-output voltage.

Table 24: Complementary Differential SelectIO DC Input and Output Levels for XPIO Banks

I/O Standard ¹	V _{ICM} (V) ²			V _{ID} (V) ³		V _{OL} (V) ⁴	V _{OH} (V) ⁵	I _{OL}	I _{OH}
	Min	Typ	Max	Min	Max	Max	Min	mA	mA
DIFF_HSTL_I	0.680	V _{CCO} /2	(V _{CCO} /2) + 0.150	0.100	–	0.400	V _{CCO} – 0.400	5.8	–5.8
DIFF_HSTL_I_12	0.400 x V _{CCO}	V _{CCO} /2	0.600 x V _{CCO}	0.100	–	0.250 x V _{CCO}	0.750 x V _{CCO}	4.1	–4.1
DIFF_HSUL_12	(V _{CCO} /2) – 0.120	V _{CCO} /2	(V _{CCO} /2) + 0.120	0.100	–	20% V _{CCO}	80% V _{CCO}	0.1	–0.1
DIFF_SSTL12	(V _{CCO} /2) – 0.150	V _{CCO} /2	(V _{CCO} /2) + 0.150	0.100	–	(V _{CCO} /2) – 0.150	(V _{CCO} /2) + 0.150	8.0	–8.0
DIFF_SSTL135	(V _{CCO} /2) – 0.150	V _{CCO} /2	(V _{CCO} /2) + 0.150	0.100	–	(V _{CCO} /2) – 0.150	(V _{CCO} /2) + 0.150	9.0	–9.0
DIFF_SSTL15	(V _{CCO} /2) – 0.175	V _{CCO} /2	(V _{CCO} /2) + 0.175	0.100	–	(V _{CCO} /2) – 0.175	(V _{CCO} /2) + 0.175	10.0	–10.0

Notes:

1. DIFF_POD10 and DIFF_POD12 XPIO bank specifications are shown in [Table 25](#), [Table 26](#), and [Table 27](#).
2. V_{ICM} is the input common mode voltage.
3. V_{ID} is the input differential voltage.
4. V_{OL} is the single-ended low-output voltage.
5. V_{OH} is the single-ended high-output voltage.

Table 25: DC Input Levels for Differential POD10, POD12, LVSTL06_12, and LVSTL_11 I/O Standards

I/O Standard ^{1, 2}	V _{ICM} (V)			V _{ID} (V)	
	Min	Typ	Max	Min	Max
DIFF_POD10	0.630	0.700	0.770	0.140	–
DIFF_POD12	0.756	0.840	0.924	0.160	–
DIFF_LVSTL06_12	0.143	0.150	0.157	0.140	–
DIFF_LVSTL_11	0.174	0.183	0.193	0.140	–

Notes:

1. Tested according to relevant specifications.
2. Standards specified using the default I/O standard configuration. For details, see the *Versal Adaptive SoC SelectIO Resources Architecture Manual* ([AM010](#)).

Table 26: DC Output Levels for Single-ended and Differential POD10, POD12, LVSTL06_12, and LVSTL_11 I/O Standards

I/O Standard	Symbol	Description ^{1, 2, 3}	V _{OUT}	Min	Typ	Max	Units
POD10 and POD12	R _{OL}	Pull-down resistance	V _{OM_DC} (as described in Table 27)	32	40	48	Ω
	R _{OH}	Pull-up resistance	V _{OM_DC} (as described in Table 27)	32	40	48	Ω
LVSTL06_12	R _{OL}	Pull-down resistance	V _{OCM_DC_LOW}	32	40	48	Ω
	R _{OH}	Pull-up resistance	V _{OCM_DC_HIGH}	32	40	48	Ω
LVSTL_11 (V _{OH} = 50)	R _{OL}	Pull-down resistance	V _{OM_DC} (as described in Table 27)	32	40	48	Ω
	R _{OH}	Pull-up resistance	V _{OM_DC} (as described in Table 27)	32	40	48	Ω

Notes:

1. Tested according to relevant specifications.
2. The tolerance limits are specified after calibration with stable voltage and temperature.
3. Standards specified using the default I/O standard configuration. For details, see the *Versal Adaptive SoC SelectIO Resources Architecture Manual* ([AM010](#)).

Table 27: Definitions for DC Output Levels for Single-ended and Differential POD10, POD12, LVSTL06_12, and LVSTL_11 I/O Standards

I/O Standard	Symbol	Description	All Speed Grades	Units
POD10 and POD12	V_{OM_DC}	DC output mid measurement level (for IV curve linearity)	$0.8 \times V_{CCO}$	V
LVSTL_11 ($V_{OH} = 50$)	V_{OM_DC}	DC output mid measurement level (for IV curve linearity)	$V_{CCO}/2$	V
LVSTL06_12	$V_{OCM_DC_LOW}$	DC output mid measurement level (for IV curve linearity), drive logic Low	$V_{CCO}/2$	V
	$V_{OCM_DC_HIGH}$	DC output measurement level (for IV curve linearity), drive logic High	$V_{CCO}/2$ to $V_{CCO}/4$	V

LVDS DC Specifications (LVDS15)

The LVDS15 standard is available in the XPIO banks. See the *Versal Adaptive SoC SelectIO Resources Architecture Manual* ([AM010](#)) for more information.

Table 28: LVDS15 DC Specifications

Symbol	DC Parameter	Conditions	Min	Typ	Max	Units
V_{CCO} ¹	Supply voltage		1.425	1.500	1.575	V
V_{ODIFF} ²	Differential output voltage: ($Q - \bar{Q}$), $Q = \text{High}$ ($\bar{Q} - Q$), $\bar{Q} = \text{High}$	$R_T = 100\Omega$ across Q and $\bar{Q}$ signals	247	350	454	mV
V_{OCM} ²	Output common-mode voltage	$R_T = 100\Omega$ across Q and $\bar{Q}$ signals	1.000	1.20	1.320	V
V_{IDIFF} ³	Differential input voltage: ($Q - \bar{Q}$), $Q = \text{High}$ ($\bar{Q} - Q$), $\bar{Q} = \text{High}$		100	350	600 ³	mV
V_{ICM_DC} ⁴	Input common-mode voltage (DC coupling)		0.300	1.200	1.320	V
V_{ICM_AC} ⁵	Input common-mode voltage (AC coupling)		200	–	330	mV

Notes:

1. In XPIO banks, when LVDS15 is used with input-only functionality, it can be placed in a bank where the V_{CCO} levels are different from the specified level only if internal differential termination is not used. In this scenario, V_{CCO} must be chosen to ensure the input pin voltage levels do not violate the Recommended Operating Condition ([Table 13](#)) specification for the V_{IN} I/O pin voltage.
2. V_{OCM} and V_{ODIFF} values are for $LVDS_PRE_EMPHASIS = \text{FALSE}$.
3. Maximum V_{IDIFF} value is specified for the maximum V_{ICM} specification. With a lower V_{ICM} , a higher V_{IDIFF} is tolerated only when the recommended operating conditions and overshoot/undershoot V_{IN} specifications are maintained.
4. Input common mode voltage for DC coupled configurations. $EQUALIZATION = \text{EQ_NONE}$ (Default).
5. AC coupling with external bias and external differential termination with $EQUALIZATION$ settings enabled. $EQUALIZATION = \text{EQ_LEVEL0}$, EQ_LEVEL1 , EQ_LEVEL2 , EQ_LEVEL3 , or EQ_LEVEL4 , any setting except EQ_NONE .

Power Supply Requirements

Versal AI Edge XQR devices are powered by multiple power supply pins that must use specific rail combinations and power sequences. Only some combinations and sequences are supported. The combinations depend upon the selected device, speed specification, and power management options. The required sequencing, power delivery options, and decoupling requirements based on design are found in the Power Design Manager (PDM) tool (download at www.amd.com/power). See also *Simplified Power Sequencing Application Note* (XAPP1375) for specific power sequencing methods.

Table 29: Power Supply Ramp Time

Symbol	Description	Min	Max	Units
T _{VCCAUX}	Ramp time from GND to 95% of V _{CCAUX}	0.2	40	ms
T _{VCCAUX_PMC}	Ramp time from GND to 95% of V _{CCAUX_PMC}	0.2	40	ms
T _{VCCAUX_SMON}	Ramp time from GND to 95% of V _{CCAUX_SMON}	0.2	40	ms
T _{VCC_FUSE}	Ramp time from GND to 95% of V _{CC_FUSE}	0.2	40	ms
T _{VCCINT}	Ramp time from GND to 95% of V _{CCINT}	0.2	40	ms
T _{VCC_IO_VCC_SOC}	Ramp time from GND to 95% of V _{CC_IO} and V _{CC_SOC}	0.2	40	ms
T _{VCCO}	Ramp time from GND to 95% of V _{CCO}	0.2	40	ms
T _{VCC_PMC}	Ramp time from GND to 95% of V _{CC_PMC}	0.2	40	ms
T _{VCC_PSPF}	Ramp time from GND to 95% of V _{CC_PSPF}	0.2	40	ms
T _{VCC_PSLP}	Ramp time from GND to 95% of V _{CC_PSLP}	0.2	40	ms
T _{VCC_RAM}	Ramp time from GND to 95% of V _{CC_RAM}	0.2	40	ms
T _{GTYP_AVCC}	Ramp time from GND to 95% of V _{GTYP_AVCC}	0.2	40	ms
T _{GTYP_AVCCAUX}	Ramp time from GND to 95% of V _{GTYP_AVCCAUX}	0.2	40	ms
T _{GTYP_AVTT}	Ramp time from GND to 95% of V _{GTYP_AVTT}	0.2	40	ms

AC Switching Characteristics

All values represented in this data sheet are based on the speed specifications in the AMD Vivado™ Design Suite as outlined in the following table.

Table 30: Speed Specification Version by Device

2024.2	Device
2.03	XQRVE2302

Notes:

1. This table lists the latest Vivado tools version. The device is supported in all subsequent versions of software including the version listed in [Table 32: Production Software and Speed Specification Release](#), no design changes are required.

Switching characteristics are specified on a per-speed-grade basis and can be designated as evaluation, engineering sample, pre-production, or production. Each designation is defined as follows:

- **Evaluation Product Specification:** These specifications are used for architecture evaluation only. They cannot be used for timing closure.
- **Engineering Sample Product Specification:** These specifications are based on simulations only and are typically available soon after dice design specifications are frozen. Although speed grades with this designation are considered relatively stable and conservative, some under-reporting might still occur.
- **Pre-production Product Specification:** These specifications are based on almost complete silicon characterization. Dice and speed grades with this designation are intended to give a better indication of the expected performance of production silicon. The probability of under-reporting delays is greatly reduced as compared to engineering sample (ES) data.
- **Production Product Specification:** These specifications are released once enough production silicon of a particular dice family member has been characterized to provide full correlation between specifications and dices over numerous production lots. There is no under-reporting of delays, and customers receive formal notification of any subsequent changes. Typically, the slowest speed grades transition to production before faster speed grades.

Testing of AC Switching Characteristics

Internal timing parameters are derived from measuring internal test patterns. All AC switching characteristics are representative of worst-case supply voltage and junction temperature conditions.

For more specific, more precise, and worst-case guaranteed data, use the values reported by the static timing analyzer and back-annotate to the simulation net list. Unless otherwise noted, values apply to all Versal AI Edge XQR devices.

Speed Grade Designations

Because individual family members are produced at different times, the migration from one category to another depends completely on the status of the fabrication process for each device. The following table correlates the current status of the Versal AI Edge XQR device on a per speed grade basis. See [Table 14: Available Speed Grades and Operating Voltages](#) for operating voltages listed by speed grade.

Table 31: Speed Grade Designations by Device

Device	Speed Grade, Temperature Ranges, Static Screen, and V _{CCINT} Operating Voltages			
	Evaluation	Engineering Sample	Pre-production	Production
XQRVE2302				-1MSB (V _{CCINT} = 0.80V)

Production Silicon and Software Status

In some cases, a particular family member (and speed grade) is released to production before a speed specification is released with the correct label (engineering sample, pre-production, production). Any labeling discrepancies are corrected in subsequent speed specification releases.

The following table lists the production released Versal AI Edge XQR device, speed grade, and the minimum corresponding supported speed specification version and Vivado software versions. The Vivado software and speed specifications listed are the minimum releases required for production. All subsequent releases of software and speed specifications are valid.

Table 32: Production Software and Speed Specification Release

Device	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT}) ^{1, 2}
	0.80V (M)
	-1
XQRVE2302	Vivado tools 2024.2 v2.03

Notes:

1. See [Table 14: Available Speed Grades and Operating Voltages](#) for the complete list of operating voltages by speed grade.
2. Blank entries indicate a device and/or speed grade in engineering sample or pre-production status.

Device Identification

The combined IDCODE and EXTENDED_IDCODE value provide the AMD device type identification as listed in the following table. For more information on the IDCODE and EXTENDED_IDCODE, see *Versal Adaptive SoC Technical Reference Manual* ([AM011](#)).

Table 33: Device Identification (IDCODE + EXTENDED_IDCODE)

Device	IDCODE[31:0] (Hex) ¹	EXTENDED_IDCODE[31:0] (Binary) ²
XQRVE2302	14CC8093	XXXX 0000 0000 0000 01XX XXXX XXXX XXXX

Notes:

1. An x at the location of the IDCODE[31:28] revision field is a placeholder until the production revision is defined.
2. Only the EXTENDED_IDCODE[27:14] bit values are relevant to device identification. The x in other bit positions indicate that they can be any value and must be ignored during device identification.

Processing System Performance Characteristics

Table 34: Processor Performance

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage	Units
		0.80V (M)	
		-1	
F _{APUMAX}	Maximum APU clock frequency where the operating voltage = V _{CC_PSF}	1300	MHz
F _{RPUMAX}	Maximum RPU clock frequency where the operating voltage = V _{CC_PSLP}	600	MHz

Table 35: PS-PL Interface Performance

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units
		0.80V (M)	
		-1	
F _{AXI_CLK}	Maximum AXI4 and ACE-LITE interface frequency	350	MHz
F _{PLAT_CLK}	Maximum PL address translation (AT) interface frequency	350	MHz
F _{PLATB_CLK}	Maximum PL-PS advanced trace bus (ATB) interface frequency	350	MHz
F _{PSACE_CLK}	Maximum AXI4 coherency extensions (ACE) interface frequency	350	MHz
F _{PSACP_CLK}	Maximum accelerator coherency port (ACP) interface frequency	350	MHz
F _{PSFCDMA_CLK}	Maximum DMA flow-control interface (FCI) frequency	350	MHz

PS and PMC Switching Characteristics

Clocks and Reset

Table 36: Reference Clock Requirements

Symbol	Description	Min	Max	Units
F_{REFCLK}	Reference clock (REF_CLK) frequency	27	60	MHz
T_{RMSJ_REFCLK}	REF_CLK input RMS clock jitter	–	3	ps
T_{INPJ_REFCLK}	REF_CLK input period jitter (peak-to-peak) Number of clock cycles = 10,000	–	50	ps
T_{DC_REFCLK}	REF_CLK duty cycle	45	55	%
T_{REFCLK}	REF_CLK rise time (20%–80%) and fall time (80%–20%) at 3.3V	–	3	ns
	REF_CLK rise time (20%–80%) and fall time (80%–20%) at 1.8V or 2.5V	–	3.5	ns
	REF_CLK rise time (10%–90%) and fall time (90%–10%) at 3.3V	–	4	ns
	REF_CLK rise time (10%–90%) and fall time (90%–10%) at 1.8V or 2.5V	–	4.66	ns

Notes:

1. The F_{REFCLK} clock frequency range and T_{DC_REFCLK} duty cycle specifications also apply to the PL alternative reference clock inputs (PL_PMC_ALT_REF_CLK, PL_LPD_ALT_REF_CLK, and PL_FPD_ALT_REF_CLK).
2. See [Table 38: Power-on Reset Assertion Timing Requirements](#) for REF_CLK operating requirements with respect to the POR_B input.

Table 37: RTC Crystal Requirements

Symbol	Description	Min	Typ	Max	Units
F_{XTAL}	Parallel resonance crystal frequency	–	32.768	–	kHz
T_{FTXTAL}	Frequency tolerance	–20	–	20	ppm
C_{XTAL}	Load capacitance for crystal parallel resonance	–	12.5	–	pF
R_{ESR}	Crystal ESR (16.8 and 19.2 MHz)	–	70	–	k Ω
C_{SHUNT}	Crystal shunt capacitance	–	1.4	–	pF

Table 38: Power-on Reset Assertion Timing Requirements

Symbol	Description	Min	Typ	Max	Units
T_{POR_B}	Required POR_B assertion time ^{1, 2}	10	–	–	μ s
$T_{MODEPOR}$	MODE[3:0] setup time to POR_B rising edge	74	–	–	ns
$T_{PORMODE}$	POR_B rising edge to MODE[3:0] hold time	74	–	–	ns

Notes:

1. The POR_B input must be asserted Low during the power-on sequence and continue to be asserted for a duration T_{POR_B} after all the required supplies of the PMC have reached minimum voltage levels. The PS, system, and PL domains can be independently powered on or off with additional power management. If the PS, system, and/or PL domains are expected to be functional at initial power-on without additional power management, then the POR_B input must be held Low until all required domain power supplies have also reached minimum voltage levels. For additional power-on sequence information, refer to the *Power Design Manager User Guide* (UG1556). For additional power management information, see the *Versal Adaptive SoC Technical Reference Manual* (AM011) or the *Versal Adaptive SoC System Software Developers Guide* (UG1304).
2. Before the deassertion of POR_B, the REF_CLK must be operating within specification.

Table 39: PS FPD Clocks Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CC_PSFP})	Units
		0.80V (M)	
		-1	
$F_{FPD_LSBUS_CLK}$	Maximum FPD LSBUS clock frequency	150	MHz
$F_{FPD_TOPSW_CLK}$	Maximum FPD top-switch clock frequency	800	MHz
$F_{DBG_FPD_CLK}$	Maximum debug FPD clock frequency	400	MHz

Table 40: PS LPD Clocks Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CC_PSLP})	Units
		0.80V (M)	
		-1	
$F_{RPLL_TO_XPD_CLK}$	Maximum RPU PLL to XPD clock frequency	1200	MHz
$F_{LPD_TOPSW_CLK}$	Maximum LPD top-switch clock frequency ¹	600	MHz
$F_{LPD_LSBUS_CLK}$	Maximum LPD LSBUS clock frequency ¹	150	MHz
$F_{IOP_SW_CLK}$	Maximum I/O peripherals (IOP) switch clock frequency	250	MHz
F_{TS_REFCLK}	Maximum time-stamp reference clock frequency	100	MHz
F_{PSM_REFCLK}	Maximum PS manager (PSM) reference clock frequency	460	MHz
$F_{DBG_LPD_CLK}$	Maximum debug LPD clock frequency	400	MHz
$F_{DBG_TS_CLK}$	Maximum debug time-stamp clock frequency	400	MHz
F_{USB_REFCLK}	Maximum USB reference clock frequency	60	MHz
$F_{CPM_TOPSW_CLK}$	Maximum CPM top-switch clock frequency	800	MHz

Notes:

1. The LPD_TOPSW_CLK operating frequency must be greater than the LPD_LSBUS_CLK operating frequency.

Table 41: PMC IRO Clock Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CC_PMC})	Units
		0.80V (M)	
		-1	
$F_{PMC_IRO_CLK}$	PMC internal clock source typical frequency	400	MHz
	PMC internal clock source tolerance	+10/-17	%

Table 42: PMC Clocks Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CC_PMC})	Units
		0.80V (M)	
		-1	
F_{EFUSE_REFCLK}	Maximum eFUSE reference clock frequency for reading	115	MHz
	Maximum eFUSE reference clock frequency for programming	60	MHz

Table 42: PMC Clocks Switching Characteristics (cont'd)

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_PMC})	Units
		0.80V (M)	
		-1	
F _{SMON_REFCLK}	Maximum system monitor reference clock frequency	300	MHz
F _{USB_SREFCLK}	Maximum USB suspend reference clock frequency	115	MHz
F _{AXI_TO_REFCLK}	Maximum AXI4 timeout reference clock frequency	115	MHz
F _{CFU_REFCLK}	Maximum configuration frame unit (CFU) reference clock frequency ¹	400	MHz
	Minimum CFU reference clock frequency	20	MHz
F _{LSBUS_REFCLK}	Maximum PMC LSBUS reference clock frequency	150	MHz
F _{NPI_REFCLK}	Maximum NoC programming interface (NPI) reference clock frequency	300	MHz
F _{HSM0_REFCLK}	Maximum horizontal super module (HSM0) reference clock frequency used with XPIO ²	200	MHz
F _{HSM1_REFCLK}	Maximum horizontal super module (HSM1) reference clock frequency used with XPIO ³	200	MHz
F _{PL0_REFCLK}	Maximum PL0 reference clock frequency	350	MHz
F _{PL1_REFCLK}	Maximum PL1 reference clock frequency	350	MHz
F _{PL2_REFCLK}	Maximum PL2 reference clock frequency	350	MHz
F _{PL3_REFCLK}	Maximum PL3 reference clock frequency	350	MHz
F _{PPLL_TO_XPD_CLK}	Maximum PMC PLL to XPD clock frequency	1200	MHz
F _{NPLL_TO_XPD_CLK}	Maximum NoC PLL to XPD clock frequency	1200	MHz

Notes:

1. The maximum configuration frame interface (CFI) clock frequency is the same as the CFU reference clock. When the programmable device image (PDI) is compressed, the compressed data rate through the CFU decompressor is limited to half of the CFI data rate.
2. When the HSM0 reference clock is used as the source clock to the AI Engine PLL, the frequency range is limited from 27 MHz to 60 MHz.
3. When the HSM1 reference clock is used as the source clock to the XPLL, the frequency range is limited from 100 MHz to 200 MHz.

Table 43: PMC PLL Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_PMC})	Units
		0.80V (M)	
		-1	
F _{PMCPLL}	PMC PLL output frequency	1600	MHz, Max
		270	MHz, Min
F _{PMCPLL} VCO	PMC PLL VCO frequency	4320	MHz, Max
		2160	MHz, Min
T _{PMCPLL} LOCK	PMC PLL lock time	100	μs, Max

Table 44: NoC PLL Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CC_PMC})	Units
		0.80V (M)	
		-1	
F_{NOCPLL}	NoC PLL output frequency	1600	MHz, Max
		270	MHz, Min
$F_{NOCPLLCO}$	NoC PLL VCO frequency	4320	MHz, Max
		2160	MHz, Min
$T_{NOCPLLLOCK}$	NoC PLL lock time	100	μ s, Max

Table 45: PS APU PLL Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CC_PSFP})	Units
		0.80V (M)	
		-1	
F_{PSAPLL}	APU PLL output frequency	1600	MHz, Max
		270	MHz, Min
$F_{PSAPLLCO}$	APU PLL VCO frequency	4320	MHz, Max
		2160	MHz, Min
$T_{PSAPLLLOCK}$	APU PLL lock time	100	μ s, Max

Table 46: PS RPU PLL Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CC_PSLP})	Units
		0.80V (M)	
		-1	
F_{PSRPLL}	RPU PLL output frequency	1600	MHz, Max
		270	MHz, Min
$F_{PSRPLLCO}$	RPU PLL VCO frequency	4320	MHz, Max
		2160	MHz, Min
$T_{PSRPLLLOCK}$	RPU PLL lock time	100	μ s, Max

PMC JTAG and SelectMAP

Table 47: JTAG/Boundary-Scan Port Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CCINT})	Units
		0.80V (M)	
		-1	
F_{TCK}^1	JTAG clock frequency	70	MHz, Max
T_{TCKL}^1	TCK Low pulse time	5.5	ns, Min
T_{TCKH}	TCK High pulse time	2.0	ns, Min
T_{TAPTCK}/T_{TCKTAP}	TMS and TDI setup and hold	3.0/2.0	ns, Min
T_{TCKTDO}	TCK falling edge to TDO output	5.5	ns, Max

Notes:

1. When using AC-JTAG, the maximum F_{TCK} frequency is 40 MHz and the minimum T_{TCKL} low pulse time is 7 ns.

Table 48: SelectMap Interface Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CCINT})	Units
		0.80V (M)	
		-1	
$F_{SMAPCLK}$	SelectMap clock frequency, write	200	MHz, Max
	SelectMap clock frequency, read	120	MHz, Max
$T_{SMAPCLKL}$	SelectMAP clock Low time	2.5	ns, Min
$T_{SMAPCLKH}$	SelectMAP clock High time	2.5	ns, Min
$T_{SMAPDCLK}/SMAPCLKD$	SelectMAP data (SMAP_IO[31:0]) setup and hold	3.0/0.0	ns, Min
$T_{SMAPCSCLK}/SMAPCLKCS$	SelectMAP chip select (SMAP_CS_b) setup and hold	3.0/0.0	ns, Min
$T_{SMAPRWCLK}/SMAPCLKRW$	SelectMAP read write (SMAP_RDWR_b) setup and hold	3.0/0.0	ns, Min
$T_{SMAPCLKO}$	SelectMAP clock to data output	13.0	ns, Max
$C_{SMAPBUSYCS}$	SelectMAP busy assertion to chip select deassertion	24	clock cycles, Max

PS and PMC Interface Specifications

PMC Quad-SPI Controller Interface

Table 49: Quad-SPI Interface

Symbol	Description ^{1, 2}	Min	Max	Units
Quad-SPI device clock frequency operating at >100 MHz up to 150 MHz. Loopback enabled.³				
F _{QSPI_CLK}	Quad-SPI device clock frequency	100	150	MHz
F _{QSPI_REFCLK} ⁴	Quad-SPI reference clock frequency	$2 \times F_{QSPI_CLK}$		MHz
T _{QSPIDCK}	Setup time, all inputs	0.77	–	ns
T _{QSPICKD}	Hold time, all inputs	1.0	–	ns
T _{QSPICKO}	Clock to output delay, all outputs	2.9	4.5	ns
T _{QSPICSCLK} ⁵	Chip select asserted to next clock edge	5.0	–	ns
T _{QSPICLKCS}	Clock edge to chip select deasserted	5.0	–	ns
T _{DCQSPICLK}	Quad-SPI clock duty cycle	45	55	%
Quad-SPI device clock frequency operating at >37.5 MHz up to 100 MHz. Loopback enabled.³				
F _{QSPI_CLK}	Quad-SPI device clock frequency	37.5	100	MHz
F _{QSPI_REFCLK} ⁴	Quad-SPI reference clock frequency	$2 \times F_{QSPI_CLK}$		MHz
T _{QSPIDCK}	Setup time, all inputs	2.0	–	ns
T _{QSPICKD}	Hold time, all inputs	0.0	–	ns
T _{QSPICKO}	Clock to output delay, all outputs	3.2	7.8	ns
T _{QSPICSCLK} ⁵	Chip select asserted to next clock edge	5.0	–	ns
T _{QSPICLKCS}	Clock edge to chip select deasserted	5.0	–	ns
T _{DCQSPICLK}	Quad-SPI clock duty cycle	45	55	%
Quad-SPI device clock frequency operating at ≤37.5 MHz. Loopback disabled.				
F _{QSPI_CLK}	Quad-SPI device clock frequency		37.5	MHz
F _{QSPI_REFCLK} ⁴	Quad-SPI reference clock frequency		300	MHz
T _{QSPIDCK}	Setup time, all inputs	19.1	–	ns
T _{QSPICKD}	Hold time, all inputs	0.0	–	ns
T _{QSPICKO}	Clock to output delay, all outputs	5.2	21.5	ns
T _{QSPICSCLK}	Chip select asserted to next clock edge	9.0	–	ns
T _{QSPICLKCS} ⁵	Clock edge to chip select deasserted	9.0	–	ns
T _{DCQSPICLK}	Quad-SPI clock duty cycle	45	55	%

Notes:

1. The test conditions are configured for the generic Quad-SPI interface with a 12 mA drive strength, fast slew rate, and load conditions (15 pF/30 pF for a Quad-SPI device clock frequency up to 100 MHz and 15 pF for a Quad-SPI device clock frequency > 100 MHz), tested at 3.3V and 1.8V.
2. 30 pF loads are for QSPI dual-stacked or QSPI dual-parallel modes.
3. When the Quad-SPI device clock frequency is >37.5 MHz, the Quad-SPI loopback clock output (QSPI_LPBK_CLK) must be enabled in the control, interface, and processing system (CIPS), and the associated MIO[6] pin must be left unconnected on the board.
4. The Quad-SPI reference clock frequency must be 2x the Quad-SPI device clock frequency when it is >37.5 MHz.
5. T_{QSPICSCLK} is only valid when two reference clock cycles are programmed between the chip select and clock.

PMC Octal-SPI Controller Interface

Table 50: Octal-SPI Interface

Symbol	Description ¹	Min	Max	Units
Octal-SPI device clock frequency operating at DDR 50 MHz up to 200 MHz.				
F _{OSPI_CLK}	Octal-SPI device clock frequency	50	200 ¹	MHz
F _{OSPI_REFCLK}	Octal-SPI reference clock frequency	= F _{OSPI_CLK} ²		MHz
T _{OSPI200IVW}	Input valid data window	0.5	–	UI
T _{OSPICKO}	Clock to output delay, all outputs	0.6	1.9	ns
T _{OSPICSCLK}	Chip select asserted to next clock edge	3.375	–	ns
T _{OSPICLKCS}	Clock edge to chip select deasserted	3.375	–	ns
T _{OSPIDCCLK}	Octal-SPI clock duty cycle	45	55	%
Octal-SPI device clock frequency operating at SDR 50 MHz to 166 MHz.				
F _{OSPI_CLK}	Octal-SPI device clock frequency	50	166	MHz
F _{OSPI_REFCLK}	Octal-SPI reference clock frequency	= F _{OSPI_CLK} ²		MHz
T _{OSPI166IVW}	Input valid data window	0.4	–	UI
T _{OSPICKO}	Clock to output delay, all outputs	1.8	3.9	ns
T _{OSPICSCLK}	Chip select asserted to next clock edge	3.375	–	ns
T _{OSPICLKCS}	Clock edge to chip select deasserted	3.375	–	ns
T _{OSPIDCCLK}	Octal-SPI clock duty cycle	45	55	%
Octal-SPI device clock frequency operating at SDR <50 MHz.				
F _{OSPI_CLK}	Octal-SPI device clock frequency	–	50	MHz
F _{OSPI_REFCLK}	Octal-SPI reference clock frequency	4 × F _{OSPI_CLK}	200	MHz
T _{OSPIDCK}	Setup time, all inputs	11	–	ns
T _{OSPICKD}	Hold time, all inputs	1.0	–	ns
T _{OSPICKO}	Clock to output delay, all outputs	2.0	18	ns
T _{OSPICSCLK}	Chip select asserted to next clock edge	3.375	–	ns
T _{OSPICLKCS}	Clock edge to chip select deasserted	3.375	–	ns
T _{OSPIDCCLK}	Octal-SPI clock duty cycle	45	55	%

Notes:

- The test conditions are configured for the Octal-SPI interface with a 12 mA drive strength, fast slew rate, and 12 pF load. The maximum Octal-SPI device clock frequency under different load conditions are 166 MHz for a 20 pF load and 100 MHz for a 40 pF load.
- The Octal-SPI reference clock frequency must be equal to F_{OSPI_CLK} when the Octal-SPI device clock frequency is ≥50 MHz.

PS SPI Controller Interface

Table 51: SPI Controller Interface

Symbol	Description ¹	Min	Max	Units
SPI Master Interface				
F _{MSPI_CLK}	SPI master device clock frequency (MIO)	–	50	MHz
	SPI master device clock frequency (EMIO)	–	25	MHz
F _{SPI_REFCLK}	SPI reference clock frequency	–	200	MHz
T _{DCMSPICLK}	SPI master mode clock duty cycle	45	55	%
T _{MSPISSCLK}	Master select asserted to first active clock edge ²	1	–	SPI_REFCLK cycles
T _{MSPISCLKSS}	Last active clock edge to slave select deasserted ²	1	–	SPI_REFCLK cycles
T _{MSPIDCK}	Input setup time for master in/slave out (MISO)	9.9	–	ns
T _{MSPICKD}	Input hold time for MISO	0.0	–	ns
T _{MSPICKO}	Master out/slave in (MOSI) and slave select clock-to-out delay	–3.7	5.0	ns
SPI Slave Interface				
F _{SSPI_CLK}	SPI slave device clock frequency	–	25	MHz
F _{SPI_REFCLK}	SPI reference clock frequency	–	200	MHz
T _{SSPISSCLK}	Slave select asserted to first active clock edge	1	–	SPI_REFCLK cycles
T _{SSPISCLKSS}	Last active clock edge to slave select deasserted	1	–	SPI_REFCLK cycles
T _{SSPIDCK}	Input setup time for MISO	5.0	–	ns
T _{SSPICKD}	Input hold time for MISO	5.0	–	ns
T _{SSPICKO}	MOSI clock-to-out delay	0.0	13	ns

Notes:

- The test conditions are configured to the LVCMOS 3.3V I/O standard with a 12 mA drive strength, fast slew rate, and 15 pF load.
- The test conditions use the SPI delay register where XSPIPS_DR_OFFSET[xspips_dr_init_mask] = 2 and XSPIPS_DR_OFFSET[xspips_dr_after_mask] = 2.

PS USB Controller Interface

Table 52: ULPI Interface

Symbol	Description ¹	Min	Max	Units
T _{ULPIDCK}	Input setup to ULPI clock, all inputs	4.5	–	ns
T _{ULPICKD}	Input hold to ULPI clock, all inputs	0.5	–	ns
T _{ULPICKO}	ULPI clock to output valid, all outputs	2.0	8.9	ns
F _{ULPICLK}	ULPI clock frequency	–	60	MHz

Notes:

- The test conditions are configured to the LVCMOS 3.3V I/O standard with a 12 mA drive strength, fast slew rate, and a 15 pF load.

PS Gigabit Ethernet MAC Controller Interface

Table 53: RGMII Interface

Symbol	Description ¹	Min	Max	Units
F _{GEMTXCLK}	RGMII_TX_CLK transmit clock frequency	–	125	MHz
F _{GEMRXCLK}	RGMII_RX_CLK receive clock frequency	–	125	MHz
F _{GEMREFCLK}	Gigabit Ethernet MAC (GEM) reference clock frequency	–	125	MHz
F _{GEMTSUREFCLK} ²	Gigabit Ethernet MAC time-stamp unit reference clock frequency	–	250	MHz
T _{DCGEMTXCLK}	Transmit clock duty cycle	45	55	%
T _{GEMTXCKO}	TXD output clock to out time	–0.5	0.5	ns
T _{GEMRXDCK}	RXD input setup time	0.8	–	ns
T _{GEMRXCKD}	RXD input hold time	0.8	–	ns
T _{MDIOCLK}	MDC output clock period	400	–	ns
T _{MDIOCKL}	MDC Low time	160	–	ns
T _{MDIOCKH}	MDC High time	160	–	ns
T _{MDIODCK}	MDIO input data setup time	80	–	ns
T _{MDIOCKD}	MDIO input data hold time	0.0	–	ns
T _{MDIOCKO}	MDIO output data delay time	–3.0	15.0	ns

Notes:

- The test conditions are configured to the LVCMOS 3.3V I/O standard with a 12 mA drive strength, fast slew rate, and a 15 pF load.
- The APB interface (LPD_LSBUS_CLK) operating frequency must be greater than or equal to the GEM_TSU_CLK operating frequency.

PMC SD/SDIO Controller Interface

Table 54: SD/SDIO Interface

Symbol	Description ¹	Min	Max	Units
F _{SD_REFCLK}	SD reference clock frequency	–	200	MHz
F _{SDDL_REFCLK}	SD DLL reference clock frequency	–	1200	MHz
SD/SDIO Interface Default Speed Mode				
F _{SDS_CLK} ²	SD standard device clock frequency	–	20	MHz
F _{SDID_CLK}	Clock frequency in identification mode	–	400	kHz
T _{SDDC_CLK}	SD standard device clock duty cycle	45	55	%
T _{SDCKO}	Clock to output delay, all outputs	–2.0	4.5	ns
T _{SDDC}	Input setup time, all inputs	2.0	–	ns
T _{SDCKD}	Input hold time, all inputs	2.0	–	ns
SD/SDIO Interface High-speed Mode				
F _{SDHS_CLK}	SD high-speed device clock frequency	25	50	MHz
T _{SDHSDC_CLK}	SD high-speed device clock duty cycle	45	55	%
T _{SDHSCKO}	Clock to output delay, all outputs	2.2	13.8	ns
T _{SDHSDIVW}	Input valid data window	0.4	–	UI
SD/SDIO Interface SDR12 Mode				
F _{SDSDR12_CLK} ²	SD SDR12 device clock frequency	–	25	MHz
T _{DCSDSDR12_CLK}	SD SDR12 device clock duty cycle	30	70	%

Table 54: SD/SDIO Interface (cont'd)

Symbol	Description ¹	Min	Max	Units
T _{SDSDR12CKO}	Clock to output delay, all outputs	1.0	36.8	ns
T _{SDSDR12DCK}	Input setup time, all inputs	10.0	–	ns
T _{SDSDR12CKD}	Input hold time, all inputs	1.5	–	ns
SD/SDIO Interface SDR25/SDR50 Mode				
F _{SDSDR_CLK}	SDR25 device clock frequency	25	50	MHz
	SDR50 device clock frequency	25	100	MHz
T _{SDSDRDC_CLK}	SD SDR50/SDR25 device clock duty cycle	30	70	%
T _{SDSDRCKO}	Clock to output delay, all outputs	1.0	6.8	ns
T _{SDDDRCKD}	Input valid data window	0.4	–	UI
SD/SDIO Interface SDR104 Mode				
F _{SDSDR104_CLK}	SDR104 device clock frequency	25	200	MHz
T _{DCSDSDR104_CLK}	SD SDR104 device clock duty cycle	30	70	%
T _{SDSDR104CKO}	Clock to output delay, all outputs	1.0	3.2	ns
T _{SDSDR104CKD}	Input valid data window	0.5	–	UI
SD/SDIO Interface DDR50 Mode				
F _{SDDDR_CLK}	SD DDR50 device clock frequency	25	50	MHz
T _{DCSDDDR_CLK}	SD DDR50 device clock duty cycle	45	55	%
T _{SDDDRCKO}	Clock to output delay, data	1.0	6.8	ns
T _{SDDDRDCK}	Input valid data window	0.5	–	UI
T _{SDDDRCKD}	Input setup time, command	4.7	–	ns
T _{SDDDRIDCLK}	Input hold time, command	1.5	–	ns
T _{SDDDRCLK}	Clock to output delay, command	1.0	13.8	ns

Notes:

- The test condition settings for SD/SDIO modes are: 12 mA drive strength, fast slew rate, and a 15 pF load. The OTAP delay (OTAP_DLY[5:0]) test condition settings are: SD high-speed mode = 0x04, SD SDR25/50 mode = 0x03, SD SDR104 mode = 0x02, and SD DDR50 mode = 0x03.
- EMIO is supported in SD default speed mode and SDR12 mode.

PMC eMMC Controller Interface

Table 55: eMMC Controller Interface

Symbol	Description ¹	Min	Max	Units
eMMC Interface Standard Mode				
F _{EMMCCLK}	eMMC standard device clock frequency ²	–	25	MHz
T _{DCEMMCCLK}	eMMC clock duty cycle	45	55	%
T _{EMMCCKO}	Clock to output delay, all outputs	–2.0	4.5	ns
T _{EMMCDCK}	Input setup time, all inputs	2.0	–	ns
T _{EMMCCKD}	Input hold time, all inputs	2.0	–	ns
eMMC Interface High-speed SDR Mode				
F _{EMMCSRCLK}	eMMC high-speed SDR device clock frequency	25	50	MHz
T _{DCEMMCSRCLK}	eMMC high-speed SDR clock duty cycle	45	55	%
T _{EMMCSRCKO}	Clock to output delay, all outputs ³	3.2	16.8	ns

Table 55: eMMC Controller Interface (cont'd)

Symbol	Description ¹	Min	Max	Units
T _{EMMCSDRDIVW}	Input valid data window ⁴	0.4	–	UI
eMMC Interface High-speed DDR Mode				
F _{EMMCDDRCLK}	eMMC high-speed DDR device clock frequency	25	50	MHz
T _{DCEMMCDRCLK}	eMMC high-speed DDR clock duty cycle	45	55	%
T _{EMMCDDRCKO1}	Data clock to output delay ³	2.7	7.3	ns
T _{EMMCDDRDIW}	Input valid data window ⁴	0.35	–	UI
T _{EMMCDDRCKO2}	Command clock to output delay	3.2	16	ns
T _{EMMCDDRCK2}	Command input setup time	3.9	–	ns
T _{EMMCDDRCKD2}	Command input hold time	2.5	–	ns
eMMC Interface HS200 Mode				
F _{EMMCHS200CLK}	eMMC HS200 device clock frequency	25	200	MHz
T _{DCEMMCHS200CLK}	eMMC HS200 clock duty cycle	30	70	%
T _{EMMCHS200CKO}	Clock to output delay, all outputs ³	1.0	3.4	ns
T _{EMMCHS200DIW}	Input valid data window ⁴	0.5	–	UI

Notes:

1. The test condition settings for the eMMC modes are: 12 mA drive strength, fast slew rate, and a 15 pF load. The OTAP delay (OTAP_DLY[5:0]) test condition settings are: eMMC high-speed SDR mode = 0x05, eMMC high-speed DDR mode = 0x05, and eMMC HS200 mode = 0x02.
2. EMIO is supported in the eMMC standard mode.
3. This specification is achieved using predetermined DLL tuning.
4. This specification is required for capturing input data using DLL tuning.

PS and PMC I2C Controller Interface

Table 56: I2C Controller Interface

Symbol	Description ¹	Min	Max	Units
I2C Fast-mode Interface				
F _{I2CF_CLK}	Serial clock line (SCL) clock frequency	–	400	kHz
F _{I2C_REFCLK} ²	I2C reference clock frequency	–	100	MHz
T _{I2CFCKL}	SCL Low time	1.3	–	μs
T _{I2CFCKH}	SCL High time	0.6	–	μs
T _{I2CFCKO}	Serial data line (SDA) clock-to-out delay	–	900	ns
T _{I2CFDCK}	SDA input setup time	100	–	ns
T _{I2CFCKD}	SDA input data hold time	0	–	ns
I2C Standard-mode Interface				
F _{I2CS_CLK}	SCL clock frequency	–	100	kHz
F _{I2C_REFCLK} ²	I2C reference clock frequency	–	100	MHz
T _{I2CSCKL}	SCL Low time	4.7	–	μs
T _{I2CSCKH}	SCL High time	4.0	–	μs
T _{I2CSCKO}	SDA clock-to-out delay	–	3450	ns
T _{I2CSDCK}	SDA input setup time	250	–	ns
T _{I2CSCKD}	SDA input data hold time	0	–	ns

Notes:

- The test conditions are configured to the LVCMOS 3.3V I/O standard with a 12 mA drive strength, fast slew rate, and a 15 pF load.
- F_{I2C_REFCLK} specification applies to PMC_I2C_REFCLK, LPD_I2C0_REFCLK, LPD_I2C1_REFCLK, and SYSMON_I2C_REFCLK.

PS CAN FD Controller Interface

Table 57: CAN FD Controller Interface

Symbol	Description ¹	Min	Max	Units
F _{CAN_FD_REFCLK} ²	CAN FD reference clock frequency 1x	–	80	MHz
	CAN FD reference clock frequency 2x	–	160	MHz
T _{CAN_FD_PWRX}	Receive pulse width	125	–	ns
T _{CAN_FD_PWTX}	Transmit pulse width	125	–	ns

Notes:

- The test conditions are configured to the LVCMOS 3.3V I/O standard with a 12 mA drive strength, fast slew rate, and a 15 pF load.
- The APB interface (LPD_LSBUS_CLK) operating frequency must be greater than or equal to the CAN FD operating frequency (which is the CAN_FD_REFCLK frequency divided by the CAN[0]1_REF_CTRL[DIVISOR]).

PS UART Controller Interface

Table 58: UART Controller Interface

Symbol	Description ¹	Min	Max	Units
F _{UART_REF_CLK}	UART reference clock frequency	–	100	MHz
BAUD _{TXMAX}	Transmit baud rate	–	6.25	Mb/s
BAUD _{RXMAX}	Receive baud rate	–	6.25	Mb/s

Notes:

- The test conditions are configured to the LVCMOS 3.3V I/O standard with a 12 mA drive strength, fast slew rate, and a 15 pF load.

PS General Purpose I/O Interface

Table 59: General Purpose I/O (GPIO) Interface

Symbol	Description ¹	Min	Max	Units
T _{GPIOH}	LPD GPIO input High pulse width	10 x 1/F _{LPD_LSBUS_CLK}	–	μs
	PMC GPIO input High pulse width	10 x 1/F _{PMC_LSBUS_CLK}	–	μs
T _{GPIOH}	LPD GPIO input Low pulse width	10 x 1/F _{LPD_LSBUS_CLK}	–	μs
	PMC GPIO input Low pulse width	10 x 1/F _{PMC_LSBUS_CLK}	–	μs

Notes:

- The test conditions are configured to the LVCMOS 3.3V I/O standard with a 12 mA drive strength, fast slew rate, and 15 pF load.

PS Trace Interface

Table 60: Trace Interface

Symbol	Description ¹	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_PSEFP})		Units
		0.80V (M)		
		-1		
		Min	Max	
F _{TCECLK}	Trace clock frequency (MIO)	–	200	MHz
	Trace clock frequency (EMIO)	–	350	MHz
F _{DBGTCECLK}	Trace debug (DBG_TRACE) clock frequency	–	400	MHz
T _{TCECKO}	Trace clock to output delay, all outputs	–0.5	0.5	ns
T _{TCECKO}	Trace clock duty cycle	45	55	%

Notes:

- The test conditions are configured to the LVCMOS 3.3V I/O standard with a 12 mA drive strength, fast slew rate, and a 15 pF load.

PS Triple-timer Counter Interface

Table 61: Triple-timer Counter Interface

Symbol	Description	Min	Max	Units
$F_{TTCX_REF_CLK}$	Triple-timer counter input clock frequency (TTCX_REF_CLK) ¹	–	See Note 1	MHz
$F_{TTCX_IO_REF_CLKX}$	Triple-timer counter MIO-EMIO input clock frequency (TTCX_IO_REF_CLKX) ²	–	50 ³	MHz
$T_{TTCICLK}$	Triple-timer counter MIO-EMIO input clock High pulse width	$2/F_{TTCX_IO_REF_CLKX}$ ²	–	μs
	Triple-timer counter MIO-EMIO input clock Low pulse width	$2/F_{TTCX_IO_REF_CLKX}$ ²	–	μs
$F_{TTCICLK}$	Triple-timer counter output clock frequency	–	$F_{TTCX_REF_CLK}/2$ ¹	MHz
			$F_{TTCX_IO_REF_CLKX}/2$ ²	MHz
$T_{TTCOCLK}$	Triple-timer counter output clock period	$2/F_{TTCX_REF_CLK}$ ¹	–	μs
		$2/F_{TTCX_IO_REF_CLKX}$ ²	–	μs

Notes:

1. TTCX_REF_CLK source is used. TTCX_REF_CLK source can be the REF_CLK (F_{REFCLK}), LPD_LSBUS_CLK ($F_{LPD_LSBUS_CLK}$), or the RPU_REF_CLK (F_{RPU_MAX}).
2. TTCX_IO_REF_CLKX source is used. TTCX_IO_REF_CLKX source can be MIO or EMIO.
3. The maximum $F_{TTCX_IO_REF_CLKX}$ is the lesser of the specified maximum $F_{TTCX_IO_REF_CLKX}$ or $F_{TTCX_REF_CLK}/2$.

PS System Watchdog Timer Interface

Table 62: System Watchdog Timer Interface

Symbol	Description	Min	Max	Units
$F_{SWDTCLK}$	System watchdog timer input clock frequency	–	100	MHz

PMC System Monitor Specifications

Table 63: PMC System Monitor Specifications

Parameter ¹	Symbol	Conditions ²	Min	Typ	Max	Units
$V_{CCAUX_SMON} = 1.5V \pm 3\%$, $V_{REFP} = 1.024V$ ³ , $V_{REFN} = 0V$, $T_j = -40^\circ C$ to $100^\circ C$, typical values at $T_j = 40^\circ C$						
ADC Accuracy⁴						
Resolution			10	–	–	Bits
Integral nonlinearity	INL		–1	–	1	LSBs
Differential nonlinearity	DNL	No missing codes, guaranteed monotonic	–1	–	1	LSBs
Offset error		Calibrated	–2	–	2	LSBs
Gain error		Calibrated	–0.4	–	0.4	%
Sample rate				0.25		MS/s
RMS code noise		External 1.024V reference	–	–	1	LSBs
		On-chip reference	–	–	1	LSBs
Analog Inputs						
ADC input ranges ($V_P - V_N$)		Unipolar operation	0		1	V
		Bipolar operation	–0.5	–	0.5	V

Table 63: PMC System Monitor Specifications (cont'd)

Parameter ¹	Symbol	Conditions ²	Min	Typ	Max	Units
ADC input common mode ranges		Unipolar common mode range (FS input)	0	–	0.25	V
		Bipolar common mode range (FS input)	0.5	–	0.6	V
Maximum external channel input ranges		Adjacent channels set within these ranges should not corrupt measurements on adjacent channels	–0.1	–	V _{CCAUX_SMON}	V
On-Chip Sensors						
Supply sensor error		Measurement range up to V _{CCAUX_SMON} ±3% T _j = –55°C to 125°C, internal or external reference	–1	–	1	%
System Monitor Reference⁵						
External reference	V _{REFP}	Externally supplied reference voltage	0.973	1.024	1.075	V
On-chip reference		Tie V _{REFP} to AGND, T _j = –40°C to 125°C	1.019	1.024	1.029	V

Notes:

1. For more information, see the *Versal Adaptive SoC System Monitor Architecture Manual* (AM006).
2. All the reference voltages required by the Versal device cannot sleep.
3. All the accuracy specs are at V_{REFP} = 1.024V (precisely). Variation in V_{REFP} will cause a proportional gain error which should be added to see the true accuracy. The temperature range is in Kelvin so the gain error is proportional.
4. Gross offset and gain errors are removed by automatic calibration. The specification quotes the net error.
5. Any variation in the reference voltage from the nominal V_{REFP} = 1.024V and V_{REFN} = 0V will result in a deviation from the ideal transfer function. This also impacts the accuracy of the internal sensor measurements (i.e., temperature and power supply). However, for external ratiometric-type applications allowing the reference to vary by ±5% is permitted. On-chip reference variation is ±0.5%.

Network on Chip Switching Characteristics

Table 64: Network on Chip Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_SOC})	Units
		0.80V (M)	
		-1	
F _{MAX}	Network on Chip (NoC) clock maximum frequency	960	MHz
F _{MAX_NMU}	NoC master unit clock maximum frequency	480	MHz

Programmable Logic Performance Characteristics

This section provides the performance characteristics of some common functions and designs implemented in the Versal AI Edge XQR devices. These values are subject to the same guidelines as the [AC Switching Characteristics](#) section.

In each of the following performance tables, the I/O bank type is either XPIO or HDIO.

Table 65: I/O Logic Performance

Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})		Units
	0.80V (M)		
	-1		
	Min	Max	
TX DDR	0	250	Mb/s
TX SDR	0	125	Mb/s
RX DDR	0	250	Mb/s
RX SDR	0	125	Mb/s

Table 66: XPHY I/O Performance

Description ^{1, 2}	Data Width	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})		Units
		0.80V (M)		
		-1		
		Min	Max	
TX DDR	8	200	1800	Mb/s
	4	200	1468	Mb/s
	2	200	734	Mb/s
RX DDR ³	8	200	1800	Mb/s
	4	200	1468	Mb/s
	2	200	734	Mb/s

Notes:

- XPHY I/O is supported through the Advanced I/O Wizard available with the Vivado Design Suite. The performance values assume a source-synchronous interface. For asynchronous interfaces, see the *Advanced I/O Wizard LogiCORE IP Product Guide (PG320)*.
- Package skews are not included and should be removed through PCB routing.
- SDR specifications are a subset of the DDR specifications.

Table 67: MIPI D-PHY Performance

Description	I/O Bank Type	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units
		0.80V (M)	
		-1	
MIPI D-PHY transmitter or receiver	XP	3200	Mb/s

Table 68: Maximum Physical Interface (PHY) Rate for Soft Memory Interface Controller

Memory Standard	DRAM Type	DIMM Slots	XPIO Bank Performance ¹	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units
				0.80V (M)	
				-1	
DDR4	Single rank component		All	2667	Mb/s
	1 rank RDIMM, DIMM ²	1	All	2667	Mb/s
	2 rank RDIMM, LRDIMM, DIMM ²	1	All	2667	Mb/s
	4 rank LRDIMM	1	All	2400	Mb/s
	1 rank RDIMM	2	All	2400	Mb/s
	1 rank DIMM ²	2	All	2133	Mb/s
	2 rank LRDIMM	2	All	2400	Mb/s
	2 rank RDIMM	2	All	2133	Mb/s
	2 rank DIMM ²	2	All	1866	Mb/s
RLDRAM3			All	1066	MHz
QDRIV	HP		All	933	MHz
	XP ³		All	1066	MHz

Notes:

1. The Versal device package pinout files specify XPIO bank performance (XPIOperf). See the *ASCII Package File* information in this data sheet.
2. Dual in-line memory module (DIMM) includes SODIMM and UDIMM.
3. The QDRIV XP performance values are for 18-bit interfaces, for 36-bit interfaces the maximum performance is 933 MHz.

Programmable Logic Switching Characteristics

Block RAM Switching Characteristics

Table 69: Block RAM Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_RAM})	Units
		0.80V (M)	
		-1	
Maximum Frequency			
F _{MAX_WF_NC}	Block RAM (WRITE_FIRST and NO_CHANGE modes)	800	MHz
F _{MAX_RF}	Block RAM (READ_FIRST mode)	675	MHz
Block RAM Clock-to-Out Delays			
T _{RCKO_DO}	Clock CLK to DOUT output (without output register)	0.967	ns, Max
T _{RCKO_DO_REG}	Clock CLK to DOUT output (with output register)	0.280	ns, Max

UltraRAM Switching Characteristics

Table 70: UltraRAM Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_RAM})		Units
		0.80V (M)		
		-1		
Maximum Frequency				
F _{MAX}	UltraRAM maximum frequency with output register	645		MHz

Accelerator RAM Switching Characteristics

The *Versal Architecture and Product Data Sheet: Overview* ([DS950](#)) lists the Versal AI Edge XQR devices that include the accelerator RAM (XRAM).

Table 71: Accelerator RAM Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CC_PSLP})	Units
		0.80V (M)	
		-1	
F_{MAX_XRAM}	Maximum accelerator RAM (XRAM) clock frequency	600	MHz
F_{PL_XRAM}	Maximum accelerator RAM PL interface frequency	300	MHz

Input/Output Delay Switching Characteristics

Table 72: Input/Output Delay Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CCINT})	Units
		0.80V (M)	
		-1	
$T_{IDELAY_RESOLUTION}/T_{ODELAY_RESOLUTION}$	XPHY IDELAY/ODELAY delay tap resolution	1.22 to 4.00	ps
$T_{IDELAY_ERROR}/T_{ODELAY_ERROR}$	XPHY calibrated delay line error (DELAY_VALUE) (REFCLK_FREQUENCY = 500 to 1800 MHz) ¹	-10 to +10	Delay Taps
$T_{IOL_IDELAY_RESOLUTION}/T_{IOL_ODELAY_RESOLUTION}$	IOL IDELAY/ODELAY uncalibrated delay tap resolution for both HD and XP IOL resources	60 to 173	ps

Notes:

- For REFCLK_FREQUENCY < 500 MHz, BISC calibration of the DELAY_VALUE_<0-5> is not guaranteed. Use the $T_{IDELAY_RESOLUTION}/T_{ODELAY_RESOLUTION}$ for delay calculations. Refer to the *Versal Adaptive SoC SelectIO Resources Architecture Manual* ([AM010](#)). IDELAY is used for alignment and ALIGN_DELAY effects the programmed DELAY_VALUE programming.

DSP58 Switching Characteristics

Table 73: DSP58 Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units
		0.80V (M)	
		-1	
Fixed Point Arithmetic			
F _{MAX_INT_MULT}	27 × 24 two’s complement multiply	984	MHz
F _{MAX_SIMD}	Single-instruction multiple-data (SIMD) arithmetic	984	MHz
Complex Arithmetic			
F _{MAX_COMPLEX_MULT}	18 × 18 two's complement complex multiply	850	MHz
Floating Point Arithmetic			
F _{MAX_FP}	Floating-point operations	700	MHz

Clock Buffers and Networks

Table 74: Clock Buffers Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units
		0.80V (M)	
		-1	
Global Clock Switching Characteristics (Including BUFGCTRL and MBUFGCTRL)			
F _{MAX}	Maximum frequency of a global clock tree (BUFG)	984	MHz
Global Clock Buffer with Input Divide Capability (BUFGCE_DIV and MBUFGCE_DIV)			
F _{MAX}	Maximum frequency of a global clock buffer with input divide capability	984	MHz
Global Clock Buffer with Clock Enable (BUFGCE)			
F _{MAX}	Maximum frequency of a global clock buffer with clock enable	984	MHz
Global Clock Buffer for the Processing System (BUFG_PS and MBUFG_PS)			
F _{MAX}	Maximum frequency of a global clock buffer with clock enable	984	MHz
GTP Clock Buffer with Clock Enable and Clock Input Divide Capability (BUFG_GT and MBUFG_GT)			
F _{MAX}	Maximum frequency of a serial transceiver clock buffer with clock enable and clock input divide capability	984	MHz

MMCM Switching Characteristics

Table 75: MMCM Specification

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_RAM})		Units
		0.80V (M)		
		-1		
F _{INMAX_MMCM}	Maximum input clock frequency ¹	984	MHz	
F _{INMIN_MMCM}	Minimum input clock frequency	10	MHz	
F _{INJITTER_MMCM}	Maximum input clock period jitter ²	< 20% of clock input period or 1 ns Max		
F _{INDUTY_MMCM}	Input duty cycle range: 10–49 MHz	25–75	%	
	Input duty cycle range: 50–199 MHz	30–70	%	
	Input duty cycle range: 200–399 MHz	35–65	%	
	Input duty cycle range: 400–499 MHz	40–60	%	
	Input duty cycle range: >500 MHz	45–55	%	
F _{MAX_PCLK_MMCM}	Maximum dynamic phase shift clock frequency	450	MHz	
F _{MIN_PCLK_MMCM}	Minimum dynamic phase shift clock frequency	0.01	MHz	
F _{VCOMAX_MMCM}	Maximum MMCM VCO frequency	4320	MHz	
F _{VCOMIN_MMCM}	Minimum MMCM VCO frequency	2160	MHz	
F _{BANDWIDTH_MMCM}	Low MMCM bandwidth at typical ³	1.00	MHz	
	High MMCM bandwidth at typical ³	4.00	MHz	
T _{STATPHAOFFSET_MMCM}	Static phase offset of the MMCM outputs ⁴	0.12	ns	
T _{OUTJITTER_MMCM}	MMCM output jitter	Note 5		
T _{OUTDUTY_MMCM}	MMCM output clock duty cycle precision ⁶	0.20	ns	
T _{LOCKMAX_MMCM}	MMCM maximum lock time (non deskew mode)	100	µs	
T _{LOCKDESKEWMAX_MMCM}	MMCM maximum lock time in deskew mode	Note 7		
F _{OUTMAX_MMCM}	MMCM maximum output clock frequency ¹	984	MHz	
F _{OUTMIN_MMCM}	MMCM minimum output clock frequency	5	MHz	
T _{EXTFDVAR_MMCM}	External clock feedback variation	< 20% of clock input period or 1 ns Max		
T _{PWRDWNMINPULSE_MMCM}	Minimum power-down pulse width	5.00	ns	
F _{PFDMAX_MMCM}	Maximum frequency at the phase frequency detector with bandwidth set to High or optimized	450	MHz	
	Maximum frequency at the phase frequency detector with bandwidth set to Low	450	MHz	
F _{PFDMIN_MMCM}	Minimum frequency at the phase frequency detector	10	MHz	
T _{FBDELAY_MMCM}	Maximum delay in the feedback path ⁸	3 ns Max or one clock cycle		

Table 75: MMCM Specification (cont'd)

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_RAM})	Units
		0.80V (M)	
		-1	
T _{DESKEWTAPDELAY_MMCM}	Nominal tap-delay of the programmable delay in the PD based deskew scheme	Note 9	

Notes:

1. The maximum input and output clock frequencies are limited by the global clock buffers. See [Table 74: Clock Buffers Switching Characteristics](#).
2. CLKIN jitter also applies to CLKIN_DESKEW and CLKFB_DESKEW in digital compensation. CLKFBIN applies only to analog compensation. This parameter is in regards to the functionality of the MMCM. Input jitter above ~1 MHz is reduced by the filtering properties of the MMCM. The magnitude of the reduction is found in the Vivado timing report.
3. The MMCM does not filter typical spread-spectrum input clocks because they are usually far below the bandwidth filter frequencies.
4. The static offset is measured between any MMCM outputs with identical phase.
5. Values for this parameter are available in the Vivado timing summary as part of the clock uncertainty equation.
6. Includes global clock buffer.
7. The maximum lock time in deskew mode is given by the following formula: Lock time in deskew mode in ms = (0.208 x (VCO_frequency in MHz) / (CLKIN_DESKEW_frequency in MHz)²) + 0.1.
8. The parameter only applies to analog compensation.
9. The value for this parameter is included in compensation delay calculations.

DPLL Switching Characteristics

Table 76: DPLL Specification

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_RAM}) ¹		Units
		0.80V (M)		
		-1		
F _{INMAX_DPLL}	Maximum input clock frequency ²	984		MHz
F _{INMIN_DPLL}	Minimum input clock frequency	50		MHz
F _{INJITTER_DPLL}	Maximum input clock jitter ³	< 20% of clock input period or 1 ns Max		
F _{INDUTY_DPLL}	Input duty cycle range: 50–399 MHz	35–65		%
	Input duty cycle range: 400–499 MHz	40–60		%
	Input duty cycle range: >500 MHz	45–55		%
F _{PSCLKMAX_DPLL}	Maximum dynamic phase shift clock frequency	450		MHz
F _{PSCLKMIN_DPLL}	Minimum dynamic phase shift clock frequency	0.01		MHz
F _{DCOMAX_DPLL}	Maximum DPLL DCO frequency	4000		MHz
F _{DCOMIN_DPLL}	Minimum DPLL DCO frequency	2000		MHz
F _{BANDWIDTH_DPLL}	DPLL bandwidth at typical ⁴	1.00		MHz
T _{STATPHAOFFSET_DPLL}	Static phase offset of the DPLL outputs ⁵	0.12		ns
T _{OUTJITTER_DPLL}	DPLL output jitter	Note 6		
T _{OUTDUTY_DPLL}	DPLL output clock duty cycle precision ⁷	0.20		ns
T _{LOCKMAX_DPLL}	DPLL maximum lock time (non-deskew mode)	Note 8		
T _{LOCKDESKEWMAX_DPLL}	DPLL maximum lock time in deskew mode	Note 9		
F _{OUTMAX_DPLL}	DPLL maximum output clock frequency ²	984		MHz
F _{OUTMIN_DPLL}	DPLL minimum output clock frequency	5		MHz
T _{PWRDWNMINPULSE_DPLL}	Minimum power-down pulse width	5.00		ns
F _{TDCMAX_DPLL}	Maximum frequency at the time to digital converter	200		MHz
F _{TDCMIN_DPLL}	Minimum frequency at the time to digital converter	50		MHz

Table 76: DPLL Specification (cont'd)

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_RAM}) ¹	Units
		0.80V (M)	
		-1	
T _{DESKEWTAPDELAY_DPLL}	Nominal tap-delay of the programmable delay in the PD based deskew scheme	Note 10	

Notes:

1. The DPLLs are powered by the V_{CC_RAM} supply, except for the DPLLs at HDIO banks are powered by the V_{CCINT} supply. See [Table 14: Available Speed Grades and Operating Voltages](#).
2. The maximum input and output clock frequencies are limited by the global clock buffers. See [Table 74: Clock Buffers Switching Characteristics](#).
3. CLKIN jitter also applies to CLKIN_DESKEW and CLKFB_DESKEW in digital compensation. CLKFBIN applies only to analog compensation. This parameter is in regards to the functionality of the DPLL. Input jitter above ~1 MHz is reduced by the filtering properties of the DPLL. The magnitude of the reduction is found in the Vivado timing report.
4. The DPLL does not filter typical spread-spectrum input clocks because they are usually far below the bandwidth filter frequencies.
5. The static offset is measured between any DPLL outputs with identical phase.
6. Values for this parameter are available in the Vivado timing summary as part of the clock uncertainty equation.
7. Includes global clock buffer.
8. The maximum lock time in non-deskew mode is given by the given formula: Lock time in non-deskew mode in ms = 153.6 × DIVCLK_DIVIDE / (CLKIN_FREQUENCY in MHz).
9. The maximum lock time in deskew mode is given by the following formula: Lock time in deskew mode in ms = (0.208 × (VCO_frequency in MHz) / (CLKIN_DESKEW_frequency in MHz)²) + (maximum lock time in non-deskew mode in ms from Note 8).
10. The value for this parameter is included in compensation delay calculations.

XPLL Switching Characteristics

Table 77: XPLL Specification

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_IO})		Units
		0.80V (M)		
		-1		
F _{INMAX_XPLL}	Maximum input clock frequency ¹	984		MHz
F _{INMIN_XPLL}	Minimum input clock frequency	100		MHz
F _{INJITTER_XPLL}	Maximum input clock jitter ²	< 20% of clock input period or 1 ns Max		
F _{INDUTY_XPLL}	Allowable input duty cycle: 100–399 MHz	35–65		%
	Allowable input duty cycle: 400–499 MHz	40–60		%
	Allowable input duty cycle: >500 MHz	45–55		%
F _{PSCLKMAX_XPLL}	Maximum dynamic phase shift clock frequency	300		MHz
F _{PSCLKMIN_XPLL}	Minimum dynamic phase shift clock frequency	0.01		MHz
F _{VCOMAX_XPLL}	Maximum XPLL VCO frequency	4320		MHz
F _{VCOMIN_XPLL}	Minimum XPLL VCO frequency	2160		MHz
F _{BANDWIDTH_XPLL}	XPLL bandwidth at typical ³	14.00		MHz
T _{STATPHAOFFSET_XPLL}	Static phase offset of the XPLL outputs ⁴	0.12		ns
T _{OUTJITTER_XPLL}	XPLL output jitter	Note 5		
T _{OUTDUTY_XPLL}	XPLL CLKOUT0, CLKOUT1, CLKOUT2, CLKOUT3 duty-cycle precision ⁶	0.15		ns
T _{LOCKMAX_XPLL}	XPLL maximum lock time (in non-deskew mode)	100		µs
T _{LOCKDESKEWMAX_XPLL}	XPLL maximum lock time in deskew mode	Note 7		
F _{OUTMAX_XPLL}	XPLL maximum output frequency at CLKOUT0, CLKOUT1, CLKOUT2, CLKOUT3 ¹	984		MHz
	XPLL maximum output frequency at CLKOUTPHY	3733		MHz
F _{OUTMIN_XPLL}	XPLL minimum output frequency at CLKOUT0, CLKOUT1, CLKOUT2, CLKOUT3	16.875		MHz
	XPLL minimum output frequency at CLKOUTPHY	200		MHz
T _{PWRDWNMINPULSE_XPLL}	Minimum power-down pulse width	5.00		ns
F _{PFDMAX_XPLL}	Maximum frequency at the phase frequency detector with bandwidth set to High optimized	667.5		MHz
F _{PFDMIN_XPLL}	Minimum frequency at the phase frequency detector	100		MHz

Table 77: XPLL Specification (cont'd)

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_IO})	Units
		0.80V (M)	
		-1	
T _{DESKEWTAPDELAY_XPLL}	Nominal tap-delay of the programmable delay in the PD based deskew scheme	Note 8	

Notes:

1. The maximum input clock frequency and output clock frequency at CLKOUT0, CLKOUT1, CLKOUT2, and CLKOUT3 are limited by the global clock buffers. See [Table 74: Clock Buffers Switching Characteristics](#).
2. CLKIN jitter also applies to CLKIN_DESKEW and CLKFB_DESKEW in digital compensation. CLKFBIN applies only to analog compensation. This parameter is in regards to the functionality of the XPLL. Input jitter above ~1 MHz is reduced by the filtering properties of the XPLL. The magnitude of the reduction is found in the Vivado timing report.
3. The XPLL does not filter typical spread-spectrum input clocks because they are usually far below the loop filter frequencies.
4. The static offset is measured between any XPLL outputs with identical phase.
5. Values for this parameter are available in the Vivado timing summary as part of the clock uncertainty equation.
6. Includes global clock buffer.
7. The maximum lock time in deskew mode is given by the following formula: Lock time in deskew mode in μs = $(208 \times (\text{VCO_frequency in MHz}) / (\text{CLKIN_DESKEW_frequency in MHz})^2 + 100$.
8. The value for this parameter is included in compensation delay calculations.

Device Pin-to-Pin Output Parameter Guidelines

The pin-to-pin numbers in the following tables are based on the clock root placement in the center of the device. The actual pin-to-pin values will vary if the root placement selected is different. Consult the Vivado Design Suite timing report for the actual pin-to-pin values.

Table 78: Global Clock Input to Output Delay With MMCM (Internal Mode)

Symbol	Description ^{1, 2}	Device	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units
			0.80V (M)	
			-1	
SSTL15 Global Clock Input to Output Delay using Output Flip-Flop, Fast Slew Rate, <i>with</i> MMCM				
T _{ICKOFMMCM}	Global clock input and output flip-flop <i>with</i> MMCM	XQRVE2302	7.11	ns

Notes:

1. This table lists representative values where one global clock input drives one vertical clock line in each accessible column, and where all accessible I/O and CLB flip-flops are clocked by the global clock net.
2. MMCM output jitter is already included in the timing calculation.

Device Pin-to-Pin Input Parameter Guidelines

The pin-to-pin numbers in the following tables are based on the clock root placement in the center of the device. The actual pin-to-pin values will vary if the root placement selected is different. Consult the Vivado Design Suite timing report for the actual pin-to-pin values.

Table 79: Global Clock Input Setup and Hold With MMCM (Internal Mode)

Symbol	Description	Device	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units	
			0.80V (M)		
			-1		
Input Setup and Hold Time Relative to Global Clock Input Signal using SSTL15 Standard. ^{1, 2, 3}					
T _{SUMMCM_XQR_VE2302}	Global clock input and input flip-flop (or latch) with MMCM	Setup	XQRVE2302	-0.48	ns
T _{HMMCM_XQR_VE2302}		Hold		3.31	ns

Notes:

1. Setup and hold times are measured over worst case conditions (process, voltage, temperature). Setup time is measured relative to the global clock input signal using the slowest process, slowest temperature, and slowest voltage. Hold time is measured relative to the global clock input signal using the fastest process, fastest temperature, and fastest voltage.
2. This table lists representative values where one global clock input drives one vertical clock line in each accessible column, and where all accessible I/O and CLB flip-flops are clocked by the global clock net.
3. Use IBIS to determine any duty-cycle distortion incurred using various standards.

Package Parameter Guidelines

The parameters in this section provide the necessary values for calculating timing budgets for clock transmitter and receiver data-valid windows.

Table 80: Package Skew

Symbol	Description	Device	Package	Value	Units
PKGSKEW	Package Skew ^{1, 2}	XQRVE2302	SSRA784	103	ps

Notes:

- 1. These values represent the worst-case skew between any two SelectIO resources in the package: shortest delay to longest delay from die pad to ball.
- 2. Package delay information is available for these device/package combinations. This information can be used to deskew the package.

DDR4 and LPDDR4/4X Memory Interface Controller

The following table provides the maximum data rates for applicable memory standards using the Versal AI Edge XQR device memory PHY. Refer to *Versal Adaptive SoC Programmable Network on Chip and Integrated Memory Controller LogiCORE IP Product Guide* (PG313) for the complete list of memory interface standards supported and detailed specifications. The final performance of the memory interface is determined through a complete design implemented in the Vivado Design Suite, following guidelines in the *Versal Adaptive SoC PCB Design User Guide* (UG863), electrical analysis, and characterization of the system.

Table 81: Maximum Physical Interface (PHY) Rate for Integrated Memory Interface Controller

Memory Standard	DRAM Type	DIMM Slots	XPIO Bank Performance ¹	Performance as a Function of Speed Grade and Operating Voltage (V _{CC_SOC})	Units
				0.80V (M)	
				-1	
DDR4	Single rank component ²		All	3200	Mb/s
	1 rank DIMM ³ , LRDIMM ⁴	1	All	3200	Mb/s
	2 rank DIMM ³	1	All	2933	Mb/s
	1 rank RDIMM, 2 rank LRDIMM	2	All	2667	Mb/s
	2 rank RDIMM	2	All	2133	Mb/s
LPDDR4 LPDDR4X ⁵	Single rank component		High	3733	Mb/s
			Medium	3733	Mb/s
			Low	3200	Mb/s
	Dual rank component		High	3733	Mb/s
			Medium	3733	Mb/s
			Low	2933	Mb/s

Notes:

1. The Versal device/package pinout files specify XPIO bank performance (XPIOPerf). See the *ASCII Package File* information in this data sheet.
2. For DDR4 DDP deep components, the maximum data rate is 2933 Mb/s for five or less DDP devices across all speed grades and temperature grades. For six or more DDP deep devices, the maximum data rate is 2133 Mb/s across all speed grades and temperature grades. For DDR4 DDP wide components, use single rank component data rates.
3. Dual in-line memory module (DIMM) includes RDIMM, SODIMM, and UDIMM.
4. Includes 2 and 4 rank LRDIMM.
5. For LPDDR4/4X, use the Vivado tool to determine the maximum performance by package and I/O bank combination.

AI Engine Switching Characteristics

The following tables provide performance characteristics for the AI Engines.

Table 82: AI Engine Switching Characteristics

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V_{CCINT})	Units
		0.80V (M)	
		-1	
F_{MAX}	AI Engine array clock maximum frequency	1150	MHz
$F_{MAX_AI_PL}$	AI Engine array to programmable logic interface	575	MHz

Table 83: AI Engine – PL Interface Performance

Connection Type	Total Bandwidth ^{2, 3} as a Function of Speed Grade and Operating Voltage (V_{CCINT})	Units
	0.80V (M)	
	-1	
PL to AI Engine array interface ¹	441.6	GB/s
AI Engine array interface to AI Engine array ⁴	368.0	GB/s
AI Engine array to AI Engine array interface ⁴	257.6	GB/s
AI Engine array interface to PL	331.2	GB/s

Notes:

1. Not all of the AI Engine interface tiles are connected to the PL.
2. Assumes all PL AXI4-Stream interfaces run at maximum frequency.
3. AI Engine array interface tiles have interfaces to both the PL and NoC. Routing to these interfaces share the same streaming interconnect resources inside the AI Engine array.
4. Total bandwidth includes horizontal interface usage.

Note: For further information on the AI Engine array interface to the PL interface, see the *AI Engine Array Interface Topology* figure and *AI Engine to PL Interface Bandwidth Performance* table in the *Versal Adaptive SoC AI Engine Architecture Manual* (AM009).

GTYP Transceiver Specifications

GTYP Transceiver DC Input and Output Levels

The following table summarizes the DC specifications of the GTYP transceivers in Versal AI Edge XQR devices. Consult the *Versal Adaptive SoC GTY and GTYP Transceivers Architecture Manual* ([AM002](#)) for further details.

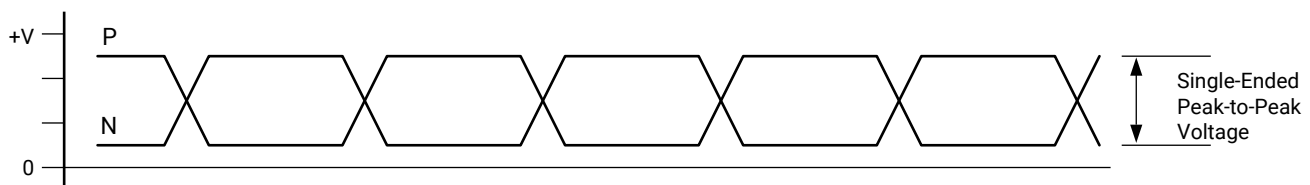
Table 84: GTYP Transceiver DC Specifications

Symbol	DC Parameter	Conditions	Min	Typ	Max	Units
DV _{PPIN}	Differential peak-to-peak input voltage (external AC coupled)	>10.3125 Gb/s	150	–	1250	mV
		6.6 Gb/s to 10.3125 Gb/s	150	–	1250	mV
		≤ 6.6 Gb/s	150	–	2000	mV
V _{IN}	Single-ended input voltage. Voltage measured at the pin referenced to GND.	DC coupled V _{GTY_AVTT} = 1.2V	–200	–	V _{GTY_AVTT}	mV
V _{CMIN}	Common mode input voltage	DC coupled V _{GTY_AVTT} = 1.2V	–	2/3 V _{GTY_AVTT}	–	mV
DV _{PPOUT}	Differential peak-to-peak output voltage ¹	Transmitter output swing is set to 11111	800	–	–	mV
V _{CMOUTDC}	Common mode output voltage: DC coupled (equation based) ²	When remote RX termination is floating	D _{VPPOUT} /2			mV
		When remote RX is terminated to V _{RX_TERM} ³	V _{RX_TERM} /2 + D _{VPPOUT} /4			mV
V _{CMOUTAC}	Common mode output voltage: AC coupled	Equation based	D _{VPPOUT} /2			mV
R _{IN}	Differential input resistance		–	100	–	Ω
R _{OUT}	Differential output resistance		–	100	–	Ω
T _{OSKEW}	Transmitter output pair (TXP and TXN) intra-pair skew		–	–	10	ps
C _{EXT}	Recommended external AC coupling capacitor ⁴		–	100	–	nF

Notes:

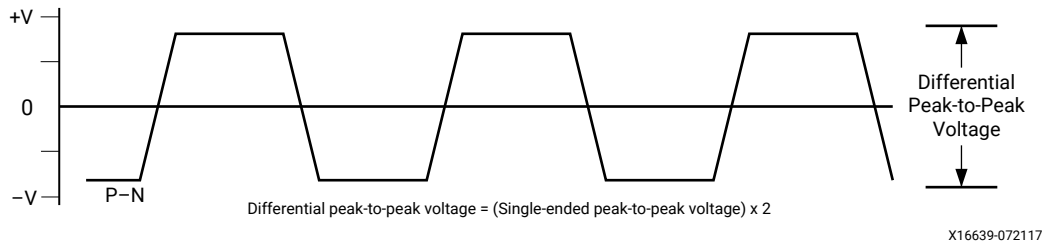
- The output swing and pre-emphasis levels are programmable using the GTYP transceiver attributes discussed in the *Versal Adaptive SoC GTY and GTYP Transceivers Architecture Manual* ([AM002](#)) and can result in values lower than reported in this table.
- Remote RX termination = GND is not supported.
- V_{RX_TERM} is the remote RX termination voltage. V_{CMOUTDC_VRXTERM} should be less than 0.92V.
- Other values can be used as appropriate to conform to specific protocols and standards.

Figure 4: Single-Ended Peak-to-Peak Voltage



X16653-072117

Figure 5: Differential Peak-to-Peak Voltage



The following tables summarize the DC specifications of the clock input/output levels of the GTYP transceivers in Versal AI Edge XQR devices. Consult the *Versal Adaptive SoC GTY and GTYP Transceivers Architecture Manual* ([AM002](#)) for further details.

Table 85: GTYP Transceiver Clock Input Level Specification

Symbol	DC Parameter	Min	Typ	Max	Units
V _{IDIFF}	Differential peak-to-peak input voltage	250	–	1600	mV
V _{ICM}	REFCLK common-mode input voltage	–	V _{GTYP_AVCC}	–	mV
R _{IN}	Differential input resistance	–	100	–	Ω
C _{EXT}	Required external AC coupling capacitor	–	10	–	nF

Table 86: GTYP Transceiver Clock Output Level Specification

Symbol	Description	Conditions	Min	Typ	Max	Units
V _{OL}	Output Low voltage for P and N	R _T = 100Ω across P and N signals	100	–	330	mV
V _{OH}	Output High voltage for P and N	R _T = 100Ω across P and N signals	500	–	700	mV
V _{DDOUT}	Differential output voltage (P–N), P = High (N–P), N = High	R _T = 100Ω across P and N signals	300	–	430	mV
V _{CMOUT}	Common mode voltage	R _T = 100Ω across P and N signals	300	–	500	mV

GTYP Transceiver Switching Characteristics

Consult the *Versal Adaptive SoC GTY and GTYP Transceivers Architecture Manual* ([AM002](#)) for further information.

GTYP Transceiver Performance

Table 87: GTYP Transceiver Performance

Symbol	Description	Subrate Divider	Output Divider	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units
				0.80V (M)	
				-1	
F _{GTYPMAX}	GTYP maximum line rate			26.5625	Gb/s
F _{GTYPMIN}	GTYP minimum line rate			1.2	Gb/s

Table 87: GTYP Transceiver Performance (cont'd)

Symbol	Description	Subrate Divider	Output Divider	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})		Units
				0.80V (M)		
				-1		
				Min	Max	
F _{GTYP} RRANGE	RINGPLL line rate range ^{1,2}		1	8	12.5	Gb/s
			2	4	8	Gb/s
			4	2	4	Gb/s
			8	1.2	2	Gb/s
				Min	Max	
F _{GTYP} PLRANGE	LCPLL line rate range ³	1	1	16	26.5625	Gb/s
			2	8	16.375	Gb/s
		2	1	8	16.375	Gb/s
			2	4	8.1875	Gb/s
		4	1	4	8.1875	Gb/s
			2	2	4.09375	Gb/s
		8	1	2	4.09375	Gb/s
			2	1.2	2.046875	Gb/s
				Min	Max	
F _{RPLL} RANGE	RINGPLL frequency range			4	8	GHz
F _{LCPLL} RANGE	LCPLL frequency range			8	16.375	GHz

Notes:

1. The values listed are the rounded results of the calculated equation $(2 \times \text{RINGPLL_Frequency}) / \text{Subrate_Divider}$.
2. The highest supported line rate when using the RPLL to drive the datapath is 12.5 Gb/s.
3. The values listed are the rounded results of the calculated equation $(2 \times \text{LCPLL_Frequency}) / (\text{Output_Divider} \times \text{Subrate_Divider})$.

GTYP Transceiver Configuration Interface Port Switching Characteristics

Table 88: GTYP Transceiver Configuration Interface Port (APB3) Switching Characteristics

Symbol	Description	All Speed Grades		Units
		Min	Max	
F _{GTYP} APB3	GTYPAPB3CLK frequency	100	300	MHz

GTYP Transceiver Reference Clock Switching Characteristics

Table 89: GTYP Transceiver Reference Clock Switching Characteristics

Symbol	Description	Conditions	All Speed Grades			Units
			Min	Typ	Max	
$F_{GT_REFCLK}^1$	Reference clock frequency range		60	–	820	MHz
T_{RCLK}	Reference clock rise time	20% – 80%	–	200	–	ps
T_{FCLK}	Reference clock fall time	80% – 20%	–	200	–	ps
T_{DCREF}	Reference clock duty cycle	Transceiver PLL only	40	50	60	%

Notes:

1. The frequency values apply when the reference clock buffer is configured as either an input or output.

GTYP Transceiver Reference Clock Oscillator Selection Phase Noise Mask

Table 90: GTYP Transceiver Reference Clock Oscillator Selection Phase Noise Mask

Symbol	Description ^{1, 2}	Offset Frequency	Min	Typ	Max	Units
LCPLL _{REFCLK} MASK	LCPLL reference clock select phase noise mask at GTY_REFCLK frequency = 156.25 MHz	10 kHz	–	–	–112	dBc/Hz
		100 kHz	–	–	–128	
		1 MHz	–	–	–145	
	LCPLL reference clock select phase noise mask at GTY_REFCLK frequency = 312.5 MHz	10 kHz	–	–	–103	dBc/Hz
		100 kHz	–	–	–123	
		1 MHz	–	–	–143	
	LCPLL reference clock select phase noise mask at GTY_REFCLK frequency = 625 MHz	10 kHz	–	–	–98	dBc/Hz
		100 kHz	–	–	–117	
		1 MHz	–	–	–140	
RINGPLL _{REFCLK} MASK	RINGPLL reference clock select phase noise mask at GTY_REFCLK frequency = 156.25 MHz	10 kHz	–	–	–112	dBc/Hz
		100 kHz	–	–	–128	
		1 MHz	–	–	–145	
		50 MHz	–	–	–145	
	RINGPLL reference clock select phase noise mask at GTY_REFCLK frequency = 312.5 MHz	10 kHz	–	–	–103	dBc/Hz
		100 kHz	–	–	–123	
		1 MHz	–	–	–143	
		50 MHz	–	–	–145	
	RINGPLL reference clock select phase noise mask at GTY_REFCLK frequency = 625 MHz	10 kHz	–	–	–98	dBc/Hz
		100 kHz	–	–	–117	
		1 MHz	–	–	–140	
		50 MHz	–	–	–144	

Notes:

1. For reference clock frequencies not in this table, use the phase-noise mask for the nearest reference clock frequency.
2. This reference clock phase-noise mask is superseded by any reference clock phase-noise mask that is specified in a supported protocol, e.g., PCIe.

GTYP Transceiver PLL/Lock Time Adaptation

Table 91: GTYP Transceiver PLL/Lock Time Adaptation

Symbol	Description	Conditions	All Speed Grades			Units
			Min	Typ	Max	
T _{LOCK}	Initial PLL lock	Reference clock frequency ≥ 150 MHz	–	–	3	ms
		Reference clock frequency < 150 MHz	–	–	5.7	ms
T _{DLOCK}	Clock recovery phase acquisition and adaptation time for decision feedback equalizer (DFE)	After the PLL is locked to the reference clock, this is the time it takes to lock the clock data recovery (CDR) to the data present at the input.	–	50,000	37 x 10 ⁶	UI
	Clock recovery phase acquisition and adaptation time for low-power mode (LPM) when the DFE is disabled		–	50,000	2.3 x 10 ⁶	UI

GTYP Transceiver Digital Monitor Clock

Table 92: GTYP Transceiver Digital Monitor Port Switching Characteristics

Symbol	Description	All Speed Grades		Units
		Min	Max	
F _{GTYDMONITORCLK_INPUT}	GTYP transceiver digital monitor port input maximum frequency	100	500	MHz
F _{GTYDMONITORCLK_OUTPUT}	GTYP transceiver digital monitor port output maximum frequency	100	500	MHz

GTYP Transceiver User Clock Switching Characteristics

Table 93: GTYP Transceiver User Clock Switching Characteristics

Symbol	Description ¹	Data Width Conditions (Bit)		Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units
				0.80V (M)	
		Internal Logic	Interconnect Logic	-1	
F _{TXOUTPMA}	TXOUTCLK maximum frequency sourced from OUTCLKPMA			664.063	MHz
F _{RXOUTPMA}	RXOUTCLK maximum frequency sourced from OUTCLKPMA			664.063	MHz
F _{TXOUTPROGDIV}	TXOUTCLK maximum frequency sourced from OUTPROGDIV			1000	MHz
F _{RXOUTPROGDIV}	RXOUTCLK maximum frequency sourced from OUTPROGDIV			1000	MHz
F _{TXIN}	TXUSRCLK maximum frequency	16	16	415.039	MHz
			32	207.520	MHz
		32	32	415.039	MHz
			64	207.520	MHz
		64	64	415.039	MHz
			128	207.520	MHz
		20	20	332.032	MHz
			40	166.016	MHz
		40	40	664.063	MHz
			80	166.016	MHz
		80	80	332.031	MHz
			160	166.016	MHz
F _{RXIN}	RXUSRCLK maximum frequency	16	16	415.039	MHz
			32	207.520	MHz
		32	32	415.039	MHz
			64	207.520	MHz
		64	64	415.039	MHz
			128	207.520	MHz
		20	20	332.032	MHz
			40	166.016	MHz
		40	40	664.063	MHz
			80	166.016	MHz
		80	80	332.031	MHz
			160	166.016	MHz

Notes:

1. Clocking must be implemented as described in the *Versal Adaptive SoC GTY and GTYP Transceivers Architecture Manual* (AM002).

GTYP Transceiver Transmitter and Receiver Switching Characteristics

Table 94: GTYP Transceiver Transmitter Switching Characteristics

Symbol	Description	Condition	Min	Typ	Max	Units
F _{GTYTX}	Serial data rate range		1.2	–	F _{GTYMAX}	Gb/s
T _{RTX}	TX rise time	20%–80%	–	21	–	ps
T _{FTX}	TX fall time	80%–20%	–	21	–	ps
T _{LLSKEW}	TX lane-to-lane skew ¹		–	–	500.00	ps
T _{J26.5625}	Total jitter ^{2, 4}	26.5625	–	–	0.28	UI
D _{J26.5625}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J16.375}	Total jitter ^{2, 4}	16.375 Gb/s	–	–	0.28	UI
D _{J16.375}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J15.0}	Total jitter ^{2, 4}	15.0 Gb/s	–	–	0.28	UI
D _{J15.0}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J14.1}	Total jitter ^{2, 4}	14.1 Gb/s	–	–	0.28	UI
D _{J14.1}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J14.1}	Total jitter ^{2, 4}	14.025 Gb/s	–	–	0.28	UI
D _{J14.1}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J13.1}	Total jitter ^{2, 4}	13.1 Gb/s	–	–	0.28	UI
D _{J13.1}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J12.5_LCPLL}	Total jitter ^{2, 4}	12.5 Gb/s	–	–	0.28	UI
D _{J12.5_LCPLL}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J12.5_RPLL}	Total jitter ^{3, 4}	12.5 Gb/s	–	–	0.33	UI
D _{J12.5_RPLL}	Deterministic jitter ^{3, 4}		–	–	0.17	UI
T _{J11.3_LCPLL}	Total jitter ^{2, 4}	11.3 Gb/s	–	–	0.28	UI
D _{J11.3_LCPLL}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J10.3125_LCPLL}	Total jitter ^{2, 4}	10.3125 Gb/s	–	–	0.28	UI
D _{J10.3125_LCPLL}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J10.3125_RPLL}	Total jitter ^{3, 4}	10.3125 Gb/s	–	–	0.33	UI
D _{J10.3125_RPLL}	Deterministic jitter ^{3, 4}		–	–	0.17	UI
T _{J9.953_LCPLL}	Total jitter ^{2, 4}	9.953 Gb/s	–	–	0.28	UI
D _{J9.953_LCPLL}	Deterministic jitter ^{2, 4}		–	–	0.17	UI
T _{J9.953_RPLL}	Total jitter ^{3, 4}	9.953 Gb/s	–	–	0.33	UI
D _{J9.953_RPLL}	Deterministic jitter ^{3, 4}		–	–	0.17	UI
T _{J8.0}	Total jitter ^{3, 4}	8.0 Gb/s	–	–	0.32	UI
D _{J8.0}	Deterministic jitter ^{3, 4}		–	–	0.17	UI
T _{J6.6}	Total jitter ^{3, 4}	6.6 Gb/s	–	–	0.30	UI
D _{J6.6}	Deterministic jitter ^{3, 4}		–	–	0.15	UI
T _{J5.0}	Total jitter ^{3, 4}	5.0 Gb/s	–	–	0.30	UI
D _{J5.0}	Deterministic jitter ^{3, 4}		–	–	0.15	UI
T _{J4.25}	Total jitter ^{3, 4}	4.25 Gb/s	–	–	0.30	UI
D _{J4.25}	Deterministic jitter ^{3, 4}		–	–	0.15	UI
T _{J3.20}	Total jitter ^{3, 4}	3.20 Gb/s	–	–	0.20	UI
D _{J3.20}	Deterministic jitter ^{3, 4}		–	–	0.10	UI

Table 94: GTYP Transceiver Transmitter Switching Characteristics (cont'd)

Symbol	Description	Condition	Min	Typ	Max	Units
T _{J2.5}	Total jitter ^{3, 4}	2.5 Gb/s	–	–	0.20	UI
D _{J2.5}	Deterministic jitter ^{3, 4}		–	–	0.10	UI
T _{J1.25}	Total jitter ^{3, 4}	1.25 Gb/s	–	–	0.15	UI
D _{J1.25}	Deterministic jitter ^{3, 4}		–	–	0.06	UI

Notes:

- Using same REFCLK input with TX phase alignment enabled for up to four consecutive transmitters (one fully populated GTYP Quad) at maximum line rate.
- Using LCPLL_FBDIV = 40, 20-bit internal data width. These values are NOT intended for protocol specific compliance determinations.
- Using RPLL_FBDIV = 10, 20-bit internal data width. These values are NOT intended for protocol specific compliance determinations.
- All jitter values are based on a bit-error ratio of 10⁻¹².

Table 95: GTYP Transceiver Receiver Switching Characteristics

Symbol	Description	Condition	Min	Typ	Max	Units
F _{GTYRX}	Serial data rate		1.2	–	F _{GTYMAX}	Gb/s
R _{XSST}	Receiver spread-spectrum tracking	Modulated at 33 kHz	–5000	–	0	ppm
R _{XRL}	Run length (CID)		–	–	256	UI
R _{XPMTOL}	Data/REFCLK PPM offset tolerance	Bit rates ≤ 6.6 Gb/s	–1250	–	1250	ppm
		Bit rates > 6.6 Gb/s and ≤ 8.0 Gb/s	–700	–	700	ppm
		Bit rates > 8.0 Gb/s and ≤ 16.3 Gb/s	–600	–	600	ppm
		Bit rates > 16.3 Gb/s and ≤ 24.0 Gb/s	–350	–	350	ppm
		Bit rates > 24.0 Gb/s	–200	–	200	ppm
SJ Jitter Tolerance ¹						
J _{T_SJ26.5625}	Sinusoidal jitter (LCPLL) ²	26.5625 Gb/s	0.30	–	–	UI
J _{T_SJ16.375}	Sinusoidal jitter (LCPLL) ²	16.375 Gb/s	0.30	–	–	UI
J _{T_SJ15.0}	Sinusoidal jitter (LCPLL) ²	15.0 Gb/s	0.30	–	–	UI
J _{T_SJ14.1}	Sinusoidal jitter (LCPLL) ²	14.1 Gb/s	0.30	–	–	UI
J _{T_SJ13.1}	Sinusoidal jitter (LCPLL) ²	13.1 Gb/s	0.30	–	–	UI
J _{T_SJ12.5}	Sinusoidal jitter (LCPLL) ²	12.5 Gb/s	0.30	–	–	UI
J _{T_SJ11.3}	Sinusoidal jitter (LCPLL) ²	11.3 Gb/s	0.30	–	–	UI
J _{T_SJ10.32_LCPLL}	Sinusoidal jitter (LCPLL) ²	10.32 Gb/s	0.30	–	–	UI
J _{T_SJ10.32_RPLL}	Sinusoidal jitter (RPLL) ²	10.32 Gb/s	0.30	–	–	UI
J _{T_SJ9.953_LCPLL}	Sinusoidal jitter (LCPLL) ²	9.953 Gb/s	0.30	–	–	UI
J _{T_SJ9.953_RPLL}	Sinusoidal jitter (RPLL) ²	9.953 Gb/s	0.30	–	–	UI
J _{T_SJ8.0}	Sinusoidal jitter (RPLL) ²	8.0 Gb/s	0.42	–	–	UI
J _{T_SJ6.6}	Sinusoidal jitter (RPLL) ²	6.6 Gb/s	0.42	–	–	UI
J _{T_SJ5.0}	Sinusoidal jitter (RPLL) ²	5.0 Gb/s	0.42	–	–	UI
J _{T_SJ4.25}	Sinusoidal jitter (RPLL) ²	4.25 Gb/s	0.44	–	–	UI
J _{T_SJ3.2}	Sinusoidal jitter (RPLL) ²	3.2 Gb/s	0.45	–	–	UI
J _{T_SJ2.5}	Sinusoidal jitter (RPLL) ²	2.5 Gb/s	0.30	–	–	UI

Table 95: GTYP Transceiver Receiver Switching Characteristics (cont'd)

Symbol	Description	Condition	Min	Typ	Max	Units
J _{T_SJ1.25}	Sinusoidal jitter (RPLL) ²	1.25 Gb/s	0.30	–	–	UI

Notes:

1. All jitter values are based on a bit error ratio of 10⁻¹².
2. The frequency of the injected sinusoidal jitter is 10 MHz.

GTYP Transceiver Electrical Compliance

The *Versal Adaptive SoC GTY and GTYP Transceivers Architecture Manual* ([AM002](#)) contains recommended use modes that ensure compliance for the protocols listed in the following table. The transceiver wizard provides the recommended settings for those use cases and for protocol specific characteristics.

Table 96: GTYP Transceiver Protocol List

Protocol	Specification	Serial Rate (Gb/s)	Electrical Compliance
OTU4 (OTL4.4) CFP	OIF-CEI-11G-MR	11.18–13.1	Compliant
CAUI-10	IEEE 802.3-2012	10.3125	Compliant
nPPI	IEEE 802.3-2012	10.3125	Compliant
10GBASE-KR ²	IEEE 802.3-2012	10.3125	Compliant
SFP+	SFF-8431 (SR and LR)	9.95328–11.10	Compliant
XFP	INF-8077i, revision 4.5	10.3125	Compliant
RXAUI	CEI-6G-SR	6.25	Compliant
XAUI	IEEE 802.3-2012	3.125	Compliant
1000BASE-X	IEEE 802.3-2012	1.25	Compliant
5.0G Ethernet	IEEE 802.3bx (PAR)	5	Compliant
2.5G Ethernet	IEEE 802.3bx (PAR)	2.5	Compliant
HiGig, HiGig+, HiGig2	IEEE 802.3-2012	3.74, 6.6	Compliant
QSGMII	QSGMII v1.2 (Cisco System, ENG-46158)	5	Compliant
OTU2	ITU G.8251	10.709225	Compliant
OTU4 (OTL4.10)	OIF-CEI-11G-SR	11.180997	Compliant
OC-3/12/48/192	GR-253-CORE	0.1555–9.956	Compliant
PCIe Gen1, 2, 3, 4, 5	PCI Express® 5.0 ³	2.5, 5.0, 8.0, 16.0, and 32.0	Compliant
SDI ⁴	SMPTE 424M-2006	0.27–2.97	Compliant
UHD-SDI ⁴	SMPTE ST-2081 6G, SMPTE ST-2082 12G	6 and 12	Compliant
Hybrid memory cube (HMC)	HMC-15G-SR	10, 12.5, and 15.0	Compliant
MoSys bandwidth engine	CEI-11-SR and CEI-11-SR (overclocked)	10.3125, 15.5	Compliant
CPRI	CPRI_v_6_1_2014-07-01	0.6144–12.165	Compliant
Passive optical network (PON)	10G-EPON, 1G-EPON, NG-PON2, XG-PON, and 2.5G-PON	0.155–10.3125	Compliant
JESD204a/b	OIF-CEI-6G, OIF-CEI-11G	3.125–12.5	Compliant
Serial RapidIO	RapidIO specification 3.1	1.25–10.3125	Compliant
DisplayPort	DP 1.2B CTS	1.62–5.4	Compliant
Fibre channel	FC-PH-4	1.0625–14.025	Compliant
SATA Gen1, 2, 3	Serial ATA revision 3.0 specification	1.5, 3.0, and 6.0	Compliant

Table 96: GTYP Transceiver Protocol List (cont'd)

Protocol	Specification	Serial Rate (Gb/s)	Electrical Compliance
SAS Gen1, 2, 3	T10/BSR INCITS 519	3.0, 6.0, and 12.0	Compliant
SFI-5	OIF-SFI5-01.0	0.625 - 12.5	Compliant
Aurora	CEI-6G, CEI-11G-LR	All rates	Compliant

Notes:

- 25 dB loss at Nyquist without FEC.
- The transition time of the transmitter is faster than the IEEE Std 802.3-2012 specification.
- For GTYP transceivers, support is up to v5.0.
- This protocol requires external circuitry to achieve compliance.

Integrated Block for MRMAC

The MRMAC in Versal AI Edge XQR devices support 1x40GE, 4x25GE, 4x10GE, and 4xFC32 data rates. More information and documentation on solutions using the multirate Ethernet MAC (MRMAC) can be found at *Versal Devices Integrated 100G Multirate Ethernet MAC (MRMAC) LogiCORE IP Product Guide* ([PG314](#)).

Table 97: Maximum Performance for MRMAC Designs

Symbol	Description	Performance as a Function of Speed Grade and Operating Voltage (V _{CCINT})	Units
		0.80V (M)	
		-1	
F _{RX_CORE_CLK}	Receive core clock	664.063	MHz
F _{TX_CORE_CLK}	Transmit core clock	664.063	MHz
F _{RX_AXIS_CLK}	Receive AXI4-Stream interface clock	420.000	MHz
F _{TX_AXIS_CLK}	Transmit AXI4-Stream interface clock	420.000	MHz
F _{RX_SERDES_CLK}	Serializer/deserializer clock	664.063	MHz
F _{RX_ALT_SERDES_CLK} ¹	Receive alternate serializer/deserializer clock	332.031	MHz
F _{TX_ALT_SERDES_CLK} ¹	Transmit alternate serializer/deserializer clock	332.031	MHz
F _{RX_TS_CLK}	Receive timestamp clock	350.000	MHz
F _{TX_TS_CLK}	Transmit timestamp clock	350.000	MHz
F _{RX_FLEXIF_CLK}	Receive flex interface clock	420.000	MHz
F _{TX_FLEXIF_CLK}	Transmit flex interface clock	420.000	MHz
F _{APB3_CLK}	AMBA® advance peripheral bus (APB3) clock	300.000	MHz

Notes:

- The ALT_SERDES_CLKs run at half the speed of the primary SERDES_CLK. See *Versal Devices Integrated 100G Multirate Ethernet MAC (MRMAC) LogiCORE IP Product Guide* ([PG314](#)) for more information.

Programmable Logic Integrated Block for PCIe

More information and documentation on solutions for PCI Express designs can be found at [PCI Express](#).

Table 98: Maximum Performance for Programmable Logic Integrated Block for PCIe Rev. 4.0

Symbol	Description ^{1, 2}	Performance as a Function of Speed Grade and Operating Voltage (V_{CCINT})	Units
		0.80V (M)	
		-1	
$F_{PIPECLK}$	Pipe clock maximum frequency	500	MHz
$F_{CORECLK}$	Core clock maximum frequency	500	MHz
F_{APBCLK}	APB clock maximum frequency	250	MHz

Notes:

1. This table only specifies the AC switching characteristics of the identified integrated block for PCIe. AMD LogiCORE™ IP solutions for PCIe that incorporate this block also integrate clocking, transceivers, logic, and block memory. LogiCORE IP solutions for PCIe must achieve timing closure during design implementation in the target device, including user-contributed application logic. For information and technical guidance on resource use and minimum device requirements, see the *Versal Adaptive SoC Integrated Block for PCI Express LogiCORE IP Product Guide* ([PG343](#)) and *Versal Adaptive SoC DMA and Bridge Subsystem for PCI Express Product Guide* ([PG344](#)).
2. PCI Express Gen4 operation is supported for x1, x2, x4, and x8 widths.

SSRA784 Package for the XQRVE2302 Device Specifications

Table 99: Package Specifications

BGA Packages	Description	Package Specifications		
		Pitch (mm)	Size (mm)	LSC Ball Grid Size (balls)
SSRA784	Ruggedized fine pitch lidless with stiffener ring	0.8	23 × 23	–

The following table lists the number of available PMC and LPD multiplexed I/Os (MIO), 3.3V-capable high-density (HD), and 1.5V-capable high-performance (XP) I/Os and the number of HD or XP differential I/O in the XQRVE2302.

Table 100: XQRVE2302-SSRA784 I/O Pins

Device	Package	MIO	Total User I/O		Differential I/O	
			XPIO	HDIO	XPIO	HDIO
XQRVE2302	SSRA784	78	216	22	108	11

Pin Definitions

The following tables list the pin definitions.

SelectIO Pin Definitions

Table 101: SelectIO Pin Definitions

Pin Name	Direction	Description
IO	Bidirectional	Pin name consists of: L[1 to 24][P or N]: Differential pair number with P (positive) and N (negative) N[0 to 8]P[0 to 5]: XPHY nibble number and pin number within the nibble (NIBBLESlice number) M[0 to number of triplets]P[0 to 161]: Triplet number and pin within the triplet [bank number]: Bank number The XP IOB block provides resources to enable high-speed interfaces between the programmable logic (PL) and the system outside the device. The XP IOB resources are designed to accommodate the signaling needs for high-speed memory and chip-to-chip interfaces that are powered between 1.0V and 1.5V. The XP IOB provides internal termination, internal reference generation, support for a diverse set of I/O standards, driver emphasis, and receiver equalization. These features allow the XP IOB to integrate with a diverse range of systems. High-density (HD) I/O banks are resources designed to support I/O standards with voltages ranging from 1.8V to 3.3V. HD I/Os support single ended and pseudo-differential I/O bidirectional signaling operating at data rates of up to 400 Mb/s. Limited support for true differential inputs (with external termination) is also available to support LVDS and LVPECL clock inputs. HD I/O banks contain interface logic (HD IOL) that includes registers, a DPLL, and static delay lines to support asynchronous, system synchronous, and clock-based source synchronous interfaces.
GC/HDGC	Bidirectional	Global clock (GC) inputs provide dedicated, high-speed access to the internal global and regional clock resources. GC inputs use dedicated routing and must be used for clock inputs where the timing of various clocking features is imperative. General-purpose I/O with local interconnects must not be used for clock signals.
IO_VR	N/A	This pin is for the DCI voltage reference resistor, one per bank row: IO_VR_700 and IO_VR_800. Tie to VCCO_700 and VCCO_800, respectively, with a reference resistor.

Power Pin Definitions

Table 102: Power Pin Definitions

Pin Name	Direction	Description
VCCINT	N/A	Power-supply pins for the programmable logic, AI Engines, and CPM
VCCINT_SENSE	Output	Device VCCINT rail sense pin
VCCAUX	N/A	Power-supply pins for the auxiliary circuits
VCC_IO	N/A	Power-supply pins for the HDIO and XPIO banks
VCCO (XPIO or HDIO)	N/A	Power-supply pins for the output drivers (per bank)
VCCO_500	N/A	PMC MIO bank 0 power supply
VCCO_501	N/A	PMC MIO bank 1 power supply
VCCO_502	N/A	LPD MIO bank power supply

Table 102: Power Pin Definitions (cont'd)

Pin Name	Direction	Description
VCCO_503	N/A	PMC dedicated I/O bank power supply
VCCAUX_PMC	N/A	Auxiliary power-supply pin for the PMC
VCC_PSFP	N/A	Power-supply for the PS full-power domain
VCC_PSLP	N/A	Power-supply for the PS low-power domain
VCC_PMC	N/A	Auxiliary power-supply pins for the PMC
VCC_SOC	N/A	Power supply pins for NoC, NPI, and the DDR
VCC_RAM	N/A	Power-supply pins for the RAM, UltraRAM, and clocking network
VCC_BATT	N/A	Battery supply voltage
VCC_FUSE	N/A	Power-supply pins for eFUSE programming
VCCAUX_SMON	N/A	Analog supply pin for the ADC and other analog circuits in the SYSMON
GND_SMON	N/A	Ground reference pin for the ADC and other analog circuits in the SYSMON
GND	N/A	Ground
GND_SENSE	Output	Device ground sense pin
RSVD	N/A	Reserved pin. Leave floating. Note: In footprint compatible devices, this pin can be labeled differently and serve different purposes. When planning migration between devices, include the functionality between all footprint compatible devices.
RSVD_VCC	N/A	Reserved pin. Leave floating. Note: In footprint compatible devices, this pin can be labeled differently and serve different purposes. When planning migration between devices, include the functionality between all footprint compatible devices.
RSVDGND	N/A	Reserved pin. Tie to GND. Note: In footprint compatible devices, this pin can be labeled differently and serve different purposes. When planning migration between devices, include the functionality between all footprint compatible devices.
NC	N/A	Not connected. Can be left floating or tied to ground. Note: In footprint compatible devices, this pin can be labeled differently and serve different purposes. When planning migration between devices, include the functionality between all footprint compatible devices.

PMCMIO, LPDMIO, PMCDIO Pin Definitions

Table 103: PMCMIO, LPDMIO, PMCDIO Pin Definitions

Pin Name	Direction	Description
PMC_MIO[0 to 51]	Bidirectional	Platform management controller (PMC) multiplexed I/O (MIO)
LPD_MIO[0 to 25]	Bidirectional	Low-power domain (LPD) MIO
MODE[0 to 3]	Input	The MIO MODE selection pins are used to select the boot mode of the device. The value of these pins is captured on the rising edge of POR_B.
ERROR_OUT	Output	The ERROR_OUT pin is open drain with a weak internal pull-up. An external pull-up is recommended. ERROR_OUT is 3-stated and pulled High when an error occurs in the device.
RTC_PADO	Output	Real-time clock (RTC) crystal output

Table 103: PMCMIO, LPDMIO, PMCDIO Pin Definitions (cont'd)

Pin Name	Direction	Description
RTC_PADI	Input	RTC crystal input
PUDC_B	Input	The pull-up during configuration (PUDC_B) pin is used to select the behavior of the XPIO or HDIO during configuration. If the PUDC_B pin is High, then the XPIO and HDIO are put into tristate mode. If the PUDC_B is Low, then internal pull-ups at each XPIO and HDIO are enabled. The PUDC_B pin does not affect the PS or PMC I/O during boot and configuration.
DONE	Bidirectional	The DONE pin is open drain with a weak internal pull-up resistor. An external pull-up is recommended. DONE is tristated and pulled High when the boot sequence is complete.
REF_CLK	Input/Output	The reference clock is required for all boot modes. The REF_CLK is also the input clock to the PMC PLL and is required for the slave boot interface software register access.
POR_B	Input	Active-Low POR_B pin is the global power-on reset for the Versal device. It must remain asserted Low until power is fully applied to at least the VCC_PMC, VCCAUX_PMC, and VCCO_503. When the reset is released, the PMC begins the initialization and boot process.
TCK	Input	JTAG test clock
TDI	Input	JTAG test data input
TDO	Output	JTAG test data output
TMS	Input	JTAG test mode select

GTYP Transceiver Quad Pin Definitions

Table 104: GTYP Transceiver Quad Pin Definitions

Pin Name	Direction	Description
GTYP_RXP, GTYP_RXN	Input	RXP and RXN are the differential input pairs for each of the receivers in the transceiver Quad
GTYP_TXP, GTYP_TXN	Output	TXP and TXN are the differential output pairs for each of the transmitters in the transceiver Quad
GTYP_REFCLKP, GTYP_REFCLKN	Input	Configured as either reference clock input pins or as RX recovered clock output pins for the transceiver Quad
GTYP_AVTTRCAL	N/A	Bias current supply for the termination resistor calibration circuit
GTYP_RREF	N/A	Calibration resistor input pin for the termination resistor calibration circuit
GTYP_AVCC	N/A	The analog supply for the internal analog circuits of the transceiver Quad tiles. This includes the analog circuits for the PLLs, transmitters, and receivers. Most packages have multiple groups of power supply connections in the package for the AVCC. Refer to the package pin definitions to identify in which power supply group a specific transceiver Quad is located.
GTYP_AVTT	N/A	AVTT is the analog supply for the transmitter and receiver termination circuits of the transceiver Quad tile. Most packages have multiple groups of power supply connections in the package for AVTT. Refer to the package pin definitions to identify in which power supply group a specific transceiver Quad is located.
GTYP_AVCCAUX	N/A	AVCCAUX is the analog LCPLL voltage supply for the transceivers. Most packages have multiple groups of power supply connections in the package for AVCCAUX. Refer to the package pin definitions to identify in which power supply group a specific transceiver Quad is located.

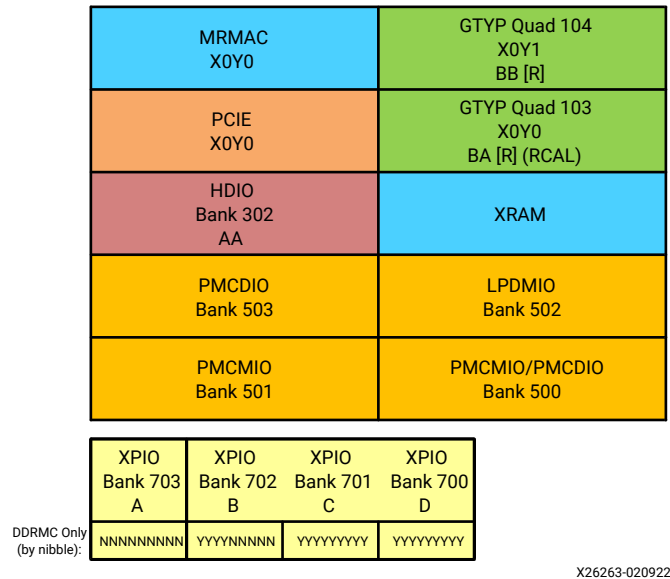
SYSMON Pin Definitions

Table 105: SYSMON Pin Definitions

Pin Name	Direction	Description
VREFP	N/A	This pin can be tied to an external 1.024V accurate reference IC for best performance of the ADC. This pin should be connected to GND_SMON if an external reference is not supplied.
VREFN	N/A	This pin should be tied to the ground pin of an external 1.024V accurate reference IC for best performance of the ADC. This pin should always be connected to GND_SMON even if an external reference is not supplied.
VP	Input	This is the positive input terminal of the dedicated differential analog input channel (VP/VN). When not used, tie to GND_SMON.
VN	Input	This is the negative input terminal of the dedicated differential analog input channel (VP/VN). When not used, tie to GND_SMON.

XQRVE2302 Bank Diagram

Figure 6: XQRVE2302 Banks in SSRA784 Package



GTYP Columns

- One GTYP Quad = Four transceivers = Four GTYP_QUAD primitives.
- Not all GTYP Quads are bonded out in every package.
- Also shown are quads labeled with RCAL. This specifies the location of the RCAL masters for each device. With respect to the package, the RCAL masters are located on the same package pin for each package, regardless of the device.
- The XY coordinates shown in each quad correspond to the transceiver channel number found in the pin names for that quad (as shown in the next section).
- An alphabetic designator is shown in each quad.
- The power supply group is shown in brackets [] for each quad.

I/O Banks

- Each user XPIO bank has a total of 54 I/Os that can be used as differential (27 differential pairs) or single-ended I/Os. All 54 pads of a bank are not always bonded out to pins.
- A limited number of XPIO banks have fewer than 54 SelectIO pins. These banks are labeled as partial.
- Each user HDIO bank has a total of 22 I/Os that can be used as pseudo-differential (11 differential pairs) or single-ended I/Os.
- Adjacent to each bank is a physical layer (PHY) containing clock buffers and clock management resources.
- XPIO banks are arranged in rows on the north and south of the device with adjacent PHY and clock management resources.
- XPIO banks are separated into groups of 1, 2, or 3 banks that signify how to group them when used for DDR memory controller applications.
- I/Os in some banks on the left and right edges of the XPIO rows do not have direct access to all I/O logic resources and can only be used for DDR memory controller (DDRMC) applications. These I/Os are specified in the following device diagrams, where each XPIO bank is designated by a Y (DDRMC only) or a N (access to all I/O logic resources) for each nibble (nine nibbles per bank and six I/Os per nibble). This is also specified in the *DDRMC Only* discussion in the [ASCII Package File](#).
- Global clock (GC) pins in *DDRMC Only* nibbles include full access to clock management resources.
- HDIO banks are arranged in columns and separated into rows that are pitch-matched with adjacent PHY, clock management resources, and GT blocks.
- An alphabetic designator is shown in each bank.

Clocking

- Each XPIO bank has four pairs of global clock (GC) inputs for four differential or four single-ended clock inputs. Single-ended clock inputs should be connected to the P-side of the differential pair.
- Each HDIO bank has two P-N pairs of global clock (HDGC) input pins. When used as a clock input, each P-N pair of global clock input pins can support a pseudo-differential pair, a true-differential pair, or a single-ended clock input that must be connected to the P-side of the pair. A limited selection of I/O standards are supported for HDGC clock inputs.
- Clock signals are distributed through global buffers driving routing and distribution networks to reach any clock region, I/O, or GTYP transceivers.
- Global clock inputs can connect to MMCMs and PLLs within the XPIO bank or adjacent clock management resources.

Processor Blocks

- PMC MIO pins are in banks 500 and 501
- LPD MIO pins are in bank 502
- PMC dedicated pins are in bank 503

PCIe and 40GE Integrated Blocks

- **PCIe:** Integrated block for PCIe®. The PCIe blocks with an additional (Tandem) label support tandem configuration.

- **MRMAC:** 40G Ethernet block

ASCII Package File

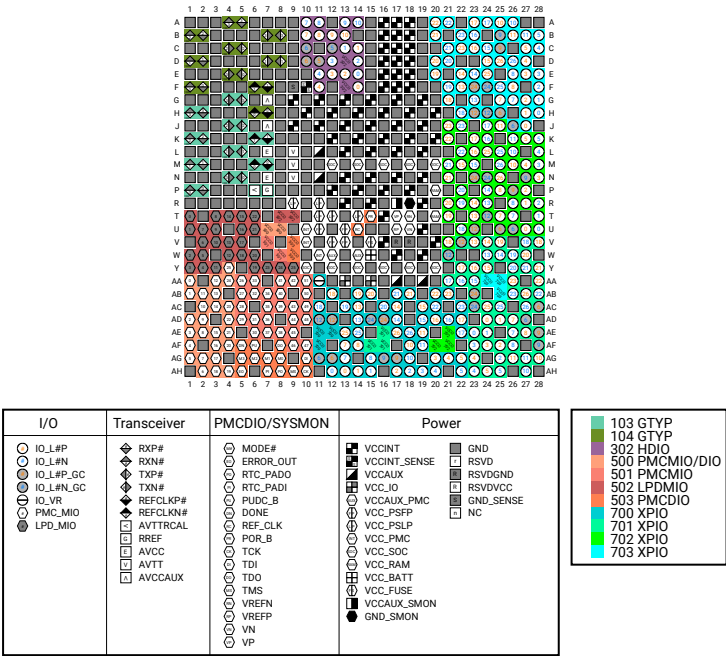
The ASCII package files for the XQRVE2302 are available at: <https://www.xilinx.com/support/package-pinout-files/versal-pkgs.html>. The information in each of the files includes:

- Header with device/package name (family-device-package), date and time of creation, package specification designator, and revision history
- Seven columns containing data for each pin:
 - Pin—Pin location on the package
 - Pin Name—The name of the assigned pin
 - Bank—Bank number
 - I/O Type—HDIO, XPIO, GTY, PMCMIO, PMCDIO, or LPDMIO depends on the I/O type. For more information on the I/O type, see *Versal Adaptive SoC SelectIO Resources Architecture Manual (AM010)*.
 - XPIOPerf—Performance that the LPDDR4 interfaces are specified. See the [DDR4 and LPDDR4/4X Memory Interface Controller](#) tables for performance definitions.
 - DDRMC Only—For XPIO pins, YES specifies that the pin is supported for use in DDR applications only and does not include full access to XPIO logic resources. YES_PLUS_GC specifies the pin is supported for use in DDR applications and additionally has full access to clock management resources. NO specifies the pin has full access to XPIO logic resources.
- Total number of pins in the package

SSRA784—XQRVE2302 Pin Map

The following figures show the top-view perspective of the package pinout of the XQRVE2302 in the SSRA784 package. The pin map shows the location of each user I/O, PMCMIO, LPDMIO, PMCDIO, and GTYP transceiver and the respective bank or GTYP quad, as well as the location of every power pin in the package.

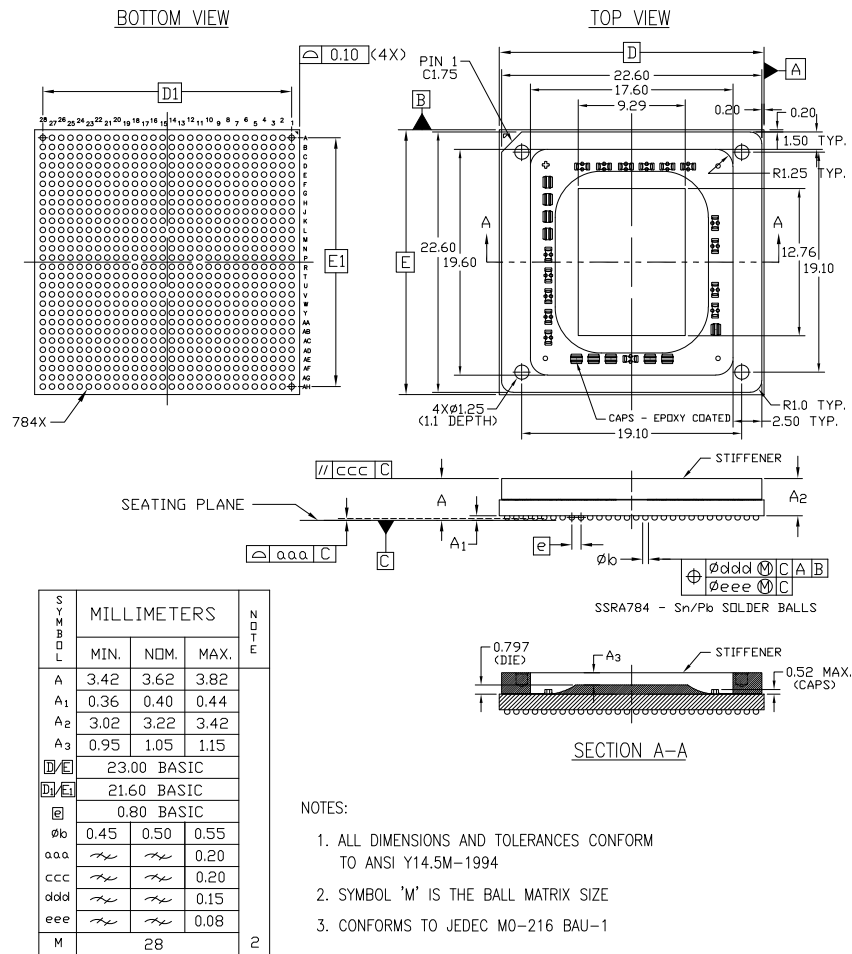
Figure 7: SSRA784 Package—XQRVE2302 Device Diagram



X26692-052322

SSRA784 Mechanical Drawing

Figure 8: Package Dimensions for SSRA784 (VE2302)



X29524-081324

Package Marking

Figure 9: AMD Versal™ XQR Devices Package Marking

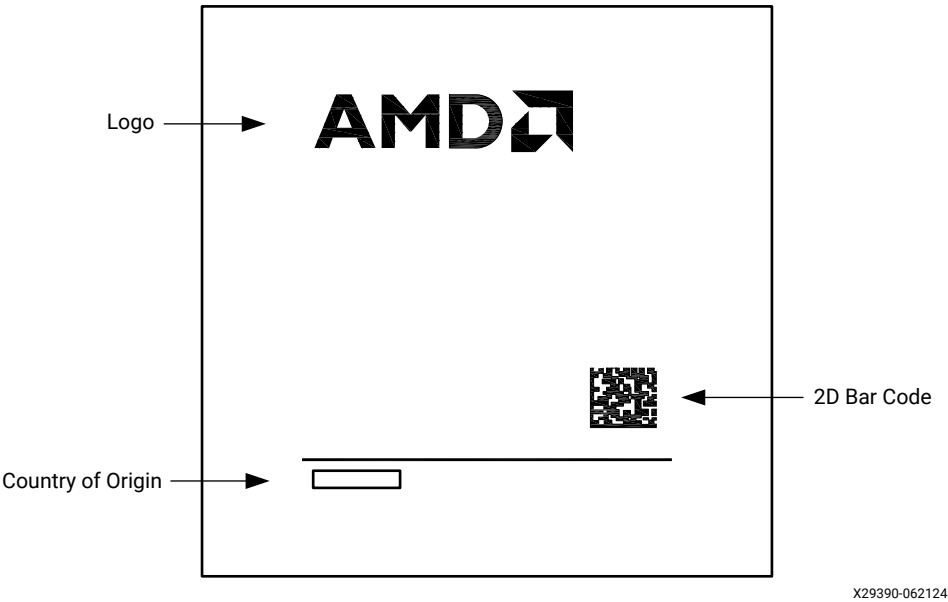


Table 106: Versal Device Marking Definition—Example

Item	Definition
AMD Logo	AMD logo, AMD name with trademark, and trademark-registered status.
Bar Code	A device-specific bar code is marked on each device.

Packing and Shipping

The XQRVE2302-SSRA784 devices are packed in trays. Trays are used to pack most of AMD surface-mount devices because they provide excellent protection from mechanical damage. In addition, they are manufactured using antistatic material to provide limited protection against ESD damage and can withstand a bake temperature of 125°C.

Table 107: Standard Device Counts per Tray and Box

Package	Maximum Number of Devices Per Tray	Maximum Number of Units In One Internal Box
SSRA784	60	300

Soldering Guidelines

To implement and control the production of surface-mount assemblies, the dynamics of the solder reflow process and how each element of the process is related to the end result must be thoroughly understood.

 **RECOMMENDED:** AMD recommends that customers qualify their custom PCB assembly processes using package samples.

The primary phases of the reflow process are:

- Melting the particles in the solder paste
- Wetting the surfaces to be joined
- Solidifying the solder into a strong metallurgical bond

The peak reflow temperature of a surface-mount component body should not be more than 220°C for eutectic-SnPb packages, and is package size dependent. For multiple BGAs in a single board and because of surrounding component differences, AMD recommends checking all BGA sites for varying temperatures.

The infrared reflow (IR) process is strongly dependent on equipment and loading. Components might overheat due to lack of thermal constraints. Unbalanced loading can lead to significant temperature variation on the board. These guidelines are intended to assist users in avoiding damage to the components; the actual profile should be determined by those using these guidelines. For complete information on package moisture / reflow classification and package reflow conditions, refer to the Joint IPC/JEDEC Standard J-STD-020C.

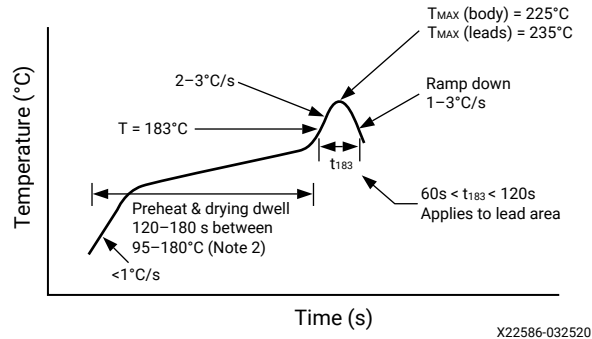
 IMPORTANT!

Following the initial placement and reflow process, devices should not be reflowed more than two additional times and should not be removed from the board. Any additional rework beyond that is likely to cause irreparable damage to the device.

Sn/Pb Reflow Soldering

The following figure shows typical conditions for solder reflow processing of Sn/Pb soldering using IR/convection. Both IR and convection furnaces are used for BGA assembly. The moisture sensitivity of surface-mount components must be verified prior to surface-mount flow.

Figure 10: Typical Conditions for IR Reflow Soldering of Sn/Pb Solder



Notes in figure:

1. Maximum temperature range = 225°C (body). Minimum temperature range before 205°C (leads/balls).
2. Preheat dwell 95–180°C for 120–180 seconds.
3. IR reflow must be performed on dry packages.

Peak Package Reflow Temperatures

For sophisticated boards with a substantial mix of large and small components, it is critical to minimize the ΔT across the board ($<10^{\circ}\text{C}$) to minimize board warpage and thus, attain higher assembly yields. Minimizing the ΔT is accomplished by using a slower rate in the warm-up and preheating stages.

It is also important to minimize the temperature gradient on the component, between top surface and bottom side, especially during the cooling down phase. The key is to optimize cooling while maintaining a minimal temperature differential between the top surface of the package and the solder joint area. The temperature differential between the top surface of the component and the solder balls should be maintained at less than 7°C during the critical region of the cooling phase of the reflow process. This critical region is in the part of the cooling phase where the balls are not completely solidified to the board yet. To efficiently cool the parts, divide the cooling section into multiple zones, with each zone operating at different temperatures.

The optimal profile must take into account the solder paste/flux used, the size of the board, the density of the components on the board, and the mix between large components and smaller, lighter components. Profiles should be established for all new board designs using thermocouples at multiple locations on the component. In addition, if there is a mixture of devices on the board, then the profile should be checked at various locations on the board, as shown in the following thermocouple pictures. Ensure that the minimum reflow temperature is reached to reflow the larger components and at the same time, the temperature does not exceed the threshold temperature that might damage the smaller, heat sensitive components.

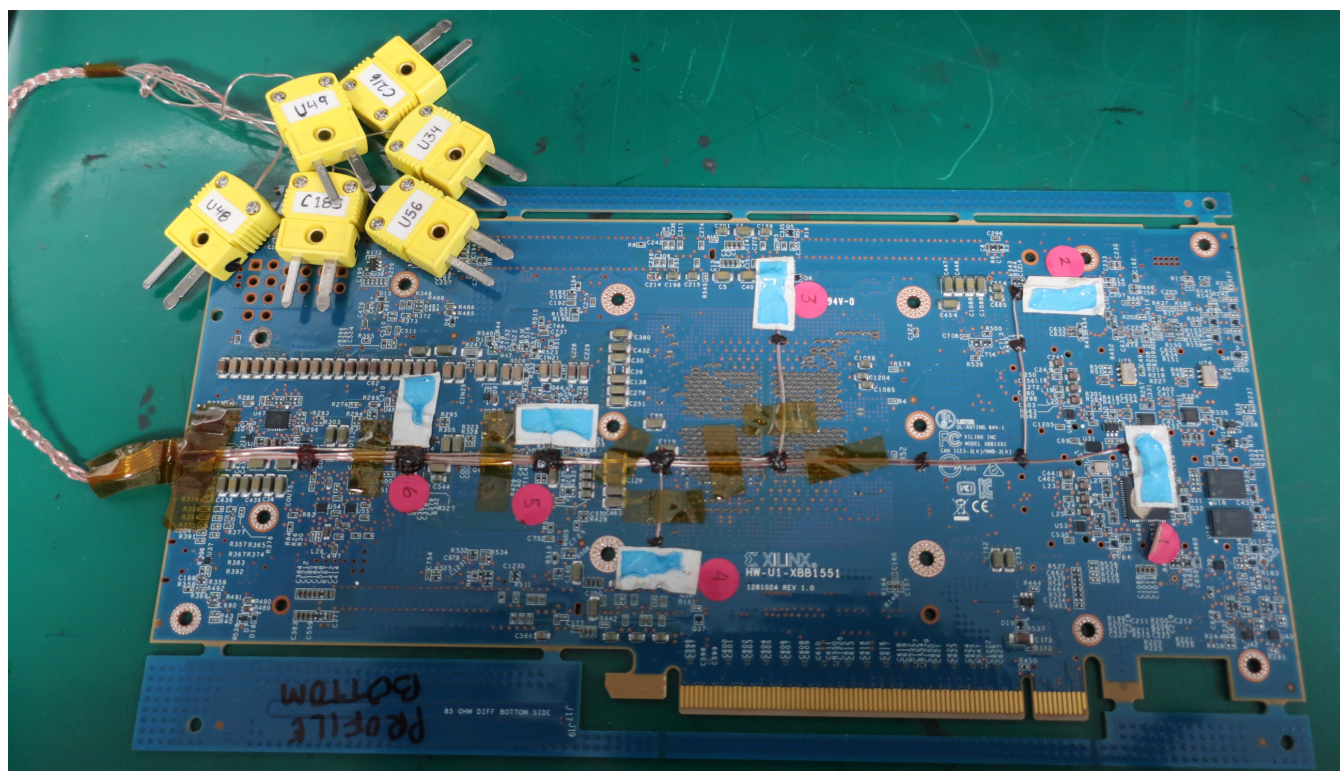
Figure 11: Thermocouple Top

TOP Profile,

- TC 1 : U17 Edge 1
- TC 2 : U17 Edge 2
- TC 3 : U17 Edge 3
- TC 4 : U17 Edge 4
- TC 5 : U17 Middle 1
- TC 6 : U17 Middle 2
- TC 7 : U17 Body
- TC 8 : Q25
- TC 9 : U25



Figure 12: Thermocouple Bottom



Post Reflow/Cleaning/Washing

Many PCB assembly subcontractors use a no-clean process in which no post-assembly washing is required. Although a no-clean process is recommended, if cleaning is required, AMD recommends a water-soluble paste and a washer using a deionized-water. Baking after the water wash is recommended to prevent fluid accumulation. Cleaning solutions or solvents are not recommended because some solutions contain chemicals that can compromise the stiffener adhesive, thermal compound, or components inside the package.

Strain Gauge Measurement

Strain gauge measurements are recommended to be done at each process step that has the potential to cause excessive board flexing leading to solder joint cracking. Assembly processes where strain gauge measurements are recommended include:

- PCB router (during PCB loading/unloading into fixture and during the routing process)
- PTH solder assembly during top-catch loading/unloading
- Press fit assembly during press base and tooling loading/unloading and during machine pressing process
- DIMM memory (during PCB loading/unloading and during insertion/removal of DIMM)
- Heat-sink assembly process (during PCB loading/unloading and during entire screw assembly process)
- X-ray fixture (during PCBA loading/unloading)

Strain gauge measurements should be in the range of $\pm 500 \mu\text{strain}$. Dye and pry analysis is required to confirm if the measured strain causes solder joint cracking. It is recommended to conduct dye and pry analysis for any strain reading greater than $500 \mu\text{strain}$. To reduce the affects of strain on a device, edge bonding can be used and is recommended for larger packages. See for implementation details.

Solder Paste

Solder paste consists of solder alloy and a flux system. A typical solder paste composition by volume is split between about 50% alloy and 50% flux. The metal load mass (solder alloy powder) is around 90%, with the remaining 10% mass a flux system. The primary purpose of the flux system is to remove the contaminants from the solder joints during the soldering process. The capability of removing contaminants is determined by the activation level of the type of solder paste. A no-clean solder paste is preferred to eliminate any risk of improper cleaning that could leave active residue beneath the device and other BTC components. The paste must be suitable for printing the solder stencil aperture dimensions. Type 4 paste is recommended for better paste release performance. When using a solder paste, you must adhere to the handling recommendations of the paste manufacturer.

Component Placement

 **IMPORTANT!** *The following component placement guidelines apply to the lidless packages described in this data sheet.*

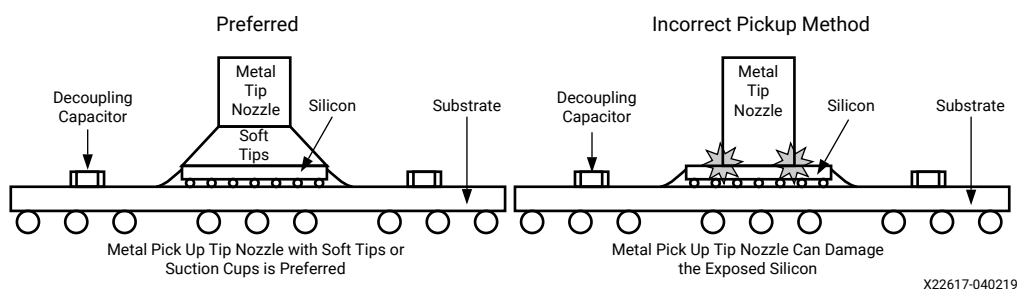
AMD device packages must be placed accurately according to their geometry outline. Positioning packages manually via hand mounting is not recommended. Typical component placement accuracies of $\pm 50\text{ }\mu\text{m}$ can be achieved using standard pick and place equipment with vision system. The PCB and components are optically checked and measured by the pick and place vision system. The pick and place machine detects the fiducial on the PCB immediately prior to the placement of the Versal device. Recognition of the packages is performed by the vision system, to ensure correct centering of the Versal device placement on the PCB pad array. BGA packages with solder balls can self-align during the reflow process because of the solder high surface tension that enables the pulling and centering of the device, and where a slight offset of the placement is still allowed. For guidance, the maximum tolerable offset of device placement is around 30% of the pad diameter on the PCB for typical non-solder mask defined pads. This means that the misalignment of solder ball device packages to the PCB pad must be better than $150\text{ }\mu\text{m}$ to assure a robust mounting process. Generally, this is achievable using a wide range of modern pick and placement systems. The following setup conditions are important for the pick and placement systems:

- The pick and place nozzle type should be selected based on the dimensions of the Versal device. The nozzle needs to firmly hold the device package during the pick and place stage. The appropriate nozzle type for the device package can be chosen from the manual provided by the pick and place equipment manufacturer.
- The ball recognition capabilities of the placement system should be used and package outline centering should be avoided. This eliminates the solder ball to package edge tolerances of the package. Refer to the specific package outline drawing for details.
- To ensure the proper identification of the device package during vision recognition, prior to its placement, a suitable lighting system and the correct choice of the features of the measuring method are essential. The most suitable settings can be chosen from the manual provided by the pick and place equipment manufacturer.
- To avoid solder bridging or solder smear, ensure the proper placement force of the device package during placement on the PCB. Excessive placement force can lead to excess solder paste and cause solder bridging. However, a slight placement force can lead to insufficient solder paste contact between the device package solder balls and the solder paste, causing solder defects including open solder joints, badly centered packages, or even head-in-pillow (HIP) defects.

Component Pick-up Tool Consideration

For pick and place machines to place lidless and bare-die flip-chip BGAs onto PCBs, AMD recommends using soft tips or suction cups for the nozzles. This prevents chipping, scratching, or even cracking of the bare die.

Figure 13: Recommended Method For Using Pick-up Tools

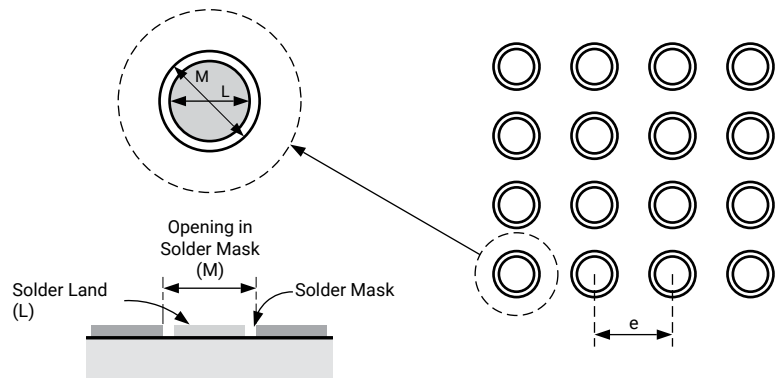


X22617-040219

Recommended PCB Design Rules for BGA

AMD provides the diameter of a land pad on the package side. This information is required prior to the start of the board layout so the board pads can be designed to match the component-side land geometry. The typical values of these land pads are shown in the following figure and summarized in the table. For AMD BGA packages, non-solder mask defined (NSMD) pads on the board are suggested to allow a clearance between the land metal (diameter L) and the solder mask opening (diameter M) as shown in the following figure.

Figure 14: Suggested Board Layout of Soldered Pads for BGA Packages



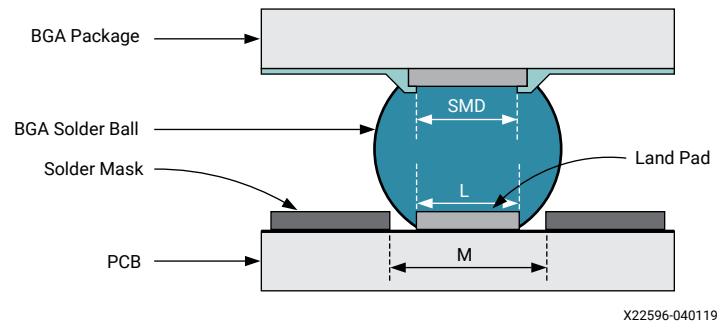
X22593-040119

Table 108: BGA Package Design Rules

Flip-Chip BGA Packages	1.0 mm Pitch	0.92 mm Pitch	0.80 mm Pitch
Design Rule	BGA packages, non-solder mask defined dimensions in mm (mils)		
Package land pad opening (SMD)	0.53 mm (20.9 mils)	0.53 mm (20.9 mils)	0.40 mm (15.7 mils)
Maximum PCB solder land (L) diameter	0.53 mm (20.9 mils)	0.51 mm (20.0 mils)	0.40 mm (15.7 mils)
Opening in PCB solder mask (M) diameter	0.63 mm (24.8 mils)	0.61 mm (24.0 mils)	0.50 mm (19.7 mils)
Solder ball land pitch (e)	1.00 mm (39.4 mils)	0.92 mm (36.2 mils)	0.80 mm (31.5 mils)

An example of an NSMD PCB pad solder joint is shown in the following figure. The space between the NSMD pad and the solder mask, as well as the actual signal trace widths, depend on the capability of the PCB vendor. The cost of the PCB is higher when the line width and spaces are smaller.

- RECOMMENDED:** AMD recommends not mixing PCB pad isolated via-in-pad plated over (VIPPO) and non-VIPPO design styles because they can cause hot-tear defects that are related to localized Z-direction thermal expansion coefficient mismatch between VIPPO and non-VIPPO vias. A VIPPO via expands less than a non-VIPPO via.
- RECOMMENDED:** AMD recommends not mixing PCB pad of solder mask defined (SMD) and non-solder mask defined (NSMD) design styles, because this can cause solder bridging defects. For violation, customers need to work with their CM to optimize the stencil design and assembly process.

Figure 15: Example of an NSMD PCB Pad Solder Joint

Stencil

Solder paste is applied to PCB metal pads by screen printing. The volume of the printed solder paste is determined by the stencil aperture and the stencil thickness. In most cases, the thickness of a stencil must be matched to the needs of all components on the PCB. Stencil apertures should be a circular shape. To ensure a uniform and high-solder paste transfer to the PCB, laser-cut stencil, made from mostly stainless steel, is typically used. Nickel Blank stencils, referring to stencils where the entire foil is laser-cut from a sheet of pure nickel material, can also be used. However, high-quality nano-coated stencils (laser cut from stainless steel) can perform as well as or better than Nickel Blanks.

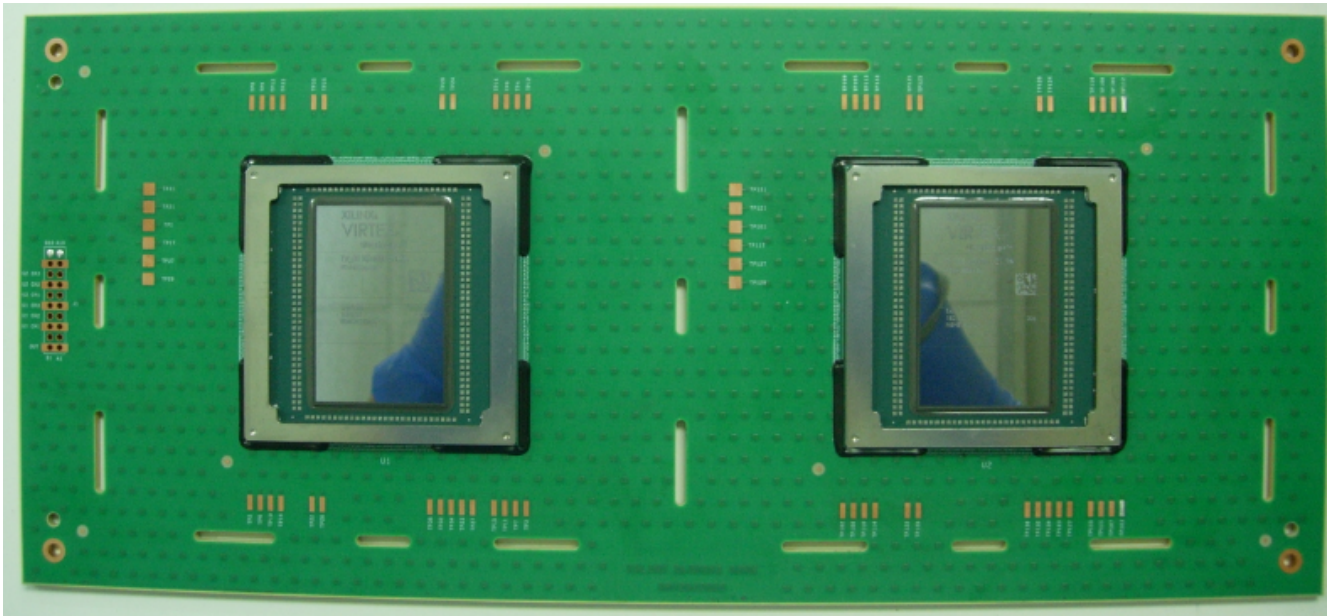
Uniform Stencil Aperture Design

- For packages with 0.92 mm ball pitch, a uniform stencil aperture opening of 19.7 mils to 20.0 mils round is recommended, matching the PCB pad size.
- AMD recommends using a 5 mil thick stencil.

Edge Bonding Guidelines

The edge bonding technique uses high-adhesion adhesives dispensed along the periphery of a component, as shown in the following figure. AMD recommends edge bonding for increased mechanical reliability in cases where the device is exposed to extensive temperature cycling or extreme shock and vibration, such as space, defense, and telecommunications applications. Designers are expected to evaluate the need for edge bonding based on the requirements of the specific application being implemented.

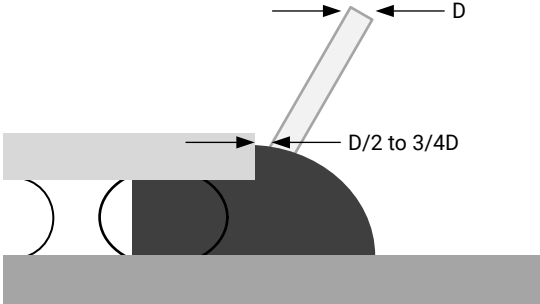
Figure 16: Edge Bonded BGA Packages



Edge Bonding Implementation

Edge bonding is the dispensing of an epoxy material around the periphery of the package after board mount. All guidelines herein are based on evaluation conducted using AMD components with Zymet UA-2605-B edge bonding material. This technique allows for component rework and improves the robustness of the mounted component by controlling the expansion and warpage of the board during normal operating conditions. To place the adhesive while using an in-line soldering robot, AMD recommends the parameters shown in the following table.

Table 109: Process Parameters for Edge Bonding

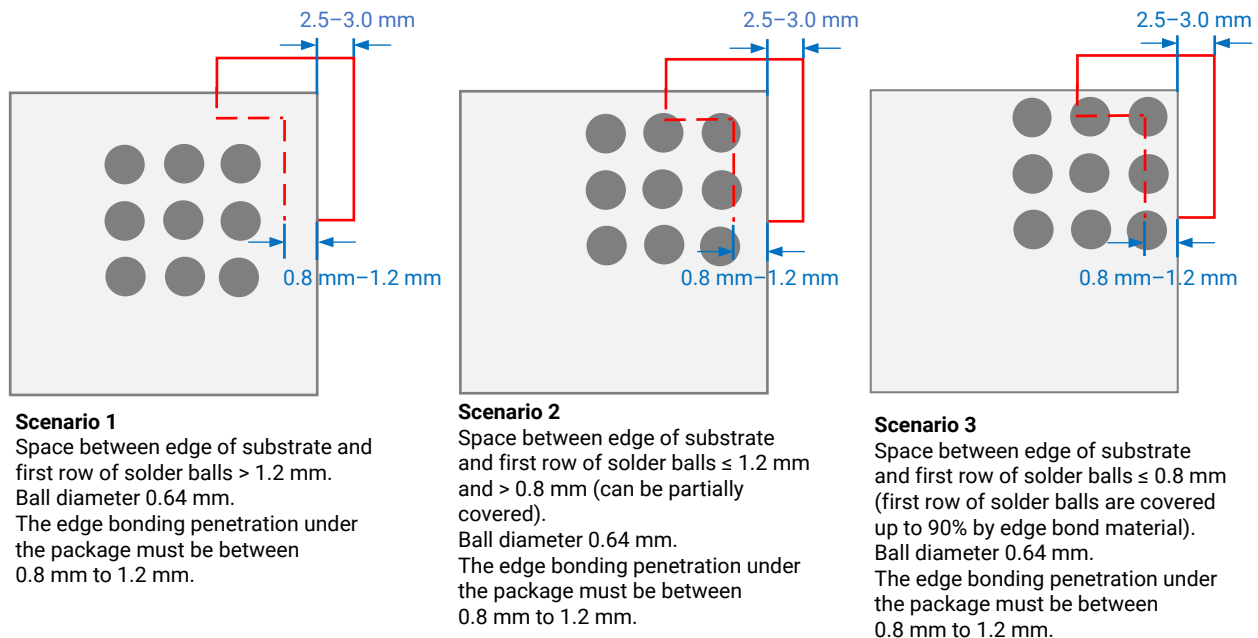
Process Parameter	Range Specification
Needle size	Range: 25—15 gauge, where needle diameter (D) is 20 gauge (0.91 mm) typical
Needle height	Range: Above the device edge midpoint, or 0 to 1.5 mm below the top surface of the device Recommended: 1.0 mm below the top surface of the device
Needle edge spacing	Range: Half to three quarters of needle outer diameter (D) Recommended: Half of needle diameter (D/2), approximately 1.0 mm 
Dispense needle speed	Range: 0.1 to 200 mm/second (6 mm/sec typical)
Value pressure	Range: 10 to 60 psi (14.5 psi typical)

The adhesive is dispensed along the perimeter of the assembled component at a width of 2.5–3.0 mm and a height of 50% to 90% the substrate height, leaving a small section at the center of each edge unbonded, as illustrated in the following figure. This is to ensure that there is an outlet for any expansion of the air during processing. AMD recommends centering the opening on each side with a width of 25–30% the length of the package substrate. The exact locations and size of the openings can be varied depending on the design and rework.

To ensure proper mechanical bonding between the package and PCB, the edge bonding feature must penetrate the edge bonding material under the package substrate by 0.8 mm. This requirement is necessary for packages with coplanarity $\geq 200\text{ }\mu\text{m}$, and BGA pitch of 1 mm or 0.92 mm, to pass physical qualification tests such as thermal cycle, power cycle, and shock and vibration requirements. It is necessary for special applications such as telecom, industrial, and data center applications.

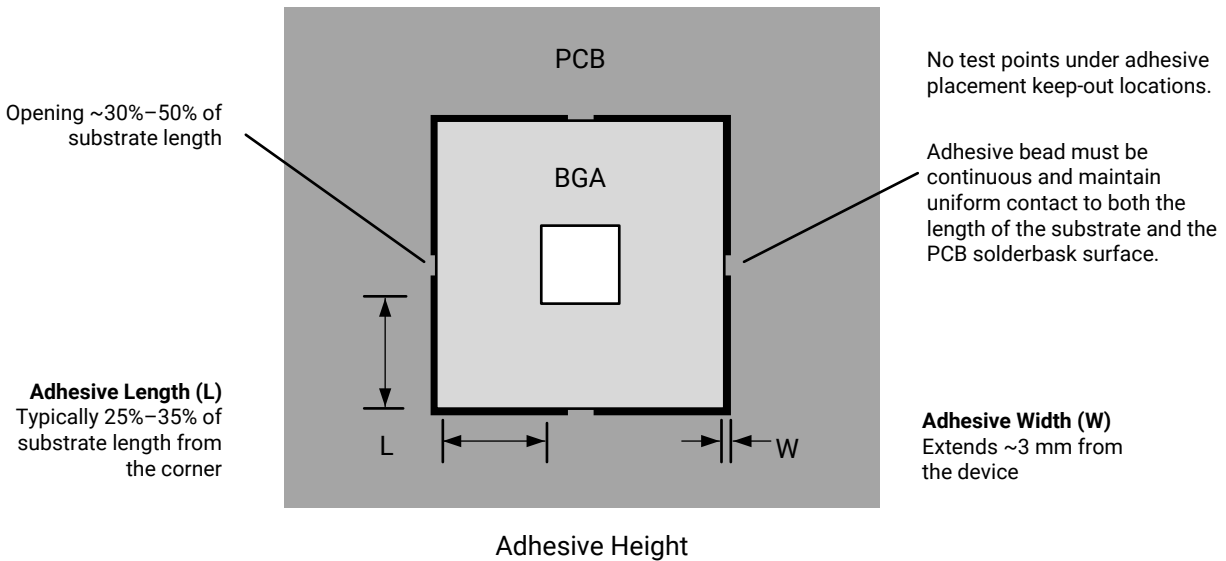
IMPORTANT! Some package devices have an overhang substrate that can result in the first row of the BGA not being covered by edge-bonding material. In other scenarios, the first BGA row could cover up to 90% of the BGA. See the following scenarios.

Figure 17: Edge Bonding Scenarios



X27880-092424

Figure 18: Edge Bonding Adhesive Placement Parameters



X23787-062322



RECOMMENDED: Curing condition of the [Zymet UA-2605-B](#) edge-bonding material is 140°C for 5 minutes.

Component Clearance Surrounding Edge Bond

An adjacent component clearance surrounding the AMD device is necessary to have the 30° to 45° angle required for the edge bond dispenser to dispense the edge bond adhesive material. The surrounding component height and distance from the device is validated based on each unique product design layout.

Edge Bond Removal

Edge bond material can be removed by heating to 170–180°C, and scraping using a stiff probe made of stable organic material such as a non-resinous wood or Teflon. Use a hot air blower on the edge bond area and slowly remove the edge bond adhesive from side to side. Do not use force to remove the edge bond adhesive. Excess adhesive on the PCB can be removed using a chisel-tip soldering iron, with sufficient precautions to limit damage to the PCB surface.

Conformal Coating

AMD does not have information regarding the reliability of flip-chip BGA packages on a board after exposure to any specific conformal coating process. Therefore, any process using conformal coating should be qualified for the specific use case to cover the materials and process steps.



RECOMMENDED: When a conformal coating is required, Parylene-based material should be used to avoid potential risk of damaging the package.

Thermal Specifications

AMD Versal™ adaptive SoCs are offered exclusively in thermally efficient flip-chip BGA packages. These flip-chip packages range in pin-count from the smaller 28 × 28 mm SSRA784 to the 52.5 × 52.5 mm VSRA2197. This suite of packages is used to address the various power requirements of the Versal adaptive SoCs. Versal devices are implemented in the 7 nm process technology. Unlike features in an ASIC, the combination of Versal adaptive SoC features used in a user application is not known to the component supplier. Therefore, it remains a challenge for AMD to predict the power requirements of a given Versal device when it leaves the factory. Accurate estimates are obtained when the board design takes shape. For this purpose, AMD offers and supports a suite of integrated device power analysis tools to help users quickly and accurately estimate their design power requirements. Versal adaptive SoCs are supported similarly to previous products. The uncertainty of design power requirements makes it difficult to apply canned thermal solutions to fit all users. Therefore, AMD devices do not come with preset thermal solutions. The operating conditions of your design dictate the appropriate solution.

Support for Thermal Models

Detailed 3D thermal models for all Versal devices are available for download on the AMD website under the [Device Models](#) tab.

AMD offers and supports a suite of integrated device power analysis tools to help you quickly and accurately estimate your design power requirements.

 **RECOMMENDED:** Download and fill out the latest version of the Xilinx Power Estimator (XPE) at <https://www.xilinx.com/power>.

The variability of design power requirements makes it difficult to apply pre-determined thermal solutions to fit all designs. The estimated power of the device using XPE, coupled with the your operating conditions and system constraints, dictate the appropriate solution. You must establish a thermal margin to account for the impact of the following (but not limited to) variations.

- Impact of the thermal sensor accuracy
- Impact of the power variation from one package to another package
- Impact of the manufacturing variation between fans
- Impact of the manufacturing variation between heat-sink fins
- Impact of the package manufacturing variation between heat-sink flatness
- Impact of the manufacturing variation of heat sink TIM contact/thickness

For example, when targeting maximum junction temperature (100°C) minus 8°C you must account for any additional system/environmental errors.

The detailed thermal model is a direct representation of the specified FCBGA package. The model provides geometric details describing the package, specifically in regards to the TIM, die, underfill, substrate, and solder balls or leads. Each specific component in the detailed model has material properties associated. When using this detailed model, several key facts must be considered:

- Two variations of the detailed thermal model are provided: Full detailed thermal model and simplified thermal model.
- The full detailed thermal model provides all the intricate geometries representing the physical package and can be used for mechanical evaluation of the device. However, it should not be used to perform thermal simulations because the detailed geometries create a large meshing size and high-computation time with little to no increase in overall accuracy.
- The simplified detailed model removes certain geometries including substrate trace layers and stiffener ring that do not have a material impact on thermal simulation accuracy and significantly increase computational time. This simplified model provides a much smaller meshing size and less computation time for simulation with less than a 5% difference in simulation accuracy from the full detailed model.
- The correct package top surface contact must be used when modeling the TIM on top of the detailed model because the top surface contact might not be the actual full package size.
- The junction temperature monitor point is at the die center by default. Junction temperature monitor points need to be positioned or added to the location of interest for your design.
- The uniform power defined in the die for the models provided is default power and does not apply to any specific cases. Power setting input is required.

Thermal Management Strategy

AMD relies on a multi-pronged approach to consuming less power and dissipating heat for systems using devices.

Flip-Chip Packages

AMD Versal™ devices are offered in flip-chip BGA packages, which present a low-resistance thermal path. Packages primarily consist of lidless with stiffener ring variations.

System Level Heat Sink Solutions

To complete a comprehensive thermal management strategy, an overall thermal budget that includes custom or OEM heat sink solutions depends on the physical and mechanical constraints of the system. A heat-sink solution, managed by the system-level designer, should be tailored to the design and specific system constraints. This includes understanding the inherent device capabilities for delivering heat to the surface.

By considering the system's physical, mechanical, and environmental constraints, the overall thermal budget is maintained and does not exceed the device's maximum operating temperature. The heat sink is an integral part of the thermal management solution to maintain a safe operating temperature. As a result, the system-level designer must be aware of the following:

- For lidless packages, the nominal stiffener height can be different from the height of the die. Therefore, the heat sink must have an island to contact the die.
- Especially for lidless packages, run thermal simulations of the system in worst-case environmental conditions using detailed thermal models, which accurately represent the device thermal performance under all boundary conditions.
- Consider the mechanical specifications of the package as well as the selection of the thermal interface between the die and the thermal management solution to ensure the lowest thermal contact resistance.

- The total thermal contact of the thermal interface material is determined based on parameters from the data sheet supplied by the thermal interface manufacturer.
- See the following applied pressure recommendation. Lower pressure runs the risk of poor thermal contact and higher pressure runs the risk of damaging the device; therefore, strict control of pressure is required. The use of a smart-torque tool is suggested when applying a heat sink to control the rate of short-term transient pressure as the thermal interface material (TIM) relaxes. Specifications from the TIM supplier should be taken into consideration.
- Consider all uncertainties in thermal modeling, including manufacturing variations from the thermal solutions (for example, fan airflow tolerance, heat pipe or vapor chamber performance tolerance, variation of the attachment of fins to heat sink base, and surface flatness).

Thermal Interface Material

When installing heat sinks for Versal devices, a suitable thermal interface material (TIM) must be used. This thermal material significantly aids the transfer of heat from the component to the heat sink.

For lidless and bare-die flip-chip BGAs, the surface of the silicon contacts the heat sink. For lidded flip-chip BGAs, the lid contacts the heat sink. The surface size of the lidless or bare-die flip-chip BGA and lidded flip-chip BGAs are different. AMD recommends a different type of thermal material for long-term use with each type of flip-chip BGAs package.

Thermal interface material is needed because even the largest heat sink and fan cannot effectively cool a Versal device unless there is good physical contact between the base of the heat sink and the top of the Versal device. The surfaces of both the heat sink and the Versal device silicon are not absolutely smooth. This surface roughness is observed when examined at a microscopic level. Because surface roughness reduces the effective contact area, attaching a heat sink without a thermal interface material is not sufficient due to inadequate surface contact.

A thermal interface material such as phase-change material, thermal grease, or thermal pads fills these gaps and allows effective transference of heat between the Versal device die and the heat sink.

The selection of the thermal interface material (TIM) between the package and the thermal management solution is critical to ensure the lowest thermal contact resistance. Therefore, the following parameters must be considered.

1. The flatness of the die and the flatness of the contact surface of the thermal solution.
2. The applied pressure of the thermal solution on the package, which must be within the allowable maximum pressure that can be applied on the package. The use of a smart-torque tool is suggested when applying a thermal solution to control the rate of short-term transient pressure as the TIM material relaxes. Specifications from the TIM supplier should be taken into consideration.
3. The total thermal contact of the thermal interface material. This value is determined based on the parameters in step 1 and step 2, which are published in the data sheet of the TIM supplier.

Types of TIM

There are many type of TIM available for sale. The most commonly used thermal interface materials are listed.

- Thermal grease

- Thermal pads
- Phase change material
- Thermal paste
- Thermal adhesives
- Thermal tape

Guidelines for Thermal Interface Materials

Five factors affect the choice, use, and performance of the thermal interface material (TIM) used between the device and the heat sink. Each factor is discussed in this section.

- Thermal conductivity of the material
- Electrical conductivity of the material
- Spreading characteristics of the material
- Long-term stability and reliability of the material
- Ease of application
- Applied pressure from heat sink to the package through the TIM

Thermal Conductivity of the Material

Thermal conductivity is the quantified ability of any material to transfer heat. The thermal conductivity of the interface material has a significant impact on its thermal performance. The higher the thermal conductivity, the more efficient the material is at transferring heat. Materials that have a lower thermal conductivity are less efficient at transferring heat, causing a higher temperature differential to exist across the interface. To overcome this less efficient heat transfer, a better cooling solution (typically, a more costly solution) must be used to achieve the desired heat dissipation.

Electrical Conductivity of the Material

Some metal-based TIM compounds are electrically conductive. Ceramic-based compounds are typically not electrically conductive. Manufacturers produce metal-based compounds with low-electrical conductivity, but some of these materials are not completely electrically inert. Metal-based thermal compounds are not hazardous to the Versal adaptive SoC die itself, but other elements on the Versal device or motherboard can be at risk if they become contaminated by the compound. For this reason, AMD does not recommend the use of electrically conductive thermal interface material.

Spreading Characteristics of the Material

The spreading characteristics of the thermal interface material determines its ability, under the pressure of the mounted heat sink, to spread and fill in or eliminate the air gaps between the Versal device and the heat sink. Because air is a very poor thermal conductor, the more completely the interface material fills the gaps, the greater the heat transference.

Long-Term Stability and Reliability of the Material

The long-term stability and reliability of the thermal interface material is described as the ability to provide a sufficient thermal conductance even after an extended time or extensive. Low-quality compounds can harden or leak out over time (the pump-out effect), leading to overheating or premature failure of the Versal device. High-quality compounds provide a stable and reliable thermal interface material throughout the lifetime of the device. Thermal greases with higher viscosity are typically more resistant to pump out effects on bare-die devices.

Ease of Application

A spreadable thermal grease requires the surface mount supplier to carefully use the appropriate amount of material. Too much or too little material can cause problems. The thermal pad is a fixed size and is therefore easier to apply in a consistent manner.

Applied Pressure from Heat Sink to the Package via Thermal Interface Materials

Measure applied pressure using a calibrated pressure sensor on multiple locations between the device and the heat sink assembly as shown in the following figure.

Figure 19: Pressure Sensor



RECOMMENDED: The following table shows the required applied pressure on the thermal interface material between the package and the heat sink. Thermocouples should not be present between the package and the heat sink, as their presence will degrade the thermal contact and result in incorrect thermal measurements. The best practice is to select the appropriate pressure for the optimum thermal contact performance between the package and the thermal system solution, and the mechanical integrity of the package (with the thermal solution to pass all mechanical stress and vibration qualification tests).

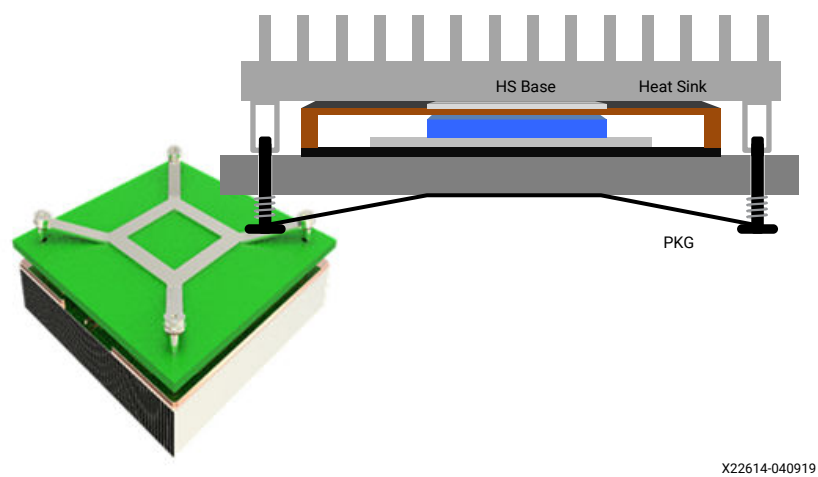
Table 110: Required Applied Pressure by Package

Package Type	Supported Pressure Range
0.80 mm pitch	20 to 50 PSI



RECOMMENDED: AMD recommends using dynamic mounting around the four corners of the device package. On the PCB, use a bracket clip as part of the heat sink attachment to provide mechanical package support. See the following figure.

Figure 20: Dynamic Mounting and Bracket Clips on Heat Sink Attachment



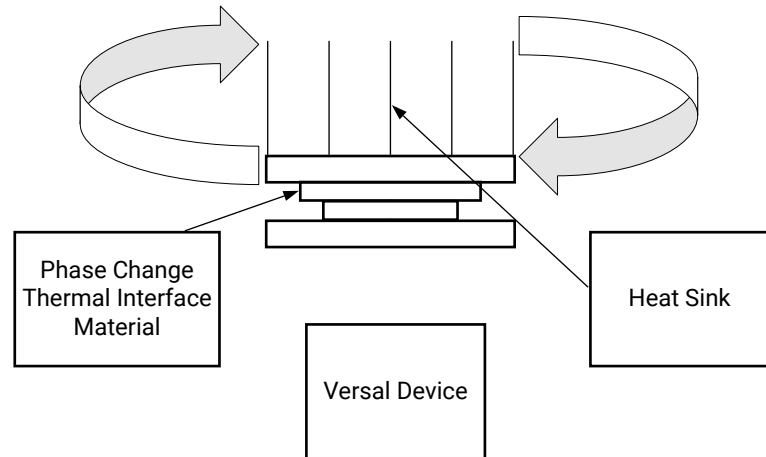
Heat Sink Removal

When removing or reworking heat sinks, the phase-change material residue must be removed from the surface of the die. Laird Technologies, Inc. provides the following guidance for complete removal of the phase-change material from the component.

Instructions for Removal of Phase-change Material

1. Separate the Components
- At room temperature, if possible, use a back and forth twisting motion to break the bond between the phase-change thermal interface material and mated components (i.e., heat sink and Versal device). See the following figure.

Figure 21: **Breaking the Bond between Thermal Interface Material and Mated Components**



X22613-032223

For smaller components (typically 15 mm x 15 mm or less), the bond usually breaks free easily at room temperature. For larger components, in situations where minimal movement is available, or if using fragile components, heat the component (preferred) or heat sink to about 40°C–60°C before removal.

The guideline is 40°C–60°C, however, you might find that for your application, heating to 35°C is adequate. You might prefer to heat to 70°C which makes the phase-change thermal interface material very soft and the components can be easily separated.

2. Scrape Away Thick Residue

For a faster clean-up once components are separated, scrape away any large residual material amounts with a plastic spatula or a wooden tongue depressor. A clean dry rag can be used to wipe away excess material.

3. Clean Remaining Residue with Solvent

Using a clean cloth/wipe, wet it with your choice of solvent (see the following list) and wipe away any remaining residue.

- Toluene (easiest)
- Acetone (very good)
- Isoparaffinic hydrocarbon: Isopar, Soltrol (trade names) (very good)
- Isopropyl alcohol (OK)

4. Working with Laird Material

Safe handling, disposal, and first-aid measures for working with phase-change material are included in the Laird Technologies material safety data sheet (MSDS). Read the MSDS before using or handling. See the [Laird Technologies, Inc.](#) website.

Measurement Debug

When performing in-system thermal testing, to ensure accurate data and not incur damage to the device, do not place a thermocouple in between the device and the heat sink. On the extreme side, it might cause additional mechanical and/or thermal stress to the device, leading to damage. Even if damage does not occur, it often leads to a thicker and or uneven thermal interface material thickness, leading to a thermal performance difference from a system without a thermocouple. To obtain the device temperature, use the System Monitor as a non-invasive means to get accurate device measurements while debugging the system.

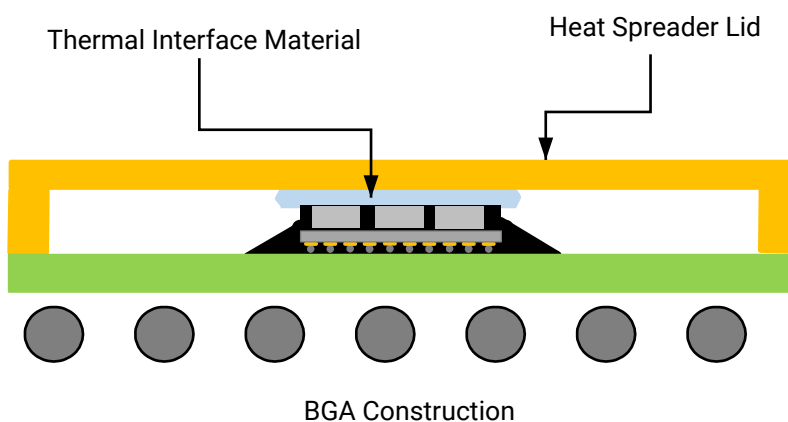
Mechanical and Thermal Design Guidelines for Lidless Flip-chip Packages

This section discusses the challenges of thermal management including reducing device thermal resistance and optimal power dissipation without an increase in junction temperature. The lidless Versal device packages target the largest devices while allowing for cooler operation temperatures (up to 10°C) with the same power dissipation. Precise mechanical design and component thermal management is vital for device and system performance. This document presents the unique thermal and mechanical design requirements for lidless devices.

Lidless Flip-Chip Packages

The AMD lidded flip-chip BGA packages exhibit a low-resistance thermal path that adequately cools devices. These packages incorporate a heat spreader lid with additional thermal interface material as shown in the following figure.

Figure 22: Lidded Flip-Chip BGA Construction with Heat Spreader and Thermal Interface Material

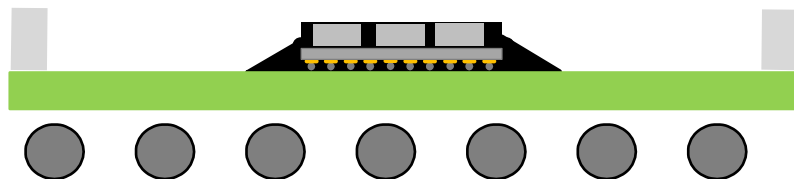


X23792-032620

Materials with high thermal conductivity and consistent process applications deliver low thermal resistance up to the heat spreader. A parallel effort to ensure optimized package electrical return paths produces an enhanced power and ground plane arrangement in the package. A boost in copper density on the planes improves the overall thermal conductivity through the laminate. The extra density and distribution via fields in the package also increases the vertical thermal conductivity.

The lidless packages (see the following figure) offer the same package substrate design with electrical and board thermal conductivity similar to the flip-chip BGA packages. However, removing the lid (heat spreader) and the thermal interface material allows for direct contact between the external heat sink and the die. This further reduces the thermal resistance and exhibits improved thermal behaviors. The use of custom passive or active heat-sink designs is facilitated by incorporating two-phase (heat pipe, vapor chamber, or even liquid) cooling methods directly adjacent to the source of the dissipated heat on the die, which allows for a more efficient means of removing the heat from the device. Consequently, the device can operate at higher ambient temperatures while in area-constrained surroundings resulting in operational power advantages.

Figure 23: Lidless Flip-Chip BGA Construction



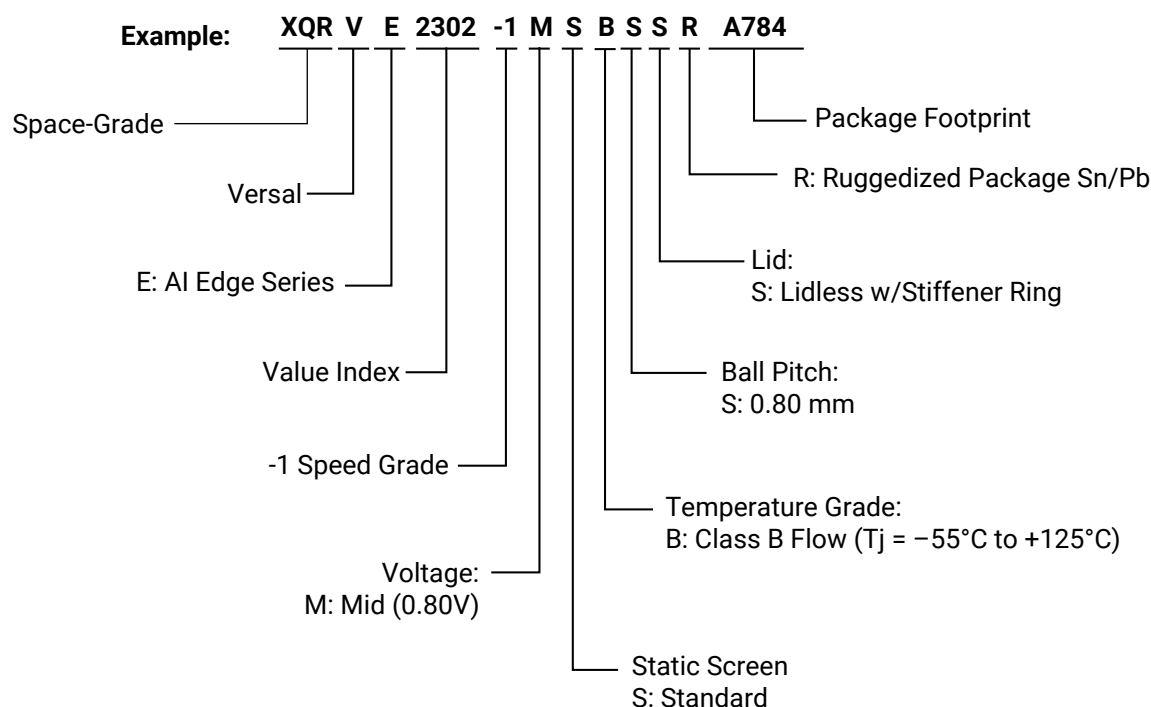
X23791-032620

A unique feature of lidless packages is the addition of a stiffener ring around the periphery of the package substrate, providing additional package rigidity and helping to improve the overall package coplanarity (flatness). It also serves as a guide for the heat sink solution applied to the device. For examples, see the [SSRA784 Mechanical Drawing](#). In lidless packages, capacitors can be placed in the area surrounding the die. Contact with electrically conductive materials must be avoided because the die-side capacitors, which are only slightly shorter than the die height, could be electrically conductive. Any thermal and mechanical solution higher than the die must not interfere with the package stiffener. Therefore, the thermal solution must have an island, see [System Level Heat Sink Solutions](#). For further guidelines on mechanical and thermal designs of lidless packages, refer to *Designing Heatsinks and Thermal Solutions for Xilinx Devices* ([XAPP1377](#)).

Ordering Information

The ordering information for the XQRVE2302 is shown in the following figure.

Figure 24: Radiation Tolerant XQRE2302-1MSBSSRA784 Ordering Information



X29562-082724

AMD class B manufacturing flows are consistent with MIL-PRF-38535 and are shown in the following table.

Table 111: AMD Class B Manufacturing Flows

Flow	AMD Class B	Comments
Wafer sort	✓	
Bumping	✓	
Assembly	✓	Commercial organic assembly
Bond pull (extended pull test)	N/A	No bond pull on flip-chip BGAs (not wire bond)
Die peel (1 unit/lot)	✓	
Die visual inspection	✓	
Pre-cap source inspections (cond A)	N/A	Lidless package, pre-cap source inspection not required
Serialization	✓	
Temperature cycling (cycles)	15	
Constant acceleration	N/A	Not required for flip-chip packages
PIND	N/A	Lidless package, no sealed cavity
Seal (fine/gross leak test)	N/A	Lidless package, no sealed cavity
X-Ray	N/A	Not required on flip-chip package
Pre burn-in electrical test	@ 25°C, 125°C, -55°C	
Dynamic burn-in (@ 125°C)	160 hours	
Post burn-in test @ 25°C with read & record	N/A	Not required for Class B

Table 111: AMD Class B Manufacturing Flows (cont'd)

Flow	AMD Class B	Comments
Static burn-in (144 hours @ 125°C)	N/A	Not required for Class B
Group A post burn-in test @ 25°C with read & record	✓	
Group A final test @ -55°C with read & record	✓	
Group A final test @ 125°C with read & record	✓	
Column attach	N/A	BGA, no columns
100% QA electrical @ 25°C	N/A	BGA balls are applied prior to electrical test and burn-in
Visual inspection	✓	
Group B lot specific	✓	
Group C sample to 44k device hours	✓	
Group D	✓	
Group E total ionizing dose	N/A	Not required for Class B
DPA/Ion milling	N/A	Not required for Class B

Revision History

The following table shows the revision history for this document.

Section	Revision Summary
1/13/2025 Version 1.0	
Initial release.	N/A

Please Read: Important Legal Notices

The information presented in this document is for informational purposes only and may contain technical inaccuracies, omissions, and typographical errors. The information contained herein is subject to change and may be rendered inaccurate for many reasons, including but not limited to product and roadmap changes, component and motherboard version changes, new model and/or product releases, product differences between differing manufacturers, software changes, BIOS flashes, firmware upgrades, or the like. Any computer system has risks of security vulnerabilities that cannot be completely prevented or mitigated. AMD assumes no obligation to update or otherwise correct or revise this information. However, AMD reserves the right to revise this information and to make changes from time to time to the content hereof without obligation of AMD to notify any person of such revisions or changes. THIS INFORMATION IS PROVIDED "AS IS." AMD MAKES NO REPRESENTATIONS OR WARRANTIES WITH RESPECT TO THE CONTENTS HEREOF AND ASSUMES NO RESPONSIBILITY FOR ANY INACCURACIES, ERRORS, OR OMISSIONS THAT MAY APPEAR IN THIS INFORMATION. AMD SPECIFICALLY DISCLAIMS ANY IMPLIED WARRANTIES OF NON-INFRINGEMENT, MERCHANTABILITY, OR FITNESS FOR ANY PARTICULAR PURPOSE. IN NO EVENT WILL AMD BE LIABLE TO ANY PERSON FOR ANY RELIANCE, DIRECT, INDIRECT, SPECIAL, OR OTHER CONSEQUENTIAL DAMAGES ARISING FROM THE USE OF ANY INFORMATION CONTAINED HEREIN, EVEN IF AMD IS EXPRESSLY ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

AUTOMOTIVE APPLICATIONS DISCLAIMER

AUTOMOTIVE PRODUCTS (IDENTIFIED AS "XA" IN THE PART NUMBER) ARE NOT WARRANTED FOR USE IN THE DEPLOYMENT OF AIRBAGS OR FOR USE IN APPLICATIONS THAT AFFECT CONTROL OF A VEHICLE ("SAFETY APPLICATION") UNLESS THERE IS A SAFETY CONCEPT OR REDUNDANCY FEATURE CONSISTENT WITH THE ISO 26262 AUTOMOTIVE SAFETY STANDARD ("SAFETY DESIGN"). CUSTOMER SHALL, PRIOR TO USING OR DISTRIBUTING ANY SYSTEMS THAT INCORPORATE PRODUCTS, THOROUGHLY TEST SUCH SYSTEMS FOR SAFETY PURPOSES. USE OF PRODUCTS IN A SAFETY APPLICATION WITHOUT A SAFETY DESIGN IS FULLY AT THE RISK OF CUSTOMER, SUBJECT ONLY TO APPLICABLE LAWS AND REGULATIONS GOVERNING LIMITATIONS ON PRODUCT LIABILITY.

Copyright

© Copyright 2025 Advanced Micro Devices, Inc. AMD, the AMD Arrow logo, Versal, Vivado, and combinations thereof are trademarks of Advanced Micro Devices, Inc. AMBA, AMBA Designer, Arm, ARM1176JZ-S, CoreSight, Cortex, PrimeCell, Mali, and MPCore are trademarks of Arm Limited in the US and/or elsewhere. PCI, PCIe, and PCI Express are trademarks of PCI-SIG and used under license. The DisplayPort Icon is a trademark of the Video Electronics Standards Association, registered in the U.S. and other countries. Other product names used in this publication are for identification purposes only and may be trademarks of their respective companies.

JEDEC standards and publications are copyrighted by the JEDEC® Solid State Technology Association. All rights reserved.